



NATIONAL UNIVERSITY OF SINGAPORE

EE4415: Integrated Digital Design

Part II - HW01

A report submitted to

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College of Design and Engineering

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Q1(a) - Id-Vd Curves

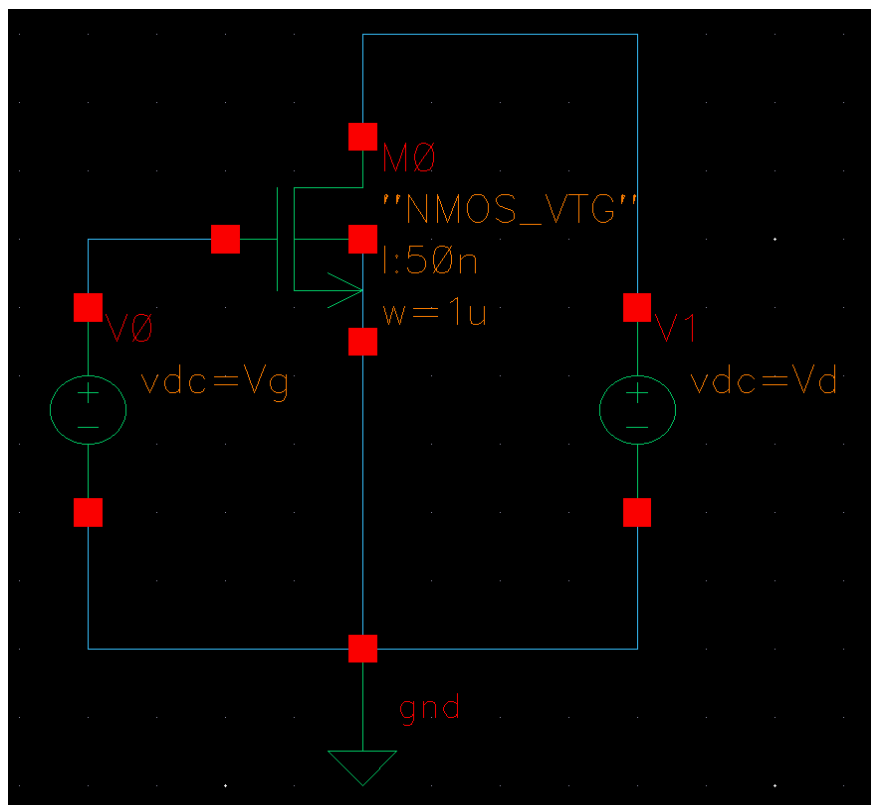
(a) NMOS & PMOS Id-Vds Curves:

Objective

To obtain the ID-VDS characteristics of an NMOS transistor for different gate voltages and analyze its operating behavior.

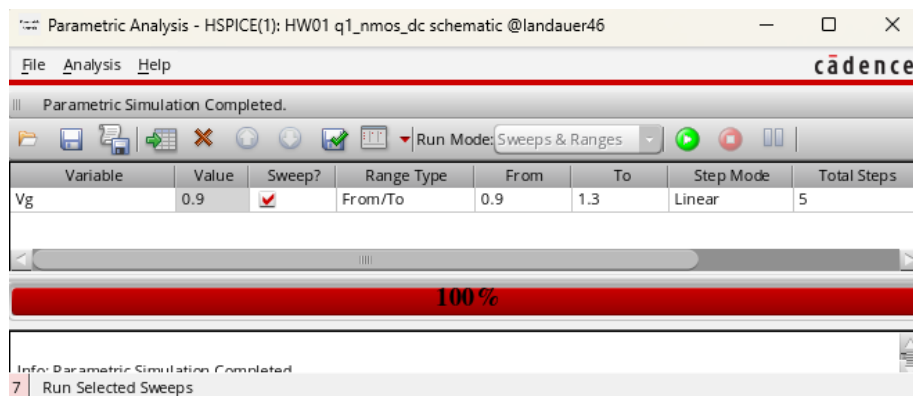
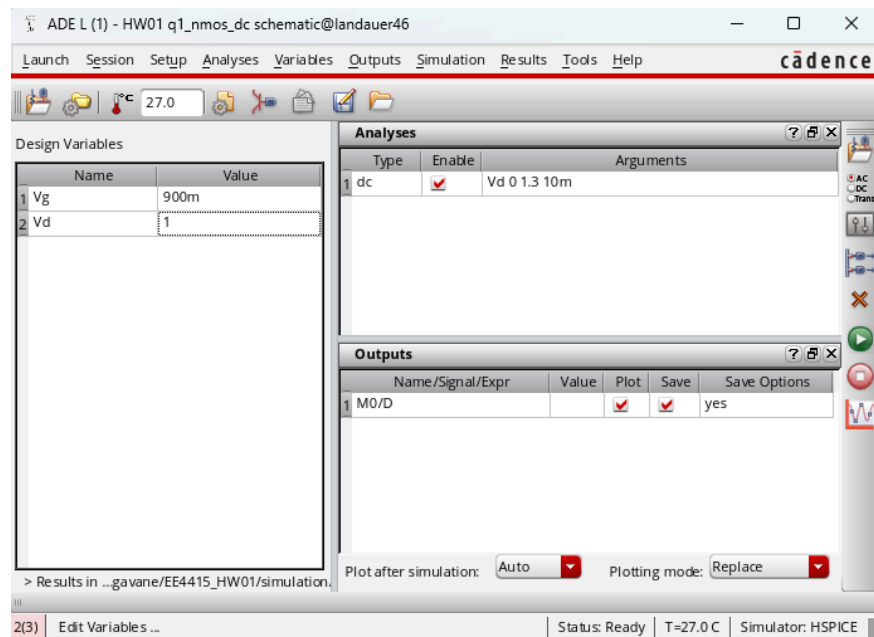
Setup

- NMOS (FreePDK45)
- $W=1\text{ }\mu\text{m}$ $L=50\text{nm}$
- Source and bulk connected to ground
- Gate biased by V_G , drain biased by V_D



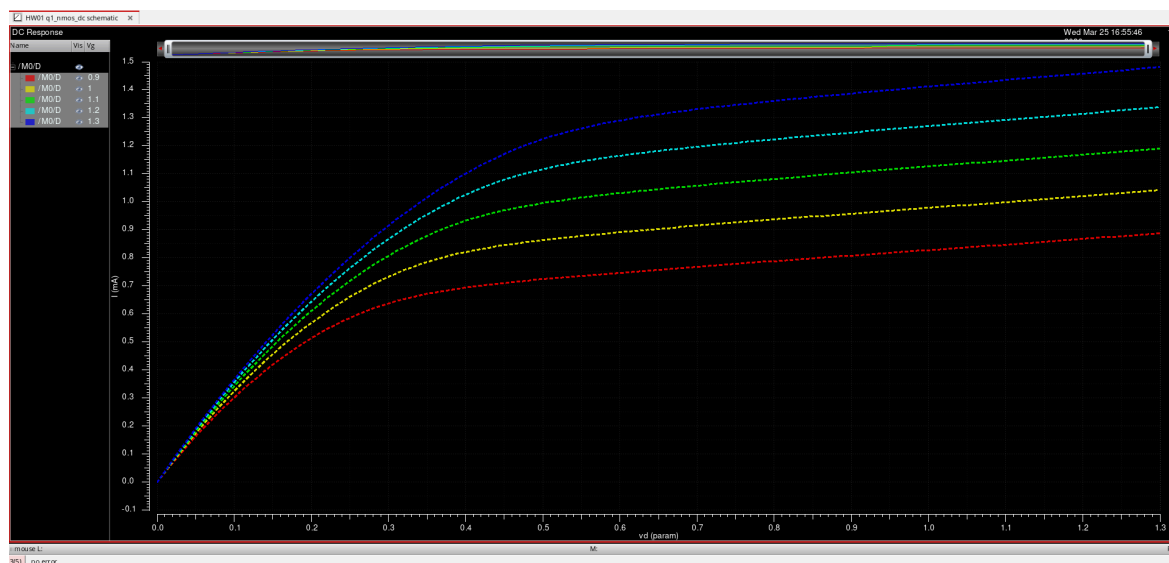
Simulation

- DC sweep:
 $V_D=0 \rightarrow 1.3\text{ V}$
- Parametric sweep:
 $V_G=0.9, 1.0, 1.1, 1.2, 1.3\text{ V}$



Results

- A single plot with 5 curves corresponding to different VGS values
- Drain current increases with increasing VGS



Discussion

- At low values of V_{DS} , the drain current increases approximately linearly, indicating operation in the linear (triode) region, where the transistor behaves like a voltage-controlled resistor.
- As V_{DS} increases, the current begins to saturate, marking the transition to the saturation region, where the channel is pinched off near the drain.
- However, the current in saturation is not perfectly constant and shows a noticeable slope. This deviation from ideal behavior is due to channel length modulation, where the effective channel length reduces with increasing V_{DS} , leading to a slight increase in current.
- Additionally, higher values of V_{GS} result in stronger channel formation, which increases the overall drain current and shifts the curves upward.

Conclusion

- The NMOS transistor exhibits expected I_D - V_{DS} characteristics with clear linear and saturation regions. The non-flat saturation behavior indicates the presence of short-channel effects.
- Since the channel length is 50 nm, the device operates as a short-channel MOSFET, and its behavior deviates from the ideal long-channel model due to effects such as channel length modulation.

PMOS

Objective

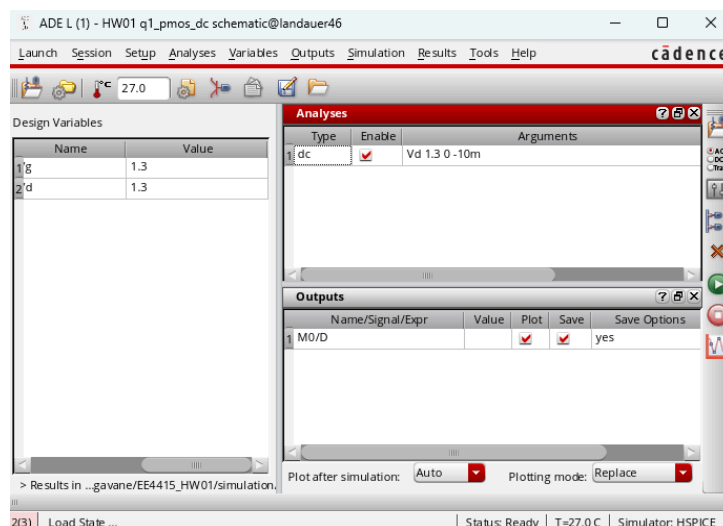
To obtain I_D - V_{DS} characteristics of a PMOS transistor for different gate voltages and analyze its operating behavior.

Setup

- PMOS (FreePDK45)
- $W = 1\ \mu\text{m}$, $L = 50\ \text{nm}$
- Source and bulk connected to $V_{DD} = 1.3\ \text{V}$
- Gate biased by V_g
- Drain biased by V_d

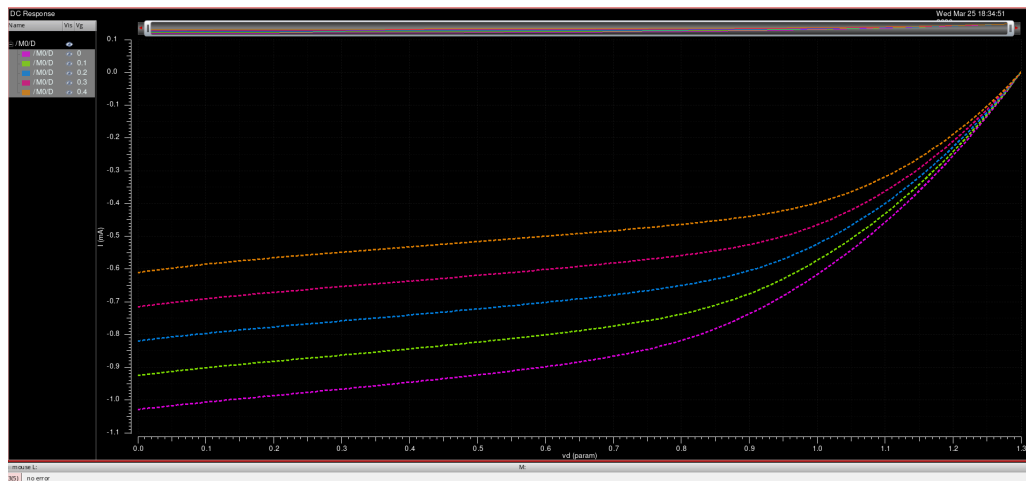
Simulation

- DC sweep:
 $V_d = 1.3 \rightarrow 0\ \text{V}$ (step = $-0.01\ \text{V}$)
- Parametric sweep:
 $V_g = 0.4, 0.3, 0.2, 0.1, 0\ \text{V}$



Results

- A single plot with 5 curves corresponding to different VGS values
- Drain current magnitude increases with increasing $|V_{GS}|$



Discussion

- At low values of V_{DS} , the drain current increases approximately linearly, indicating operation in the linear (triode) region, where the transistor behaves like a voltage-controlled resistor.
- As V_{DS} increases, the current begins to saturate, marking the transition to the saturation region, where the channel is pinched off near the drain.
- The saturation region is not perfectly flat and shows a noticeable slope. This is due to channel length modulation, where the effective channel length decreases with increasing V_{DS} , leading to a gradual increase in current.
- Higher values of $|V_{GS}|$ result in stronger channel formation, increasing the magnitude of the drain current and shifting the curves further downward (or upward if plotted as positive current).

Conclusion

- The PMOS transistor exhibits expected I_D - V_{DS} characteristics with clear linear and saturation regions. The presence of a non-flat saturation region indicates short-channel effects.
 - Since the channel length is 50 nm, the device operates as a short-channel MOSFET, and its behavior deviates from the ideal long-channel model due to effects such as channel length modulation.
-

Q1(b) - ON Resistance Determination

Objective

To determine the equivalent ON resistance of NMOS and PMOS transistors for $V_{DD} = 0.9\text{ V}$, 1.0 V , 1.1 V , 1.2 V , and 1.3 V using the two-end point method from the lecture notes and DC simulations in Cadence Virtuoso.

Setup

Technology used: FreePDK45

NMOS and PMOS dimensions: $W = 1\text{ }\mu\text{m}$, $L = 50\text{ nm}$

For NMOS:

- Source connected to ground
- Bulk connected to ground
- Gate biased by V_G
- Drain biased by V_D

For PMOS:

- Source connected to 1.3 V
- Bulk connected to 1.3 V
- Gate biased by V_G
- Drain biased by V_D

Method

The equivalent ON resistance was calculated using the two-end point method.

$$R(V_{DD}) = V_{DD} / |I_D \text{ at } V_{DD}|$$

$$R(V_{DD}/2) = (V_{DD}/2) / |I_D \text{ at } V_{DD}/2|$$

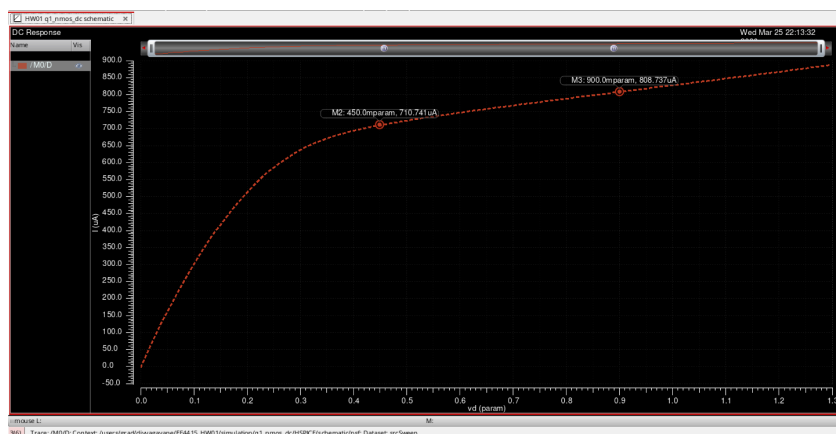
$$R_{eq} = [R(V_{DD}) + R(V_{DD}/2)] / 2$$

For PMOS, the simulated drain current was negative due to polarity convention, so the magnitude of current, $|I_D|$, was used in all calculations.

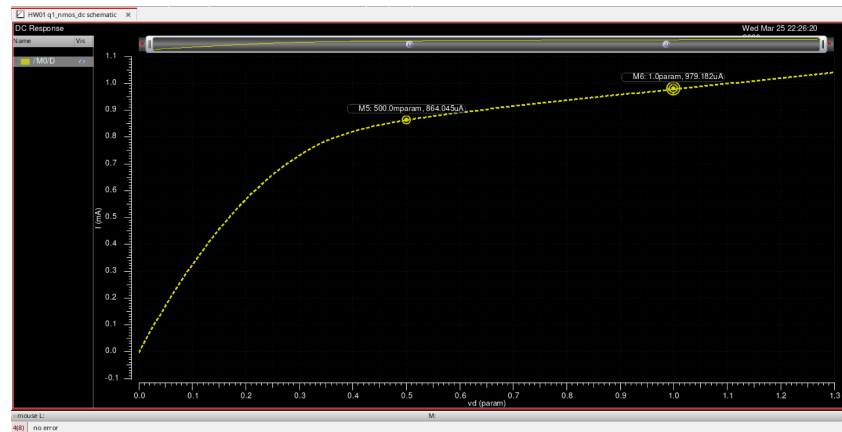
Results

NMOS:

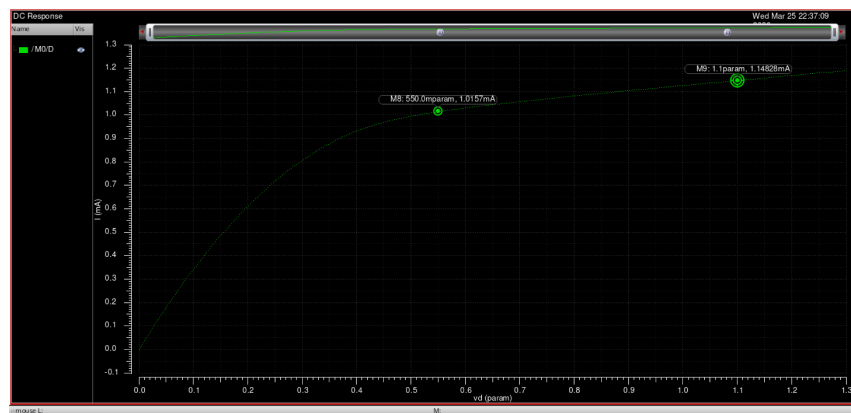
$$V_g = 0.9\text{ V}$$



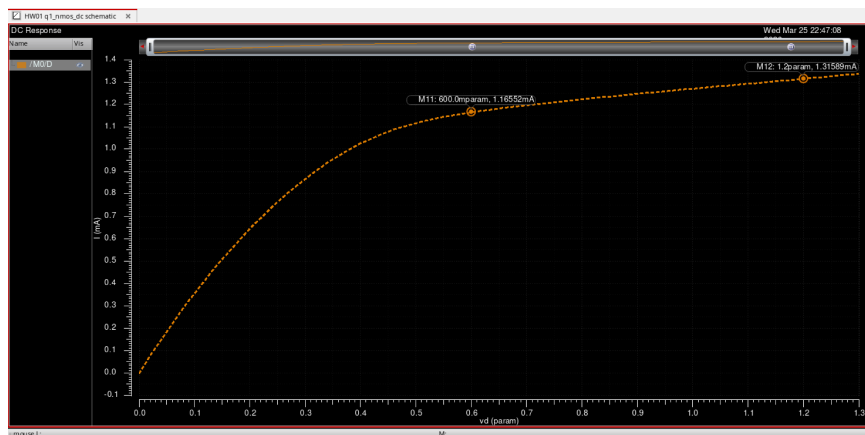
$$V_g = 1\text{ V}$$



$V_g = 1.1V$



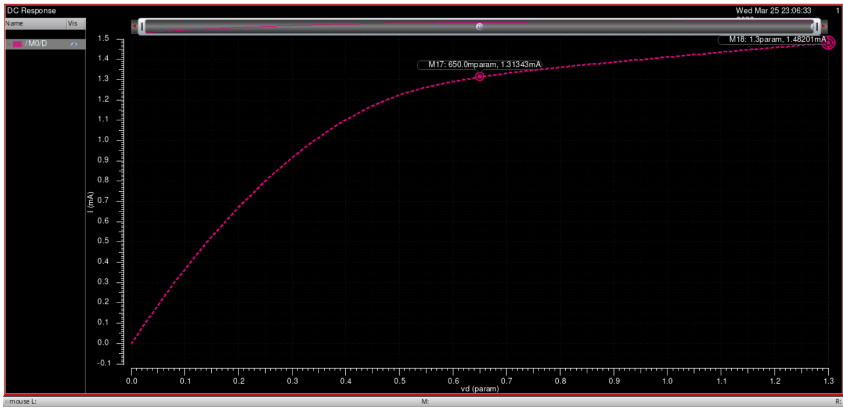
$V_g = 1.2V$



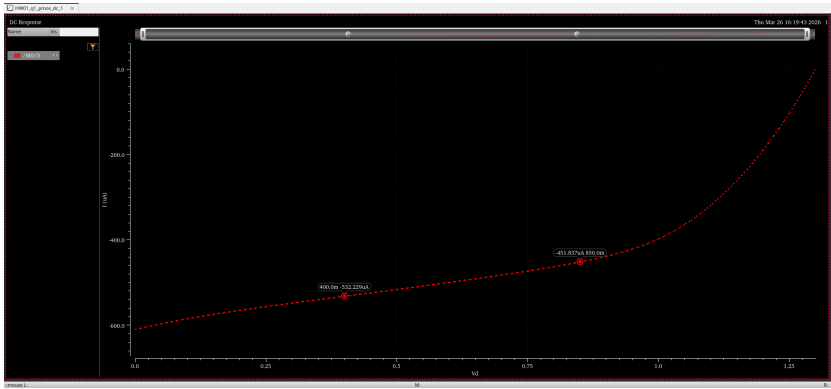
$V_g = 1.3V$

VDD	VDD/2	ID @ VDD	ID @ VDD/2	R(VDD)	R(VDD/2)	Req_n	4 significant figures
0.9	0.45	8.09E-04	7.1072E-04	1.1128e+03	6.3314E+02	8.7299E+02	8.730E+02

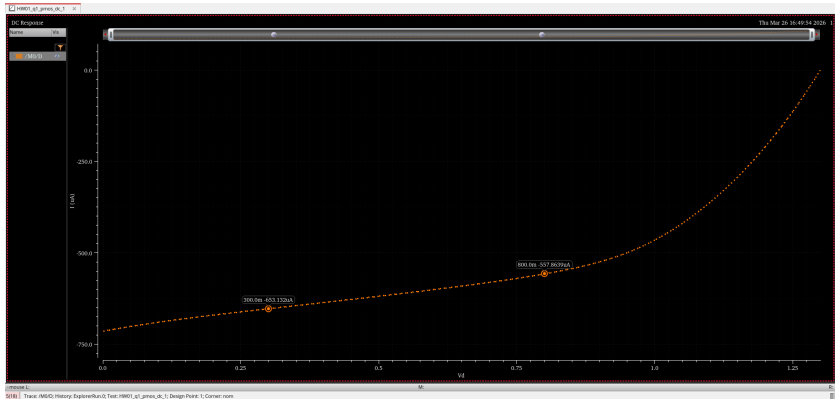
1	0.5	8.79E-04	8.6405E-04	1.0213E+03	5.7867E+02	7.9997E+02	8.000E+02
1.1	0.55	1.0157E-03	1.1483E-03	1.0830E+03	4.7898E+02	7.8099E+02	7.810E+02
1.2	0.6	1.32E-03	1.16552E-03	9.12E+02	5.15E+02	7.13E+02	7.134E+02
1.3	0.65	1.48E-03	1.31E-03	8.77E+02	4.95E+02	6.86E+02	6.860E+02



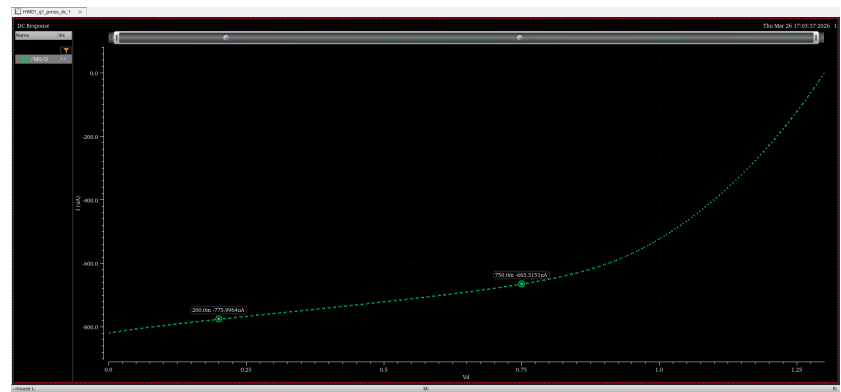
PMOS:
Vdd=0.9V | Vg = 0.4V



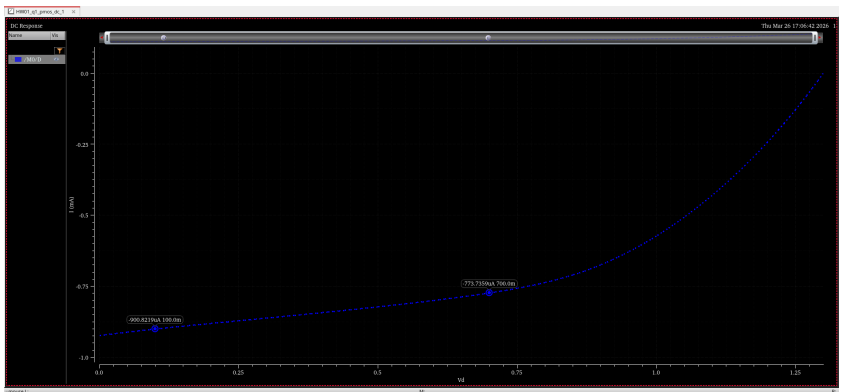
Vd=1V | Vg = 0.3V



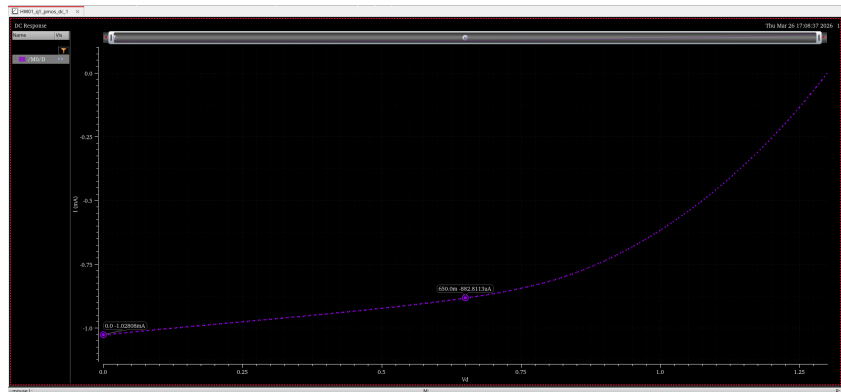
$$V_d = 1.1V \mid V_g = 0.2V$$



$$V_d = 1.2V \mid V_g = 0.1V$$



$$V_d = 1.3V \mid V_g = 0V$$



V _g	VDD	VDD/2	ID @ VDD	ID @ VDD/2	R(VDD)	R(VDD/2)	Req_n (in ohms)	4 significant figures
0.4	0.9	0.45	5.3220E-04	4.5180E-04	1.6911E+03	9.9602E+02	1.3436E+03	1.344E+03
0.3	1	0.5	6.5313E-04	5.5786E-04	1.5311E+03	8.9628E+02	1.2137E+03	1.214E+03

0.2	1.1	0.55	7.7600E-04	6.6532E-04	1.4175E+03	8.2668E+02	1.1221E+03	1.122E+03
0.1	1.2	0.6	9.0082E-04	7.7374E-04	1.3321E+03	7.7546E+02	1.0538E+03	1.054E+03
0	1.3	0.65	1.0281E-03	8.8181E-04	1.2645E+03	7.3628E+02	1.0004E+03	1.000E+03

Discussion

- For both NMOS and PMOS, the equivalent ON resistance decreases as VDD increases. This happens because a larger gate overdrive creates a stronger inversion channel and reduces channel resistance.
- The NMOS transistor shows lower equivalent ON resistance than the PMOS transistor for all VDD values. This is expected because electron mobility in NMOS is higher than hole mobility in PMOS.
- For NMOS, Req_n decreases from 8.730E+02 ohm at VDD = 0.9 V to 6.860E+02 ohm at VDD = 1.3 V.
- For PMOS, Req_p decreases from 1.344E+03 ohm at VDD = 0.9 V to 1.000E+03 ohm at VDD = 1.3 V.
- The PMOS resistance is higher than the NMOS resistance for the same transistor dimensions. Therefore, in CMOS design, PMOS devices are usually made wider to balance the pull-up and pull-down strengths.

Conclusion

The equivalent ON resistance of NMOS and PMOS transistors was determined successfully using the two-end point method and Cadence simulation results.

The results show that:

- Req_n decreases as VDD increases.
- Req_p also decreases as VDD increases.
- NMOS has lower ON resistance than PMOS for the same device dimensions.

Hence, the two-end point method gives a practical estimate of transistor ON resistance and clearly shows that NMOS is stronger than PMOS for the same W and L.

Q2 DC Characterization of CMOS Inverter

Objective

To construct and characterize a CMOS inverter using DC simulations in Cadence Virtuoso. The inverter was designed for $V_{DD} = 1.2$ V, $L_{CH} = 50$ nm, $W_n = 1$ μ m, and an optimized W_p such that the inverter output is within 1% of 0.6 V when $V_{in} = 0.6$ V. The Voltage Transfer Characteristic (VTC) and supply current characteristics were then used to extract V_{IL} , V_{IH} , V_{OL} , V_{OH} , and V_M .

Setup

Technology used: FreePDK45

CMOS inverter dimensions:

NMOS: $W_n = 1$ μ m, $L_n = 50$ nm

PMOS: W_p optimized, $L_p = 50$ nm

NMOS source and bulk connected to ground

PMOS source and bulk connected to $V_{DD} = 1.2$ V

NMOS and PMOS gates connected together as input V_{in}

NMOS and PMOS drains connected together as output V_{out}

Method

Step 1: A CMOS inverter schematic was created in Cadence Virtuoso using one NMOS and one PMOS transistor.

Step 2: The NMOS width was fixed at 1 μ m and both channel lengths were fixed at 50 nm.

Step 3: The PMOS width was adjusted iteratively so that when $V_{in} = 0.6$ V, the output voltage satisfied the required 1% condition around 0.6 V.

Acceptable output range:

$0.6 \pm 1\%$ of 0.6

0.6 ± 0.006

Hence, acceptable V_{out} range = 0.594 V to 0.606 V

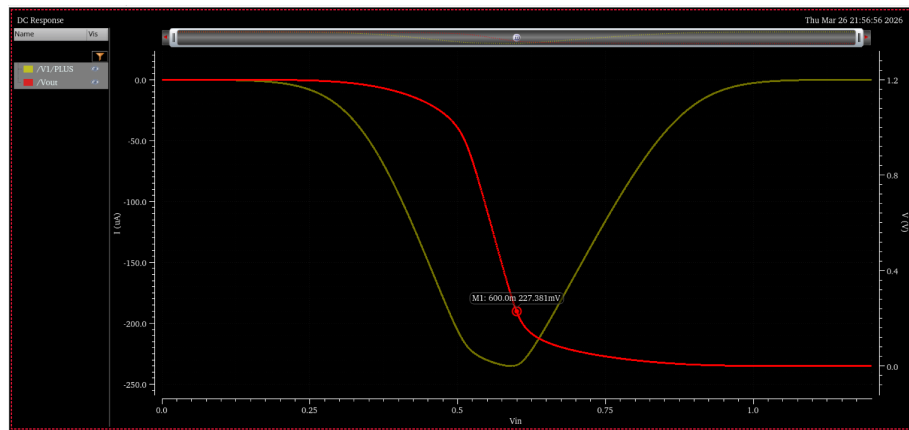
Step 4: DC operating point simulations were performed for different PMOS widths around the expected value.

Rounds performed for PMOS sizing:

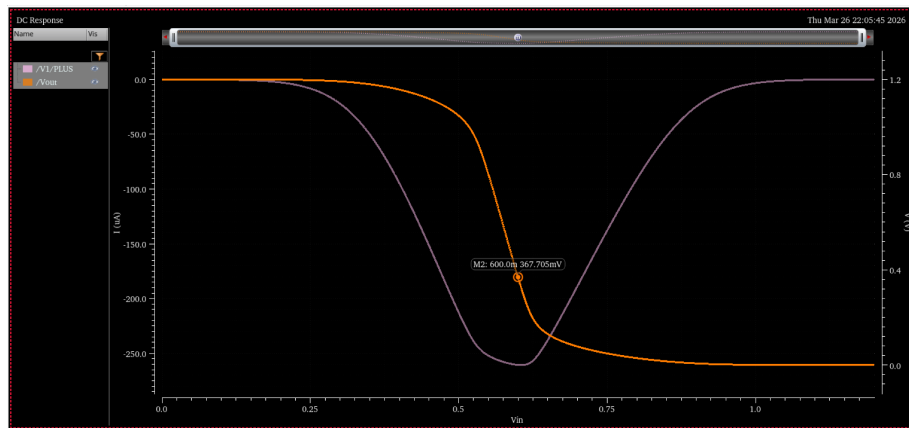
Round 1: A coarse sweep of W_p was performed around 1.5 μ m to identify the region where V_{out} approaches 0.6 V at $V_{in} = 0.6$ V.

wp (μm)	Vout at Vin=0.6v	Status
1	0.227381	Too Low
1.2	0.367705	Too Low
1.4	0.502532	Too Low
1.6	0.620608	Too High
2	0.810879	Too High

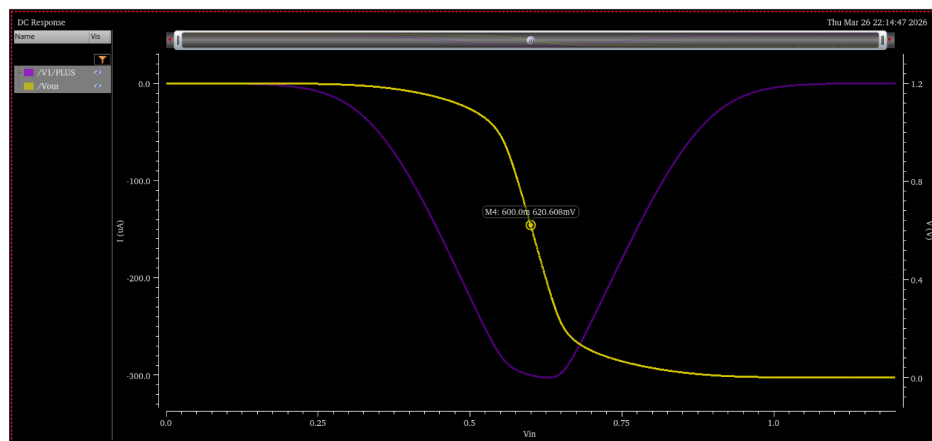
1. $w_p=1\mu$



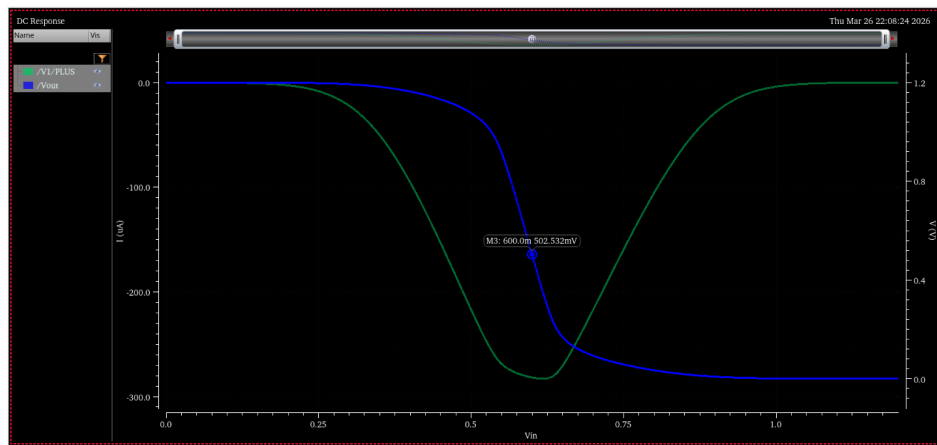
2. $w_p=1.2\mu$



3. $w_p=1.4\mu$



4. $w_p=1.6\mu$



Round 2: The sweep was narrowed near the valid range and the following values were tested:

$W_p = 1.500 \mu m$

$W_p = 1.520 \mu m$

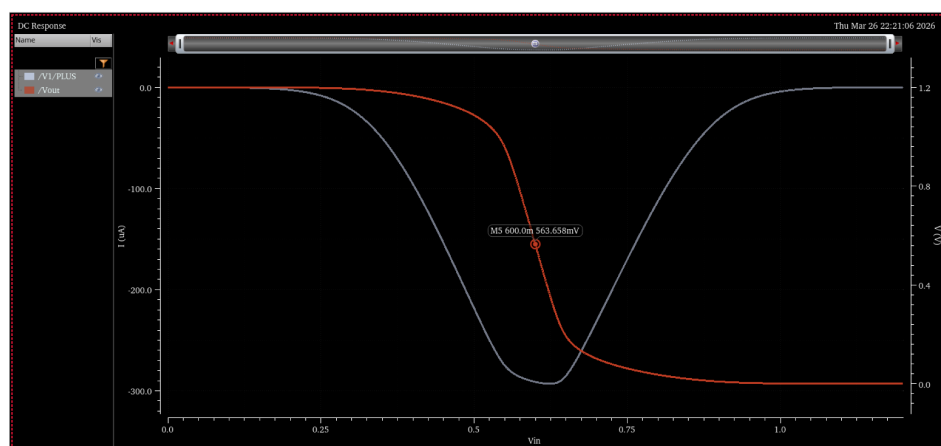
$W_p = 1.540 \mu m$

$W_p = 1.560 \mu m$

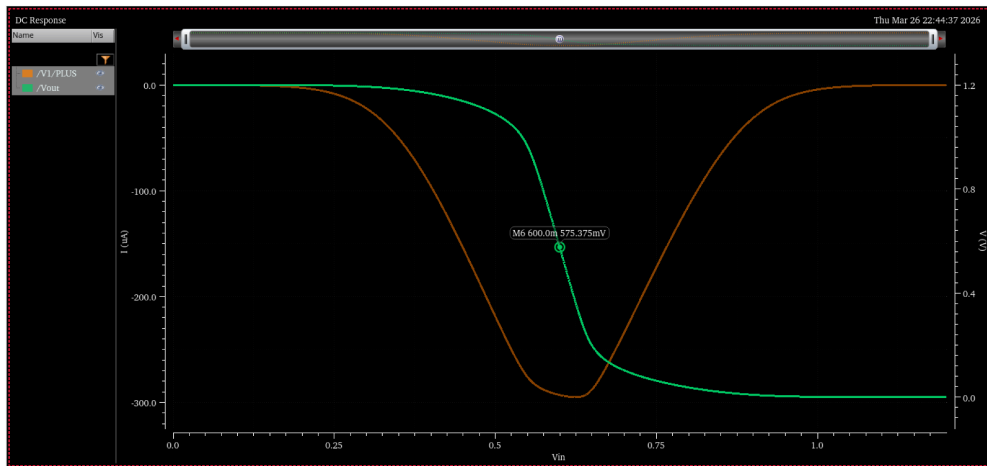
$W_p = 1.580 \mu m$

wp (um)	Vout at Vin=0.6v	Status
1.5	5.6366E-01	Too low
1.52	5.7538E-01	Too low
1.54	5.8693E-01	Too low
1.56	5.9832E-01	Acceptable
1.58	6.0954E-01	High

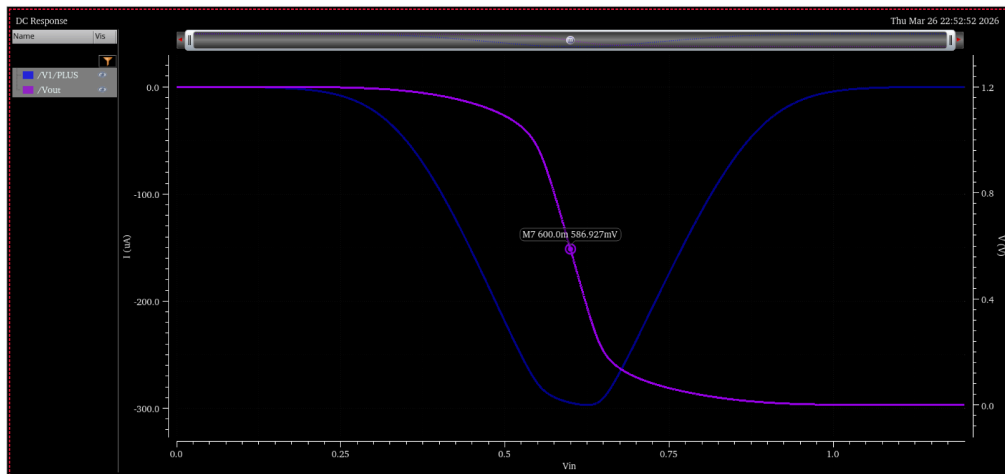
1. $w_p = 1.5u$



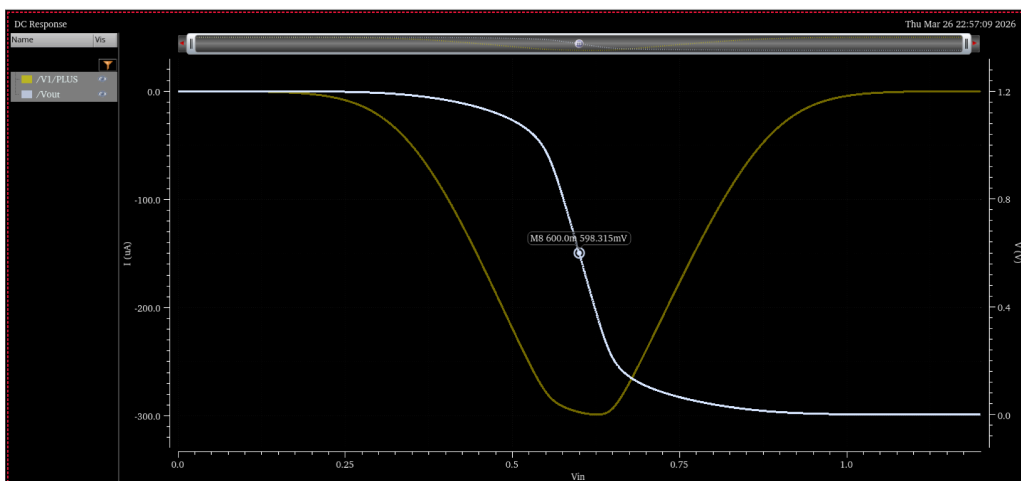
2. $w_p=1.52u$



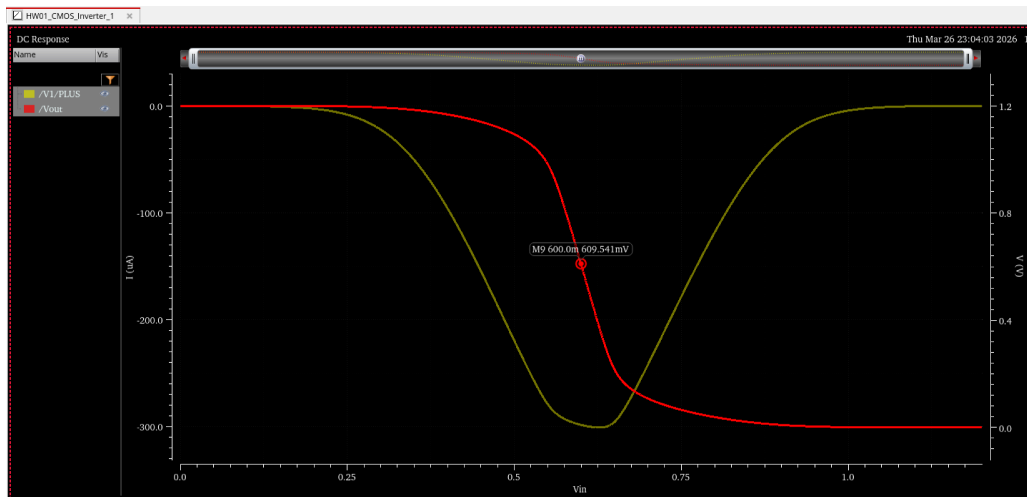
3. $w_p=1.54u$



4. $w_p=1.56u$



5. $w_p=1.58u$



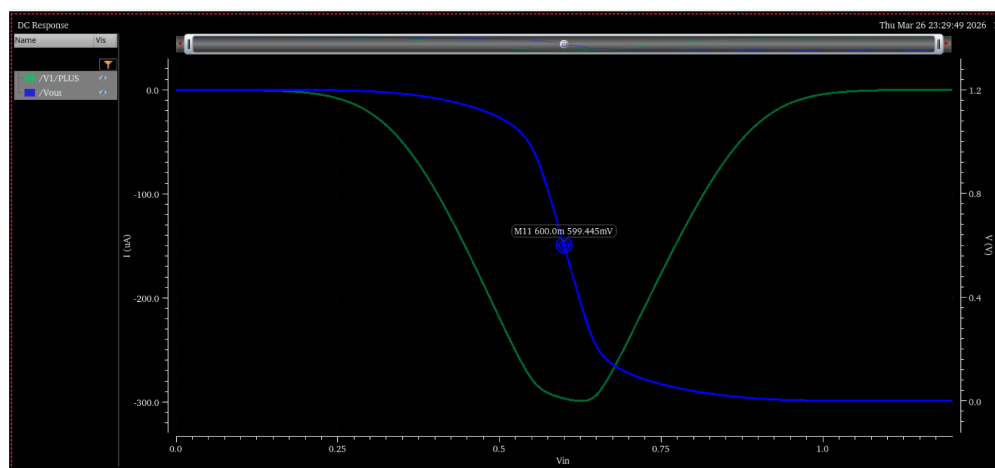
Round 3: A finer sweep was then performed around the transition point:

$W_p = 1.562 \mu m$

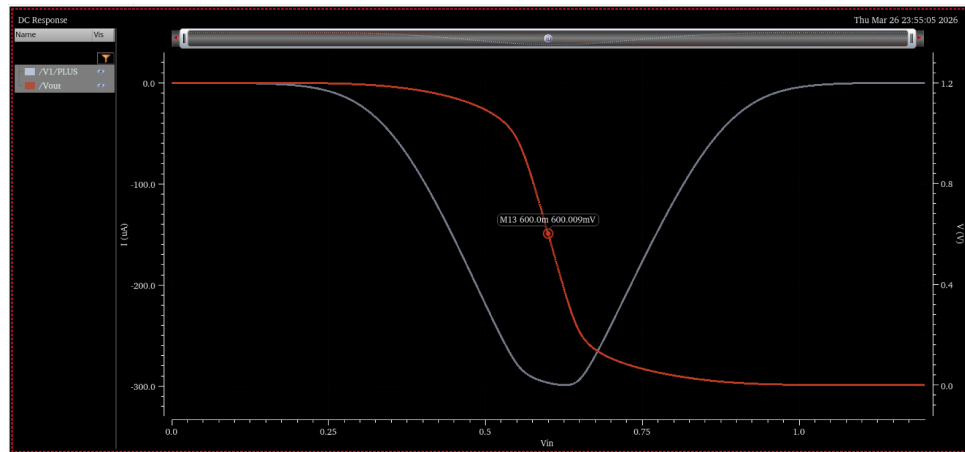
$W_p = 1.564 \mu m$

wp (um)	Vout at Vin=0.6v	Status
1.562	599.445E-03	Slightly low
1.563	600.009E-03	Adequate
1.564	600.573E-03	Slightly high

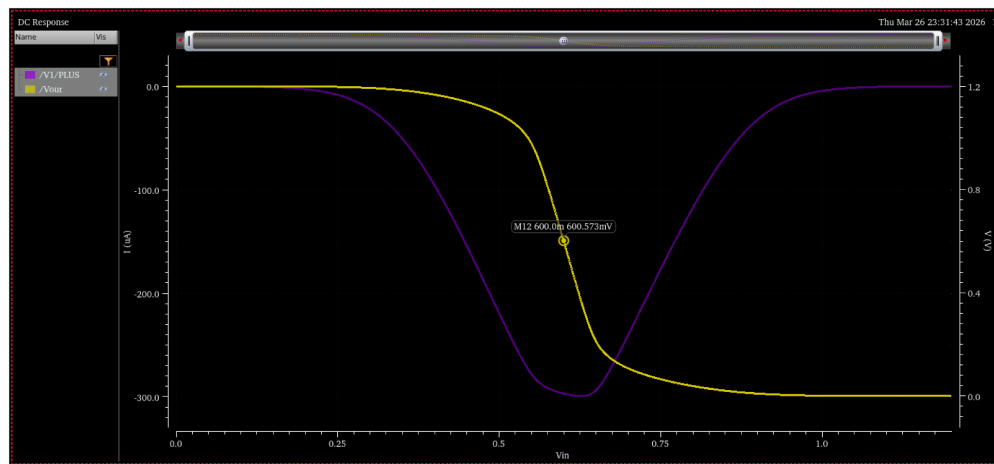
1. $w_p=1.562 \mu m$



2. $w_p=1.563 \mu m$



3. $w_p=1.564\mu$



These two values bracketed the target condition. Therefore, the final PMOS width was chosen as:

$$W_p = 1.563 \mu\text{m}$$

Final inverter sizing:

$$W_n = 1.000 \mu\text{m}$$

$$W_p = 1.563 \mu\text{m}$$

$$L_n = L_p = 50 \text{ nm}$$

Step 5: Using the final sizing, a DC sweep of V_{in} from 0 V to 1.2 V was performed to obtain the VTC, that is, V_{out} versus V_{in} .

Step 6: In the same DC sweep, the current supplied by the power source was plotted as a function of V_{in} .

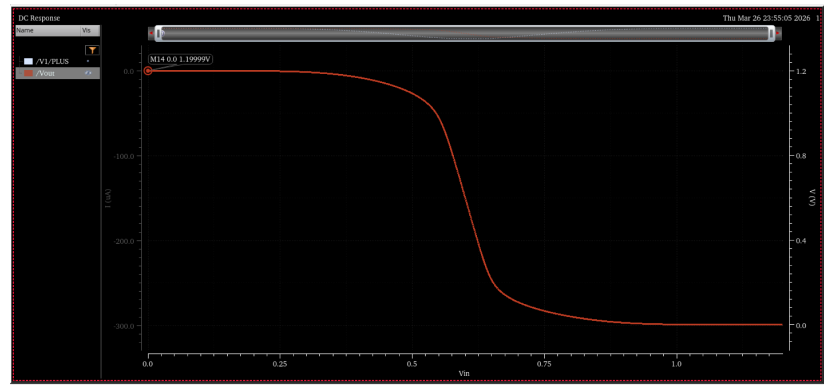
Step 7: From the VTC

V_{OH} was obtained at $V_{in} = 0 \text{ V}$

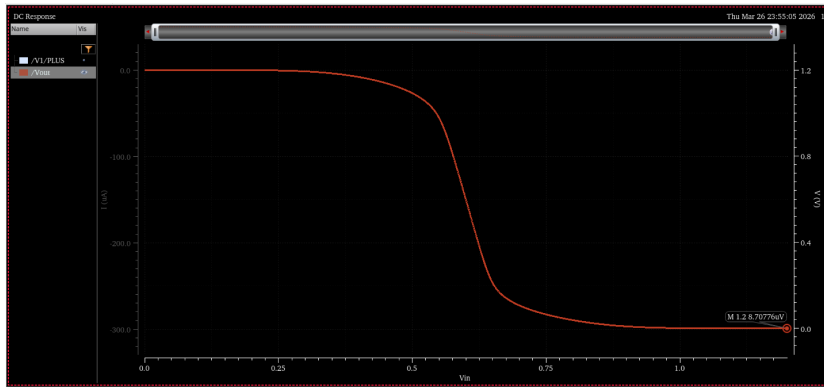
V_{OL} was obtained at $V_{in} = 1.2 \text{ V}$

V_M was obtained at the point where $V_{out} = V_{in}$

$$V_{OH} = 1.199 \text{ V}$$

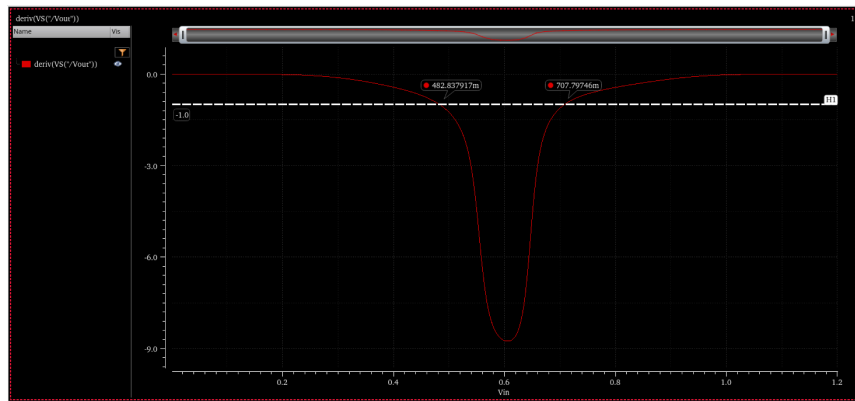


$VOL = 8.70776 \text{ uV}$

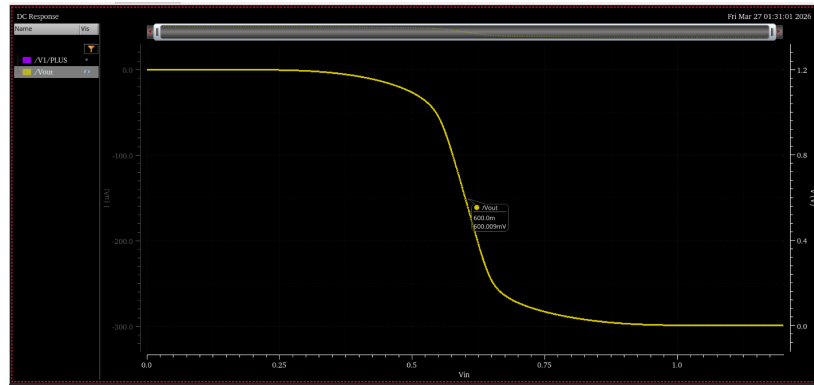


$VIL = 482.837917 \text{ mV} = 0.4828 \text{ V}$

$VIH = 707.79746 \text{ mV} = 0.7078 \text{ V}$



$V_m = 600 \text{ mV}$



Step 8: The derivative of the VTC, dV_{out}/dV_{in} , was plotted. The two points where $dV_{out}/dV_{in} = -1$ were used to extract: V_{IL} from the left crossing & V_{IH} from the right crossing.

Results

Final inverter size:

$W_n = 1.000 \text{ } \mu\text{m}$

$W_p = 1.563 \text{ } \mu\text{m}$

$L_n = L_p = 50 \text{ nm}$

Extracted DC parameters:

$V_{OH} = 1.200 \text{ V}$

$V_{OL} = 8.708\text{e-}6 \text{ V}$

$V_{IL} = 0.4828 \text{ V}$

$V_{IH} = 0.7078 \text{ V}$

$V_M = 0.6000 \text{ V}$

Discussion

- The PMOS width had to be increased above the NMOS width to balance the pull-up and pull-down strengths and obtain a switching point near 0.6 V. After iterative refinement, $W_p = 1.563 \text{ } \mu\text{m}$ satisfied the design requirement.
- The VTC shows correct inverter operation: at low V_{in} , V_{out} remains close to V_{DD} & at high V_{in} , V_{out} falls close to 0 V. The transition occurs near 0.6 V
- The derivative plot provided the values of V_{IL} and V_{IH} from the points where the slope equals -1. The supply current plot showed maximum current near the switching region, where both NMOS and PMOS conduct simultaneously.

Conclusion

A CMOS inverter was successfully designed and characterized using Cadence Virtuoso DC simulations. By iterative PMOS width tuning, the final value $W_p = 1.563 \text{ } \mu\text{m}$ was obtained so that V_{out} was within 1% of 0.6 V at $V_{in} = 0.6 \text{ V}$.

The extracted values were:

$V_{OH} = 1.200 \text{ V}$

$V_{OL} = 8.708\text{e-}6 \text{ V}$

$V_{IL} = 0.4828 \text{ V}$

$V_{IH} = 0.7078 \text{ V}$

$V_M = 0.6000 \text{ V}$

These results confirm correct DC operation of the CMOS inverter and verify the required switching behavior.

Q3 Transient DC Characterization of CMOS Inverter

Objective

To perform transient characterization of the 1X CMOS inverter obtained in Question 2 using Cadence Virtuoso. The propagation delays t_{pHL} and t_{pLH} were measured for different output load capacitances, and the variation of delay with CLOAD was studied.

Setup

Technology used: FreePDK45

Reference inverter used from Question 2:

NMOS: $W_n = 1.000 \text{ um}$, $L_n = 50 \text{ nm}$

PMOS: $W_p = 1.563 \text{ um}$, $L_p = 50 \text{ nm}$

Supply voltage: $V_{DD} = 1.2 \text{ V}$

Testbench connections:

Input applied using a pulse voltage source

Output connected to load capacitor CLOAD

NMOS source and bulk connected to ground & PMOS source and bulk connected to VDD

Pulse input parameters:

$V_1 = 0 \text{ V}$

$V_2 = 1.2 \text{ V}$

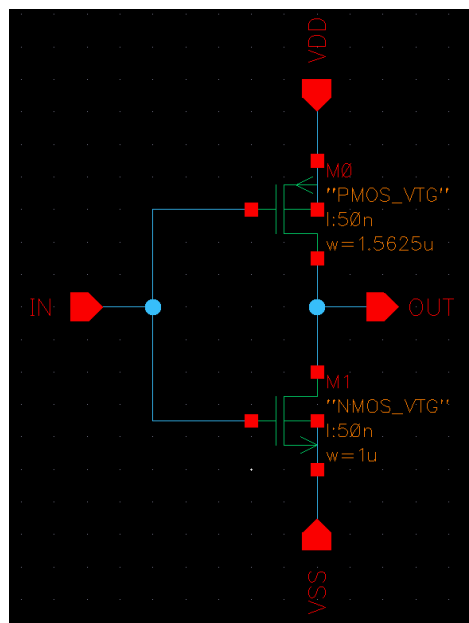
Delay = 1 ns

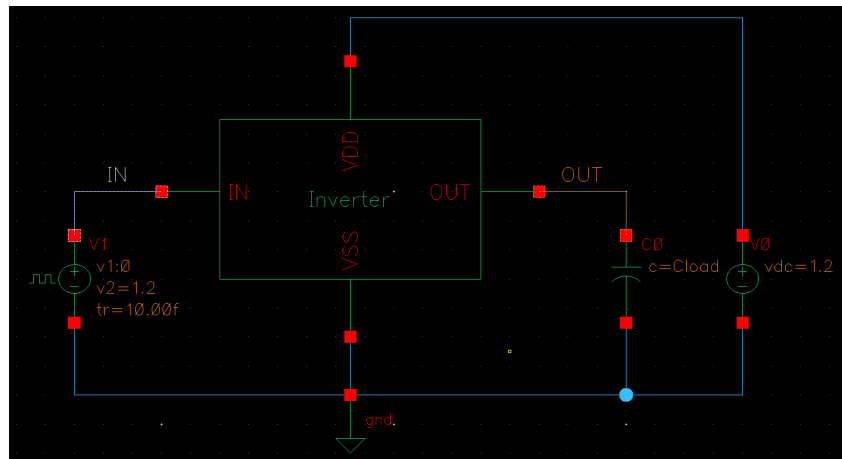
Rise time = 0.01 ps

Fall time = 0.01 ps

Pulse width = 4 ns

Period = 8 ns





Method

Step 1: The CMOS inverter designed in Question 2 was used as the 1X reference inverter.

Step 2: A transient testbench was created in Cadence Virtuoso with the inverter, a pulse input source, a 1.2 V supply, and an output load capacitor.

Step 3: Transient analysis was performed for the following load capacitances:

CLOAD = 20 fF, 40 fF, 60 fF, 80 fF, and 100 fF

Step 4: For each load capacitance, input and output waveforms were plotted.

Step 5: The propagation delays were measured using the 50 percent VDD criterion.

Since $V_{DD} = 1.2$ V, the 50 percent level is:

$$0.5 \times V_{DD} = 0.6 \text{ V}$$

Step 6: t_{pHL} was measured as the time difference between:

the input rising through 0.6 V

and the output falling through 0.6 V

t_{pLH} was measured as the time difference between:

the input falling through 0.6 V

and the output rising through 0.6 V

Step 7: For the 100 fF load case, the measured crossing times were:

Input rising at 0.6 V = 1.0000 ns

Output falling at 0.6 V = 1.0509 ns

Input falling at 0.6 V = 5.00001 ns

Output rising at 0.6 V = 5.04795 ns

Hence,

$$t_{pHL} = 1.0509 \text{ ns} - 1.0000 \text{ ns} = 0.0509 \text{ ns} = 50.90 \text{ ps}$$

$$t_{pLH} = 5.04795 \text{ ns} - 5.00001 \text{ ns} = 0.04794 \text{ ns} = 47.94 \text{ ps}$$

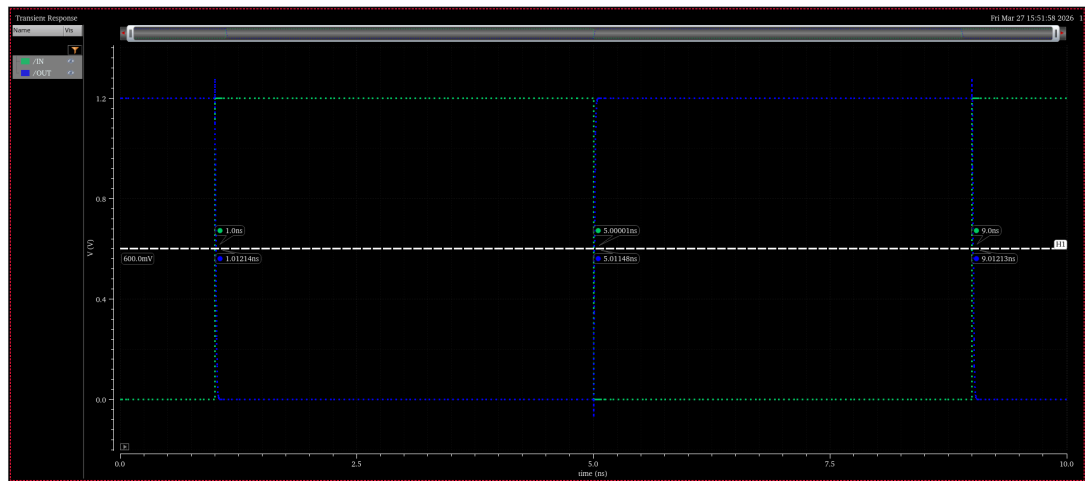
Results

For CLOAD = 100 fF:

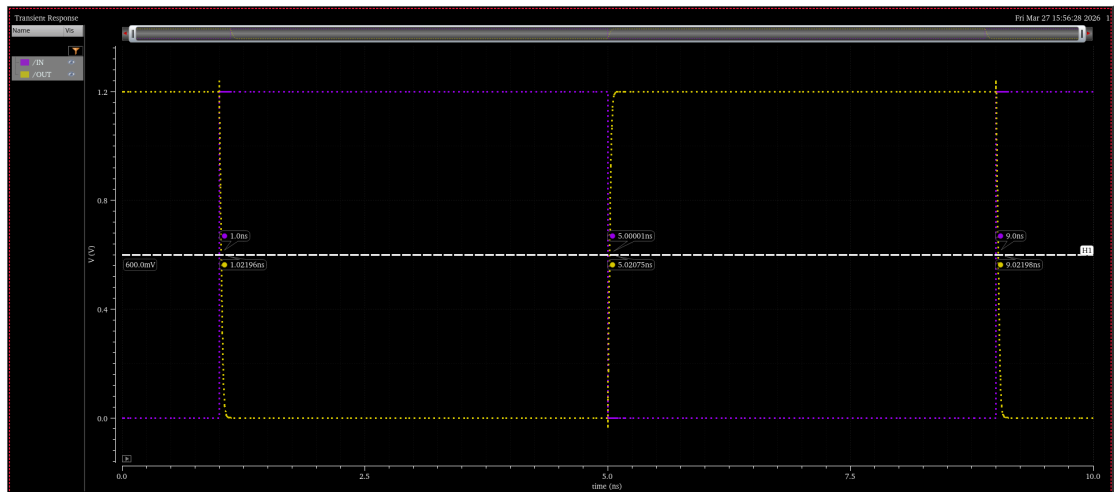
$t_{pHL} = 50.90 \text{ ps}$

$t_{pLH} = 47.94 \text{ ps}$

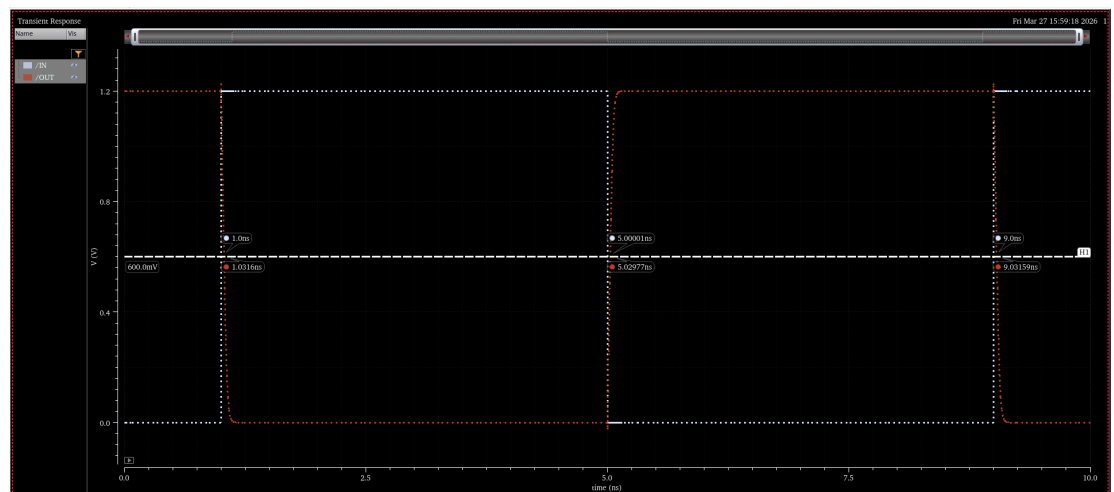
1. Cload = 20f F



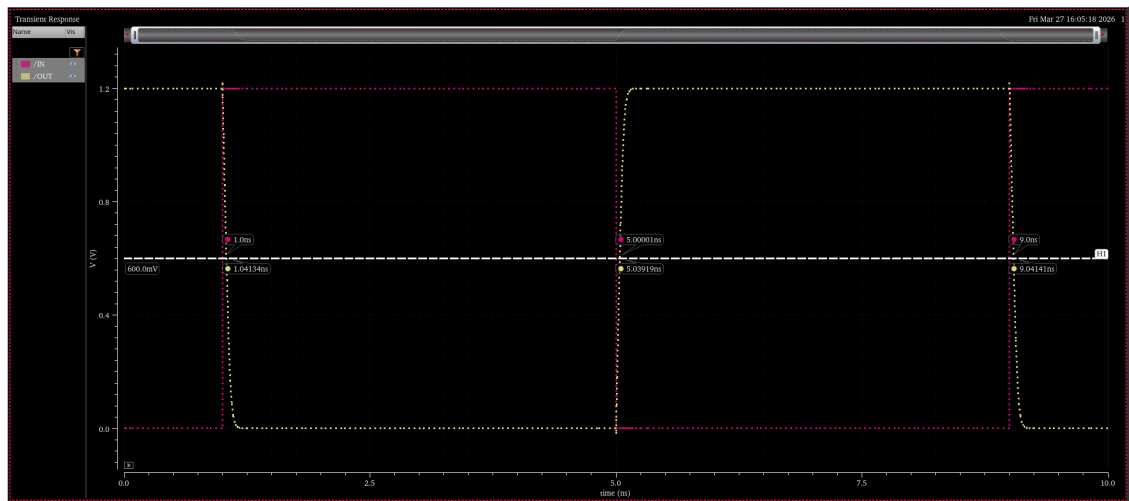
2. 40fF



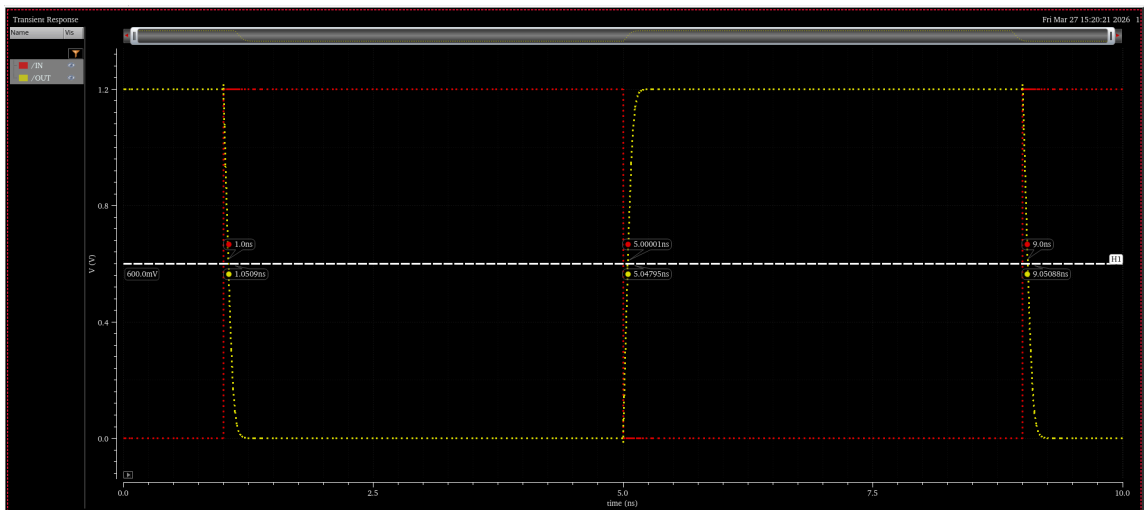
3. 60fF



4. 80fF

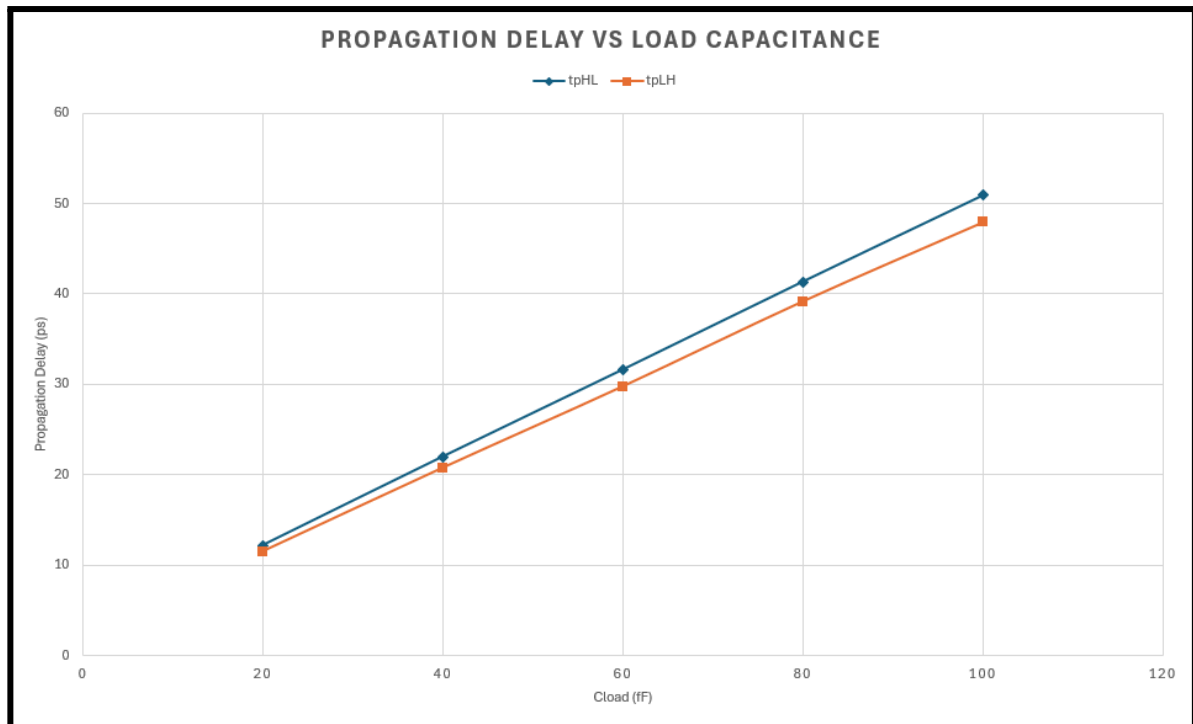


5. 100fF



Result Table:

Clload (fF)	tpHL (ps)	tpLH (ps)
20	12.14	11.47
40	21.96	20.74
60	31.60	29.76
80	41.34	39.18
100	50.90	47.94



Discussion

- The transient response showed correct inverter operation, with full output swing between approximately 0 V and 1.2 V. The propagation delays were extracted using the standard 50 percent input and output crossing method, that is, at 0.6 V.
- From the measured results, both tpHL and tpLH increase as CLOAD increases. This is expected because a larger output load capacitor requires more time to charge and discharge.
- The measured tpHL values increased from 12.14 ps at 20 fF to 50.90 ps at 100 fF, while the measured tpLH values increased from 11.47 ps at 20 fF to 47.94 ps at 100 fF.
- For all load capacitance values, tpHL is slightly larger than tpLH. This indicates that the high-to-low output transition is slightly slower than the low-to-high output transition for the given inverter sizing.
- The plot of propagation delay versus load capacitance shows an approximately linear increase in both tpHL and tpLH with CLOAD.

Conclusion

- The transient characterization of the 1X CMOS inverter was successfully carried out in Cadence Virtuoso. The propagation delays tpHL and tpLH were measured for output load capacitances of 20 fF, 40 fF, 60 fF, 80 fF, and 100 fF using the 0.6 V crossing points of the input and output waveforms.
 - The results show that both tpHL and tpLH increase with increasing load capacitance, and tpHL remains slightly larger than tpLH for all measured cases.
 - Thus, the transient analysis confirms the expected delay dependence of the CMOS inverter on output load capacitance.
-

Q4 Unloaded Transient Characterization of CMOS Inverter

Objective

To determine the unloaded propagation delays of the 1X CMOS inverter and estimate the combined drain capacitances of the MOSFETs by comparing the unloaded delay results with the delay versus load characteristics obtained in Question 3.

Setup

Technology used: FreePDK45

Reference inverter used from Question 2:

NMOS: $W_n = 1.000 \text{ } \mu\text{m}$, $L_n = 50 \text{ nm}$

PMOS: $W_p = 1.563 \text{ } \mu\text{m}$, $L_p = 50 \text{ nm}$

Supply voltage: $V_{DD} = 1.2 \text{ V}$

The same 1X inverter testbench used in Question 3 was used here, except that the external output load capacitor was removed.

Pulse input parameters:

$V_1 = 0 \text{ V}$

$V_2 = 1.2 \text{ V}$

Delay = 1 ns

Rise time = 0.01 ps

Fall time = 0.01 ps

Pulse width = 4 ns

Period = 8 ns

Method

Step 1: The transient testbench of the 1X CMOS inverter from Question 3 was reused, and the external output load capacitor was removed to obtain the unloaded condition.

Step 2: Transient simulation was performed and the input and output waveforms were plotted.

Step 3: The unloaded propagation delays were measured using the 50 percent crossing method. Since $V_{DD} = 1.2 \text{ V}$, the 50 percent level is 0.6 V.

t_{pHL} was measured from the time difference between the input rising through 0.6 V and the output falling through 0.6 V.

t_{pLH} was measured from the time difference between the input falling through 0.6 V and the output rising through 0.6 V.

The average propagation delay is defined as

$$t_p = (t_{pHL} + t_{pLH}) / 2$$

Step 4: From the lecture, the inverter propagation delay can be approximated using the RC switching model as

$$t_{pHL} \approx 0.69 R_n C_{eff}$$

$$t_{pLH} \approx 0.69 R_p C_{eff}$$

where C_{eff} is the effective capacitance seen at the output node. For the unloaded case, the external load capacitor is removed, so the effective output capacitance is mainly the internal drain capacitances of the NMOS and PMOS transistors. Thus,

$$C_{eff,unloaded} \approx C_{dn} + C_{dp}$$

Step 5: From the delay versus load results obtained in Question 3, the delay was assumed to vary approximately linearly with load capacitance. Hence,

$$tp_{HL} = m_{HL} C_{LOAD} + b_{HL}$$

$$tp_{LH} = m_{LH} C_{LOAD} + b_{LH}$$

where m_{HL} and m_{LH} are the slopes of the tp_{HL} versus C_{LOAD} and tp_{LH} versus C_{LOAD} plots, respectively.

Using the values from Question 3,

$$m_{HL} = (50.90 - 12.14) / (100 - 20) = 0.4845 \text{ ps/fF}$$

$$m_{LH} = (47.94 - 11.47) / (100 - 20) = 0.4559 \text{ ps/fF}$$

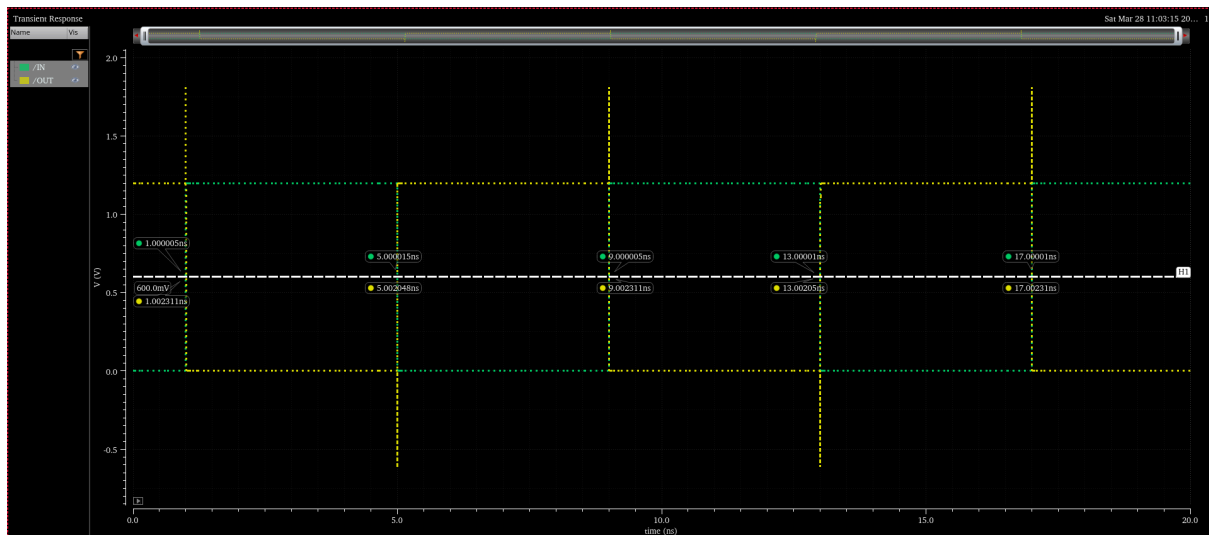
Step 6: Since the unloaded delay is caused mainly by the internal output capacitance, the equivalent internal capacitances were estimated by rearranging the proportional delay-capacitance relation:

$$C_{int,HL} = tp_{HL,unloaded} / m_{HL}$$

$$C_{int,LH} = tp_{LH,unloaded} / m_{LH}$$

The combined drain capacitance was then estimated as

$$C_{drain,total} = (C_{int,HL} + C_{int,LH}) / 2$$



Measured crossing times:

Input rising at 0.6 V = 1.000005 ns

Output falling at 0.6 V = 1.002311 ns

Input falling at 0.6 V = 5.000015 ns

Output rising at 0.6 V = 5.002048 ns

Hence,

$$tp_{HL,unloaded} = 1.002311 \text{ ns} - 1.000005 \text{ ns} = 0.002306 \text{ ns} = 2.306 \text{ ps}$$

$$tp_{LH,unloaded} = 5.002048 \text{ ns} - 5.000015 \text{ ns} = 0.002033 \text{ ns} = 2.033 \text{ ps}$$

Results

Unloaded propagation delays:

$$tp_{HL,unloaded} = 2.306 \text{ ps}$$

$$tp_{LH,unloaded} = 2.033 \text{ ps}$$

Equivalent internal capacitances:

$$C_{int,HL} = 2.306 / 0.4845 = 4.760 \text{ fF}$$

$$C_{int,LH} = 2.033 / 0.4559 = 4.460 \text{ fF}$$

Estimated combined drain capacitance:

$$C_{drain,total} = (4.760 + 4.460) / 2 = 4.610 \text{ fF}$$

Discussion

- The unloaded inverter showed much smaller propagation delays than the loaded cases in Question 3, as expected, because the external output load capacitor was removed. Only the internal parasitic capacitances of the inverter remained at the output node.
- The inverter propagation delay is proportional to the effective output capacitance in the RC switching model. Therefore, the unloaded delay can be interpreted as the delay caused by the intrinsic output capacitance of the inverter. By comparing the unloaded delay with the delay versus CLOAD trend from Question 3, the equivalent internal capacitance was estimated.
- The capacitance values obtained from tpHL and tpLH were close to each other, giving an average combined drain capacitance of 4.610 fF. This value represents the combined drain capacitances of the NMOS and PMOS transistors at the inverter output node.

Conclusion

- The unloaded propagation delays of the 1X CMOS inverter were successfully determined using transient simulation in Cadence Virtuoso.
 - The measured unloaded delays were:
 $tp_{HL,unloaded} = 2.306 \text{ ps}$
 $tp_{LH,unloaded} = 2.033 \text{ ps}$
 - Using the RC delay model and the delay versus load results from Question 3, the combined drain capacitance of the MOSFETs was estimated to be: $C_{drain,total} = 4.610 \text{ fF}$
 - Thus, the unloaded delay analysis confirms that the intrinsic parasitic capacitances at the inverter output can be estimated from the propagation delay versus load relationship.
-

Q5a Transient Characterization of 2-stage Inverter Chain

Objective

To simulate a 2-stage inverter chain consisting of 1X CMOS inverters operating at $V_{DD} = 1.2$ V, and to determine the input source rise/fall times that model the drive of a 1X inverter by matching the output rise/fall times of the first stage within 1%. The propagation delays t_{pHL} and t_{pLH} of the first stage were then measured.

Setup

Technology used: FreePDK45

Reference 1X inverter used from Question 2:

NMOS: $W_n = 1.000$ μm , $L_n = 50$ nm

PMOS: $W_p = 1.563$ μm , $L_p = 50$ nm

Supply voltage: $V_{DD} = 1.2$ V

Testbench configuration:

A 2-stage inverter chain was constructed using two identical 1X CMOS inverters.

The input source was connected to the first stage input node VIN.

The output of the first stage was denoted as V1 and connected to the input of the second stage.

The output of the second stage was denoted as VOUT.

Pulse input parameters:

Initial value = 0 V

Final value = 1.2 V

Delay = 1 ns

Rise/fall time = adjusted iteratively

Pulse width = 8 ns

Period = 16 ns

Method

- A 2-stage inverter chain consisting of two identical 1X CMOS inverters was constructed in Cadence Virtuoso and simulated at $V_{DD} = 1.2$ V. The first-stage input node was denoted as VIN and the first-stage output node was denoted as V1.
- For the rising-input case, the pulse source was set to start from 0 V and rise to 1.2 V after a delay of 1 ns. The input rise time t_r was adjusted iteratively so that the fall time of the output of the first stage, $t_f(V1)$, was within 1% of the rise time of the input waveform, $t_r(VIN)$, while keeping t_r as small as possible.
- The input rise time was measured from the 10% to 90% levels of V_{DD} , that is, from 0.12 V to 1.08 V. The first-stage output fall time was measured from the 90% to 10% levels of V_{DD} , that is, from 1.08 V to 0.12 V.

The percentage difference was calculated using:

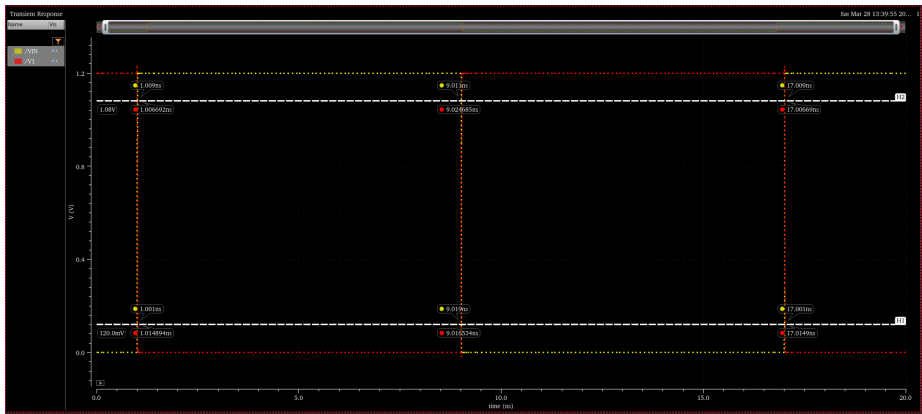
$$\% \text{ Difference} = |t_f(V1) - t_r(VIN)| / t_r(VIN) \times 100$$

A round-based iterative search was used to determine the minimum acceptable input rise time.

Round 1: A refined sweep was first performed around 10 ps using the following applied tr values:
10.0 ps, 10.5 ps, and 11.0 ps

Applied tr (ps)	Measured tr(VIN) (ps)	Measured tf(V1) (ps)	%Difference	Within 1% ?
10	8.000	8.202	2.525	No
10.5	8.400	8.303	1.155	No
11	8.790	8.407	4.357	No

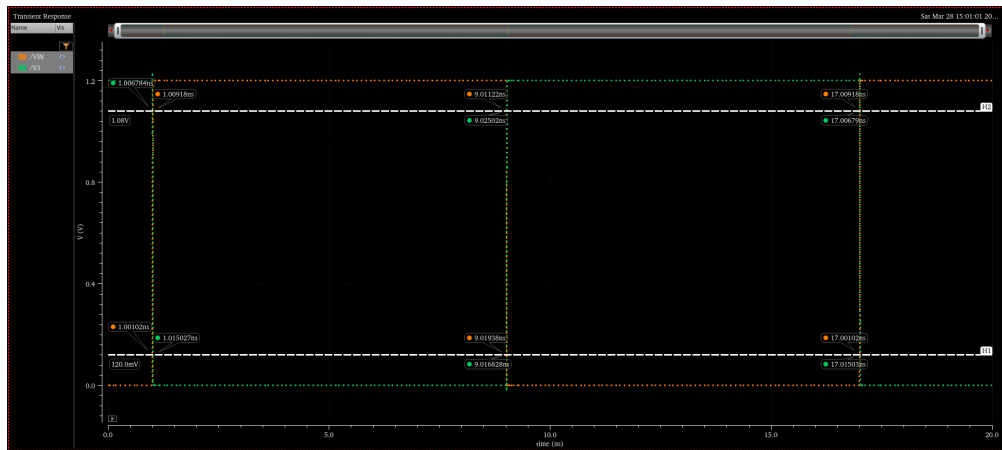
tr=10ps:



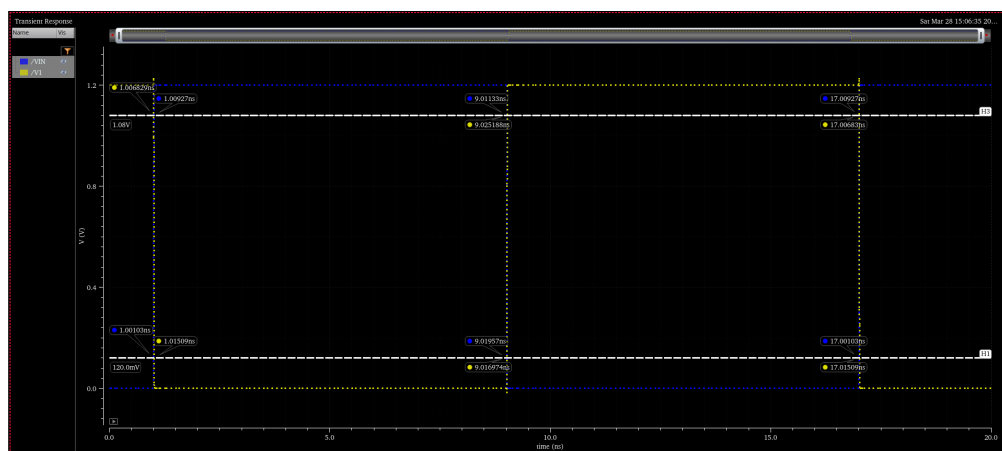
tr=10.5ps



tr=11ps



$t_r=10.3$



$t_r=10.4$



From Round 2, both 10.3 ps and 10.4 ps satisfied the 1% requirement. Since the objective was to choose the smallest acceptable input rise time, the final selected value was $t_r = 10.30$ ps

Final selected case:

For the final selected applied rise time of 10.30 ps, the measured values were

$t_r(VIN) = 8.240$ ps

$$t_f(V1) = 8.261 \text{ ps}$$

The percentage difference was

$$\% \text{ Difference} = |8.261 - 8.240| / 8.240 \times 100 = 0.2549\%$$

Thus, the selected input rise time satisfied the required 1% matching condition.

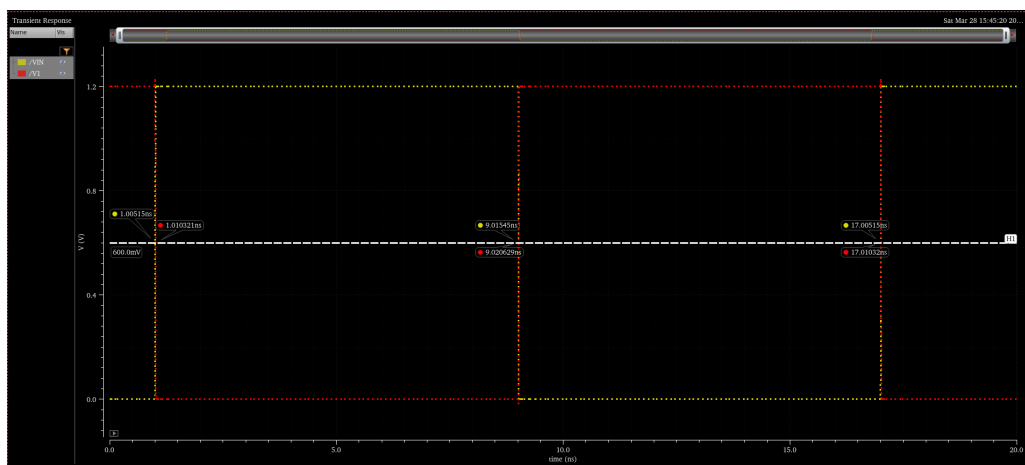
For the final selected applied rise time of 10.30 ps, the propagation delay of the first stage for the rising-input case was measured using the 50% crossing points at 0.6 V.

Measured values:

$$V_{IN} \text{ rising at } 0.6 \text{ V} = 1.00515 \text{ ns}$$

$$V1 \text{ falling at } 0.6 \text{ V} = 1.010321 \text{ ns}$$

$$\text{Hence, } t_{pHL} = 1.010321 \text{ ns} - 1.00515 \text{ ns} = 0.005171 \text{ ns} = 5.171 \text{ ps}$$



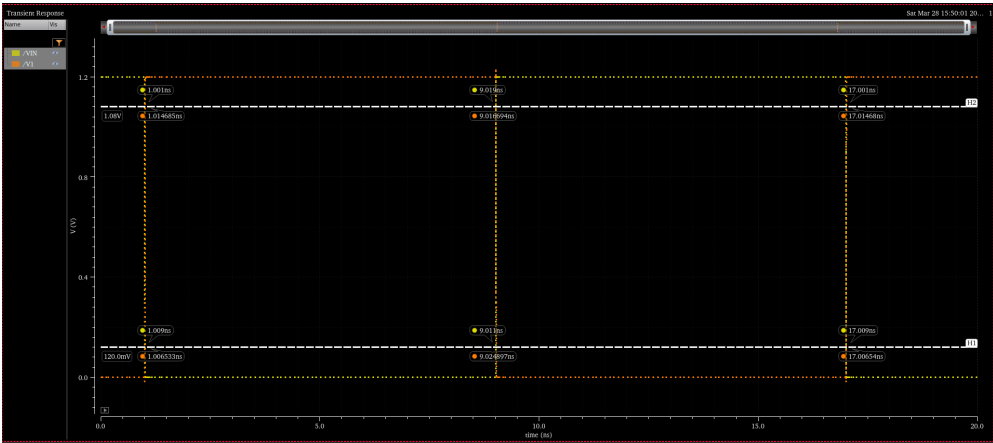
The same procedure was then repeated with reversed input polarity. The pulse source was set to start from 1.2 V and fall to 0 V after 1 ns. The input fall time t_f was adjusted iteratively so that the rise time of the output of the first stage, $t_r(V1)$, was within 1% of the fall time of the input waveform, $t_f(VIN)$. The corresponding propagation delay t_{pLH} of the first stage was then measured using the 50% crossing points at 0.6 V.

For the reversed-polarity case, the input fall time was measured from 1.08 V to 0.12 V, and the first-stage output rise time was measured from 0.12 V to 1.08 V. The percentage difference was calculated using

$$\% \text{ Difference} = |t_r(V1) - t_f(VIN)| / t_f(VIN) \times 100$$

Applied t_f (ps)	Measured $t_f(VIN)$ (ps)	Measured $t_r(V1)$ (ps)	%Difference	Within 1% ?
10.0	8.000	8.152	1.900	No
10.1	8.080	8.174	1.163	No
10.2	8.160	8.195	0.4289	Yes
10.3	8.240	8.217	0.2791	Yes
10.4	8.320	8.238	0.9856	Yes

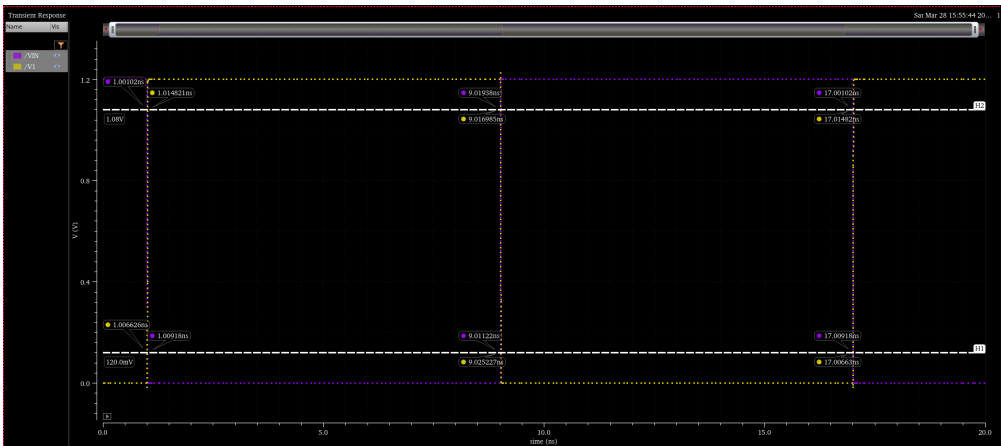
tf=10ps



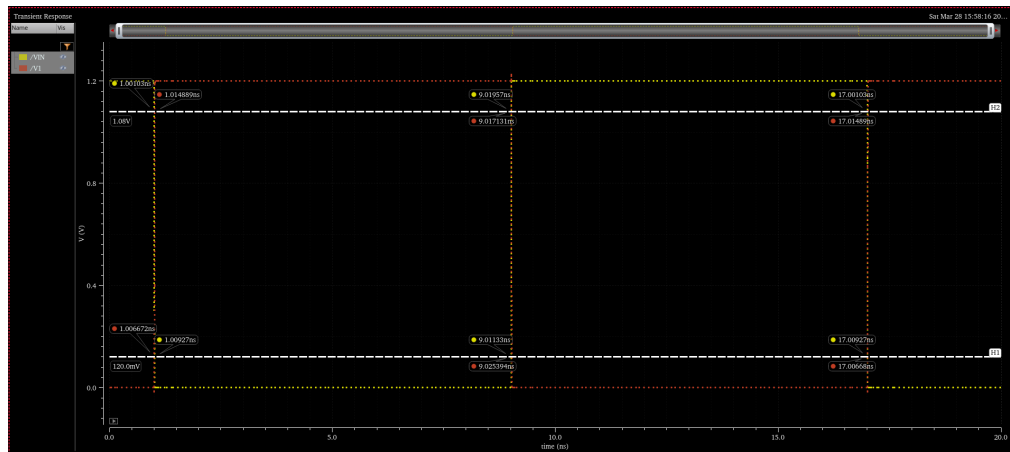
tf=10.1ps



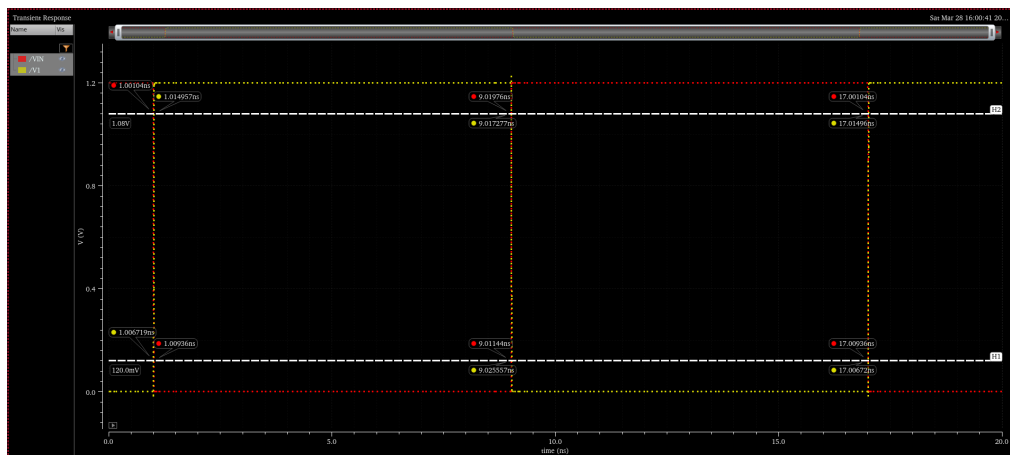
tf=10.2ps



tf=10.3ps



$t_f = 10.4\text{ps}$



From Round 2, both 10.2 ps, 10.3 ps and 10.4 ps satisfied the 1% requirement. Since the objective was to choose the smallest acceptable input fall time, the final selected value was $t_f = 10.30\text{ ps}$

Final selected case: For the final selected applied fall time of 10.30 ps, the measured values were:

$t_f(V_{IN}) = 8.240\text{ ps}$

$t_r(V_1) = 8.217\text{ ps}$

The percentage difference was

% Difference = $|8.217 - 8.240| / 8.240 \times 100 = 0.2791\%$

Thus, the selected input fall time satisfied the required 1% matching condition.

For the final selected applied fall time of 10.30 ps, the propagation delay of the first stage for the reverse-polarity case was measured using the 50% crossing points at 0.6 V.

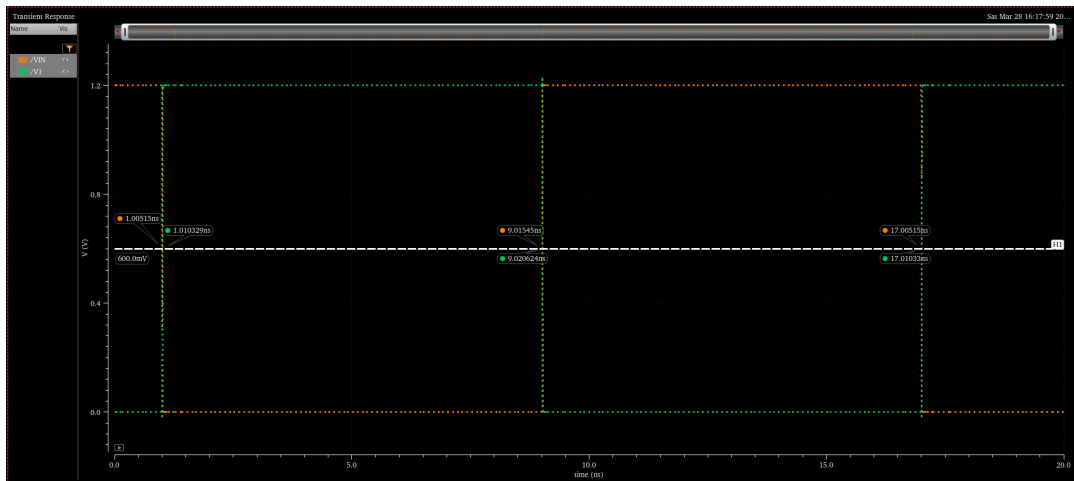
Measured values:

VIN falling at 0.6 V = 1.00515 ns

V1 rising at 0.6 V = 1.010329 ns

Hence,

$t_{pLH} = 1.010329\text{ ns} - 1.00515\text{ ns} = 0.005179\text{ ns} = 5.179\text{ ps}$



Results

For the rising-input case:

$t_r = 10.30 \text{ ps}$

$t_r(V_{IN}) = 8.240 \text{ ps}$

$t_f(V_1) = 8.261 \text{ ps}$

% Difference = 0.2549%

$t_{pHL} = 5.171 \text{ ps}$

For the reverse-polarity case:

$t_f = 10.30 \text{ ps}$

$t_f(V_{IN}) = 8.240 \text{ ps}$

$t_r(V_1) = 8.217 \text{ ps}$

% Difference = 0.2791%

$t_{pLH} = 5.179 \text{ ps}$

Discussion

- The selected source rise and fall times produced close agreement between the input transition and the first-stage output transition, with mismatch values below 1% for both cases.
- For the rising-input case, the minimum acceptable source rise time was 10.30 ps, giving $t_{pHL} = 5.171 \text{ ps}$.
- For the reverse-polarity case, the minimum acceptable source fall time was 10.30 ps, giving $t_{pLH} = 5.179 \text{ ps}$.
- The values of t_{pHL} and t_{pLH} are very close, indicating nearly symmetric switching behavior of the 1X CMOS inverter.

Conclusion

The 2-stage inverter chain using 1X CMOS inverters was successfully simulated at $V_{DD} = 1.2 \text{ V}$.

The minimum acceptable source transition times were:

$t_r = 10.30 \text{ ps}$

$t_f = 10.30 \text{ ps}$

The corresponding first-stage propagation delays were:

$t_{pHL} = 5.171 \text{ ps}$

$t_{pLH} = 5.179 \text{ ps}$

Thus, the required source waveform parameters for modeling the drive of the 1X CMOS inverter were successfully obtained.

Q5b Transient Characterization of 2-stage Inverter Chain with strength 2X

Objective

To simulate a 2-stage inverter chain in which all 1X CMOS inverters are replaced by 2X CMOS inverters, and to determine the adjusted input rise time and fall time such that the first-stage output transition time matches the input transition time within 1%. The propagation delays t_{pHL} and t_{pLH} of the first stage are then extracted and compared with the 1X case.

Setup

A 2-stage CMOS inverter chain was simulated using FreePDK45 at $V_{DD} = 1.2$ V. Both stages were changed from 1X to 2X. The transistor dimensions used for each 2X inverter were:

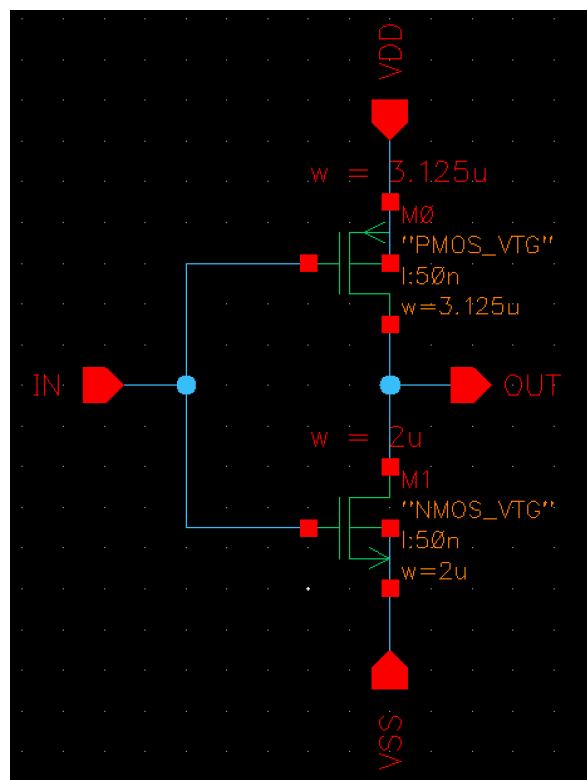
- NMOS: $W = 2$ μm , $L = 50$ nm
- PMOS: $W = 3.125$ μm , $L = 50$ nm

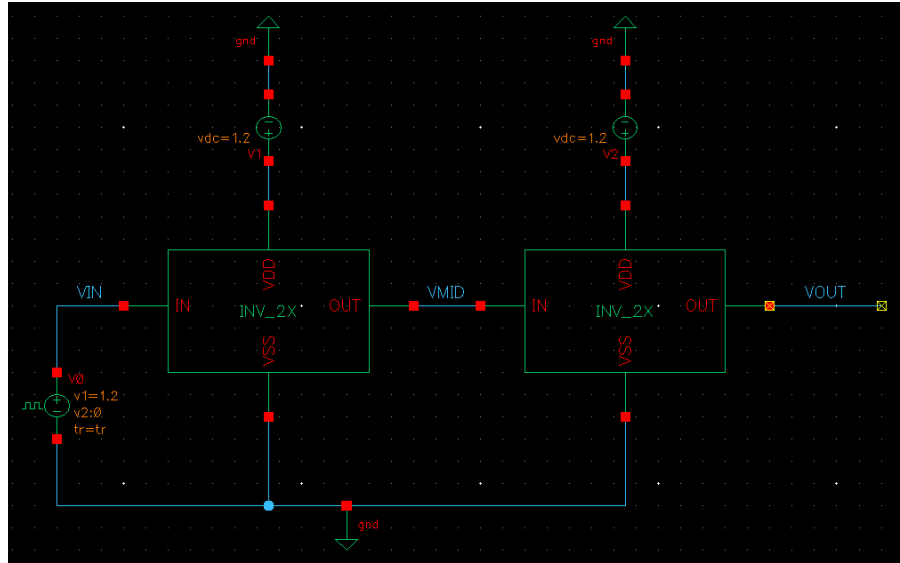
The testbench consisted of:

- input pulse source at VIN
- first-stage output node VMID
- second-stage output node VOUT
- $V_{DD} = 1.2$ V
- transient analysis stop time = 20 ns
- no significant external output loading used for this part

For the rising-input case, the input was switched from 0 V to 1.2 V after 1 ns delay.

For the falling-input case, the polarity was reversed and the input was switched from 1.2 V to 0 V after 1 ns delay.





Method

- For the rising-input case, the rise time t_r of the input source was varied until the input rise time at VIN matched the output fall time at VMID within 1%. After the best t_r was found, the propagation delay t_{pHL} of the first stage was measured using the 50% crossing point, i.e. 0.6 V.
- For the falling-input case, the fall time t_f of the input source was varied until the input fall time at VIN matched the output rise time at VMID within 1%. After the best t_f was found, the propagation delay t_{pLH} of the first stage was measured using the 50% crossing point, i.e. 0.6 V.

The percentage difference was calculated as:

$$\% \text{ Difference} = |\text{Input transition time} - \text{Stage 1 output transition time}| / \text{Input transition time} \times 100$$

Results

(a) Rising-input case

Round 1: An initial sweep was first performed around the expected matching region using the following applied t_r values: 10.0 ps and 10.5 ps.

Applied t_r (ps)	Measured $t_r(\text{VIN})$ (ps)	Measured $t_f(\text{V1})$ (ps)	%Difference	Within 1% ?
10	8.000	8.14	1.75	No
10.5	8.400	8.25	1.79	No

Round 2: A finer sweep was then performed between 10.0 ps and 10.5 ps using the following applied t_r value: 10.25 ps.

Applied t_r (ps)	Measured $t_r(\text{VIN})$ (ps)	Measured $t_f(\text{V1})$ (ps)	%Difference	Within 1% ?
10.25	8.20	8.20	0	Yes

Calculation of t_{pHL}

For the rising-input case, t_{pHL} of the first stage is measured using the 50% voltage crossing point, i.e. 0.6 V.

From the waveform: VIN crosses 0.6 V at 1.00512 ns
VMID crosses 0.6 V at 1.01025 ns

Therefore,
 $t_{pHL} = t_{VMID(50\%)} - t_{VIN(50\%)}$
 $t_{pHL} = 1.01025 \text{ ns} - 1.00512 \text{ ns}$
 $t_{pHL} = 0.00513 \text{ ns} = 5.13 \text{ ps}$

(b) Falling-input case
Round 1: An initial sweep was first performed around the expected matching region using the following applied tf values: 10.0 ps, 9.75 ps, and 9.5 ps.

Applied tf (ps)	Measured tf(VIN) (ps)	Measured tr(V1) (ps)	%Difference	Within 1% ?
10	8.30	8.12	1.75	No
9.75	7.81	8.07	3.33	No
9.5	7.60	8.02	5.53	No

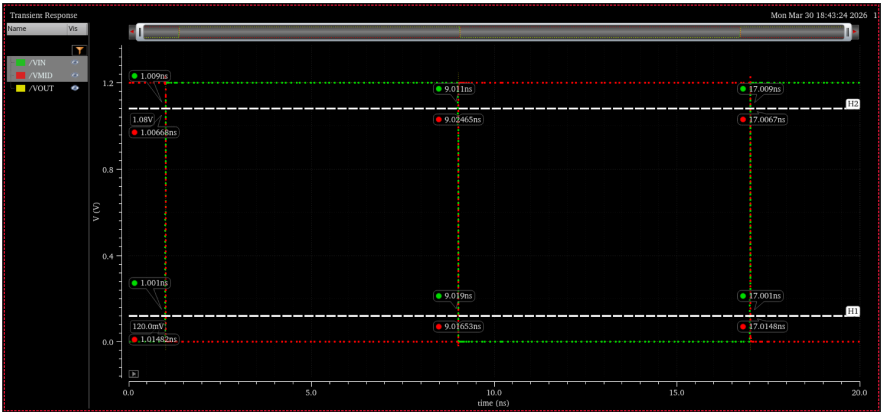
Round 2: A finer sweep was then performed near the matching point using the following applied tf value: 10.25 ps.

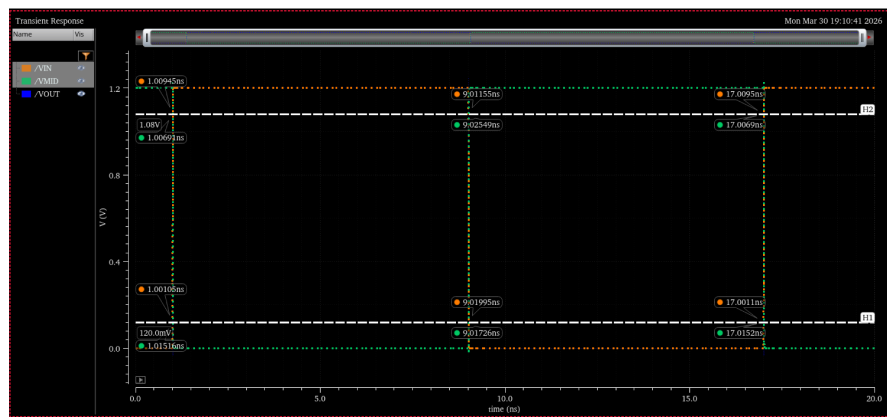
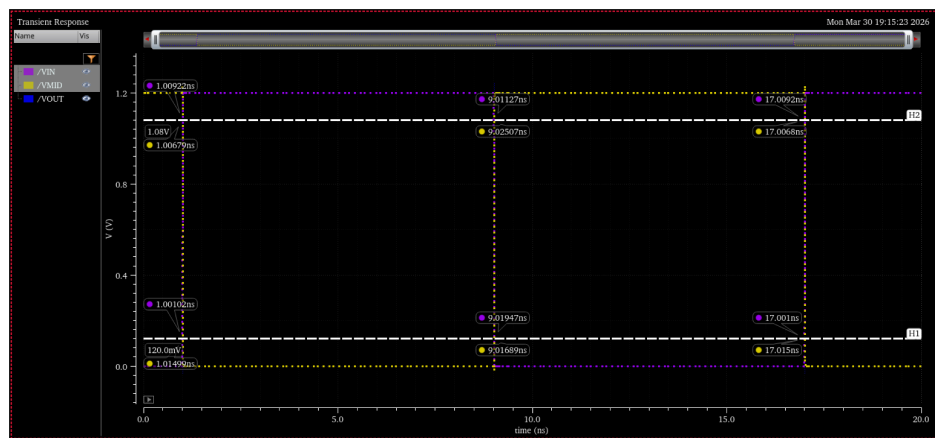
Applied tf (ps)	Measured tf(VIN) (ps)	Measured tr(V1) (ps)	%Difference	Within 1% ?
10.25	8.20	8.18	0.24	Yes

Calculation of t_{pLH}
For the falling-input case, t_{pLH} of the first stage is measured using the 50% voltage crossing point, i.e. 0.6 V.
From the waveform: VIN crosses 0.6 V at 1.00512 ns
VMID crosses 0.6 V at 1.01027 ns

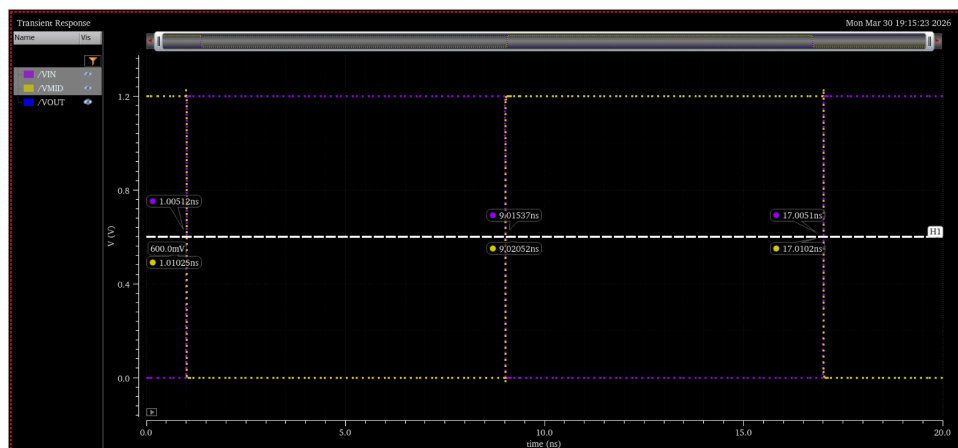
Therefore,
 $t_{pLH} = t_{VMID(50\%)} - t_{VIN(50\%)}$
 $t_{pLH} = 1.01027 \text{ ns} - 1.00512 \text{ ns}$
 $t_{pLH} = 0.00515 \text{ ns} = 5.15 \text{ ps}$

$t_r=10p$



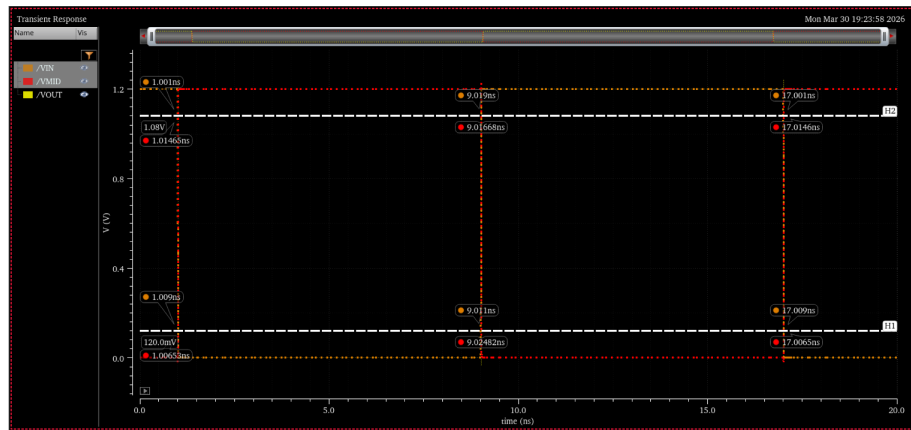
tr=10.5p
| tr=10.25p |


tpHL:

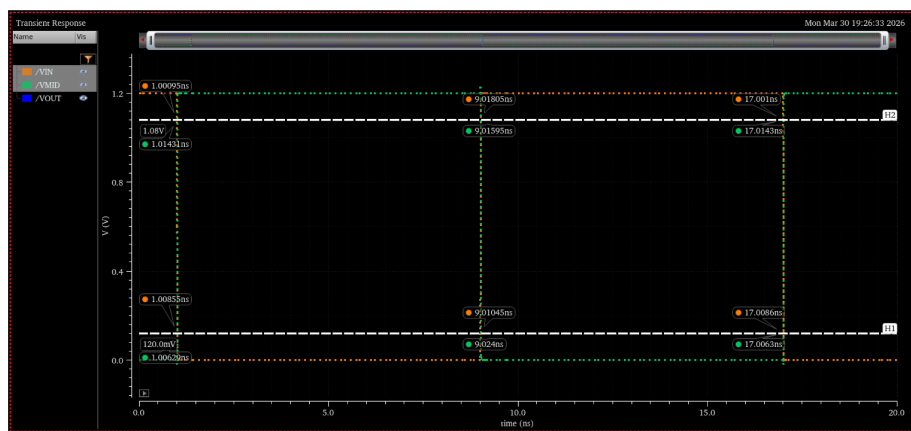


b) Falling-input case:

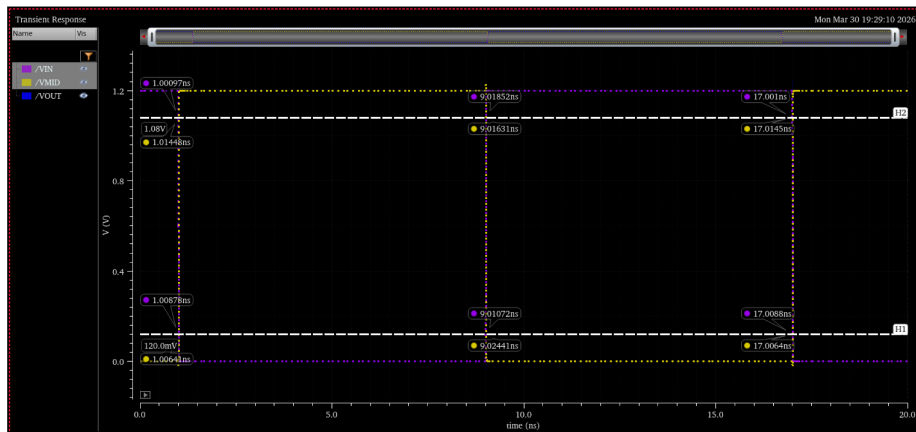
tf=10p



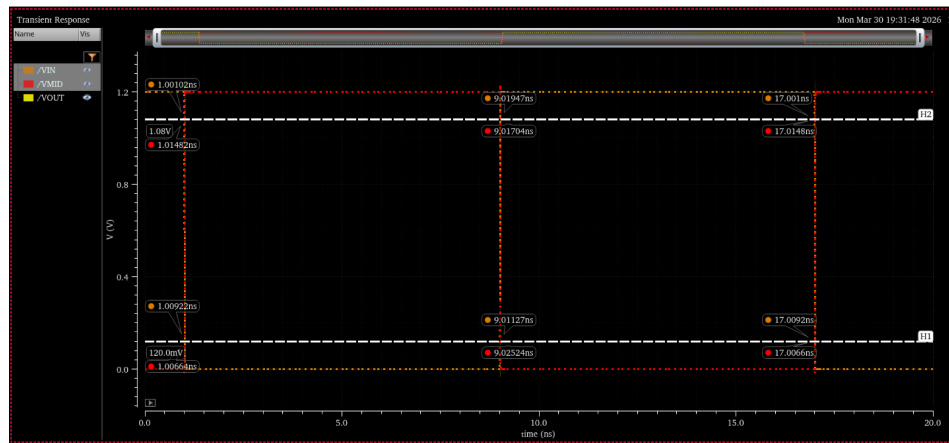
tf=9.5p



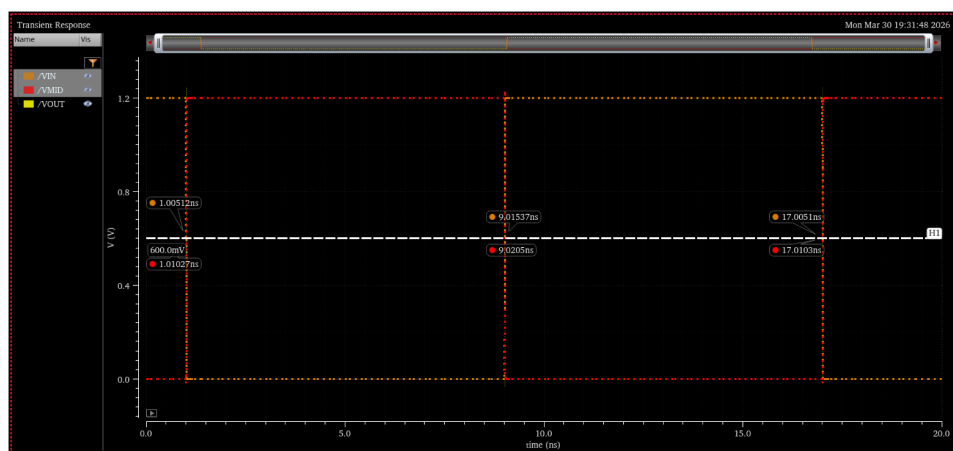
tf=9.75p



tf=10.25p



tpLH:



Discussion

- When both inverters in the 2-stage chain are changed from 1X to 2X, the effective pull-up and pull-down strengths increase because the transistor widths are larger. As a result, the inverters switch faster and the first-stage propagation delays decrease.
- The simulations confirm that the adjusted input rise and fall times required for 1% matching are both approximately 10.25 ps, giving matched input/output transition times of about 8.2 ps.
- The extracted first-stage delays are $t_{pHL} = 5.13$ ps and $t_{pLH} = 5.15$ ps, which are slightly smaller than the corresponding values expected for the 1X case. This behavior is consistent with inverter delay theory, since larger transistor sizes reduce the equivalent on-resistance and allow faster charging and discharging of internal nodes.
- Although the second-stage input capacitance also increases when the inverter is scaled to 2X, the increase in drive strength dominates, so the overall stage delay still reduces.

Conclusion

The 2-stage inverter chain with all stages changed from 1X to 2X was successfully simulated. By tuning the input transition times, the best matching values were found to be $t_r = 10.25$ ps for the rising-input case and $t_f = 10.25$ ps for the falling-input case. The extracted first-stage propagation delays were $t_{pHL} = 5.13$ ps and $t_{pLH} = 5.15$ ps. Compared to a 1X inverter chain, the 2X chain is expected to switch faster and exhibit smaller propagation delays, which agrees with the theoretical expectation for stronger CMOS inverters.

Q5c Power Parameters Extractions

Objective

To simulate a 2-stage inverter chain of 1X CMOS inverters in Cadence Virtuoso at $V_{DD} = 1.2$ V, characterize its transient response, and estimate the total, leakage, switching, and short-circuit power of the first-stage inverter.

Setup

Technology used: FreePDK45

CMOS inverter dimensions:

NMOS: $W_n = 1.000$ μm , $L_n = 50$ nm

PMOS: $W_p = 1.563$ μm , $L_p = 50$ nm

Input pulse parameters:

Voltage 1 = 0 V

Voltage 2 = 1.2 V

Delay time = 1 ns

Rise time = 10.3 ps

Fall time = 10.3 ps

Pulse width = 8 ns

Period = 16 ns

Method

Step 1: A 2-stage inverter chain was created using two identical 1X CMOS inverters.

Step 2: Transient simulation was performed with the given pulse input, and the first-stage supply current was measured over one steady-state cycle from 1 ns to 17 ns.

Step 3: Total average power was calculated using:

$$P_{\text{total}} = V_{DD} \times |I_{DD,\text{avg}}|$$

Step 4: Leakage power was estimated using two static transient runs with $V_{IN} = 0$ V and $V_{IN} = 1.2$ V.

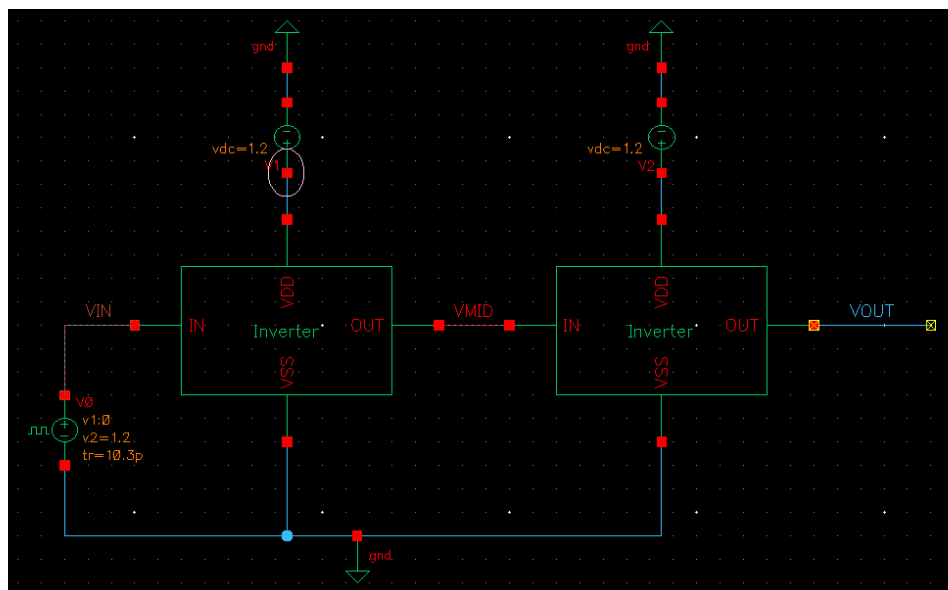
Step 5: Switching power was estimated using:

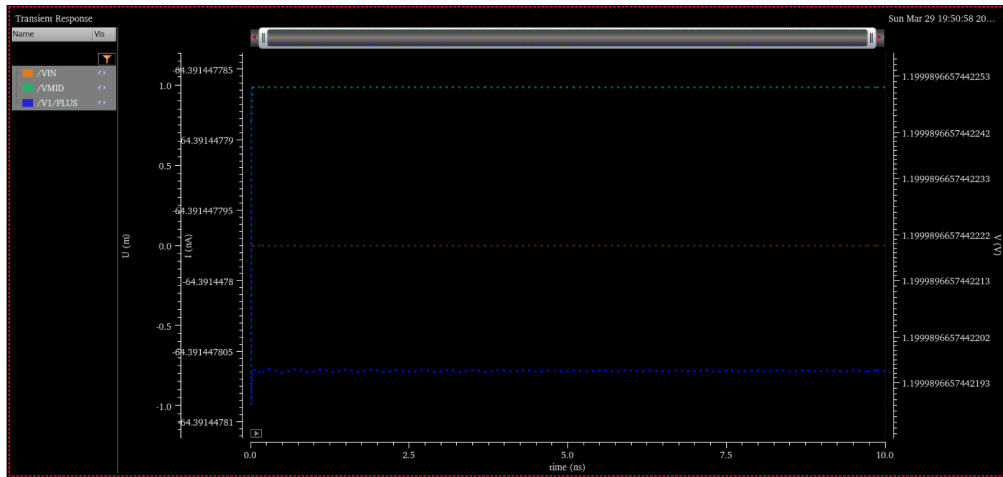
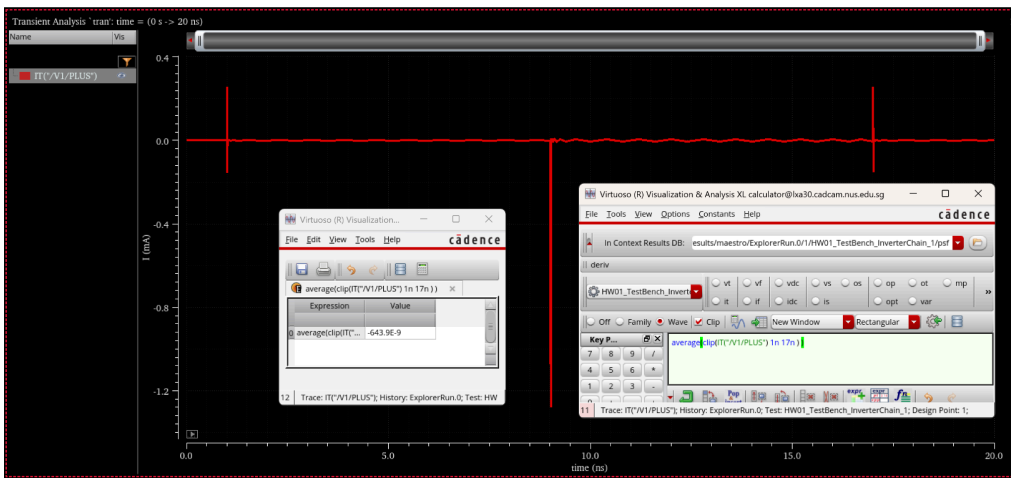
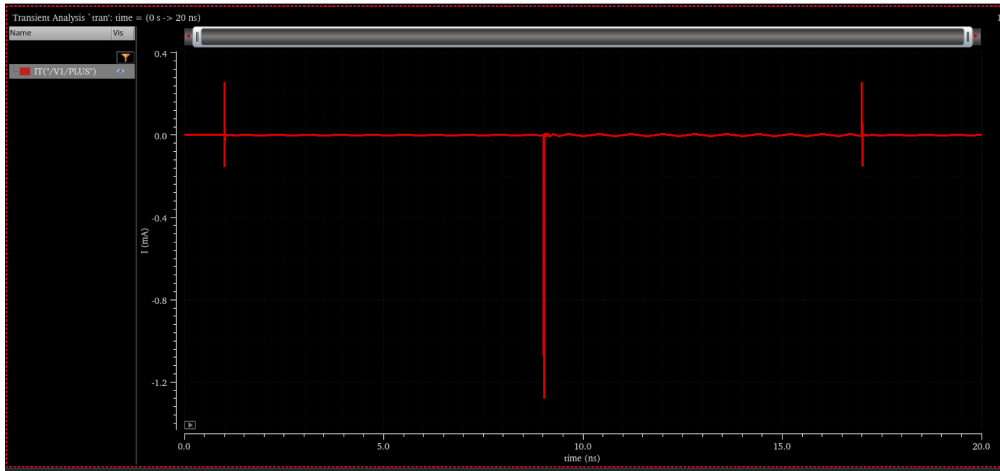
$$P_{\text{sw}} = C_L \times V_{DD}^2 \times f$$

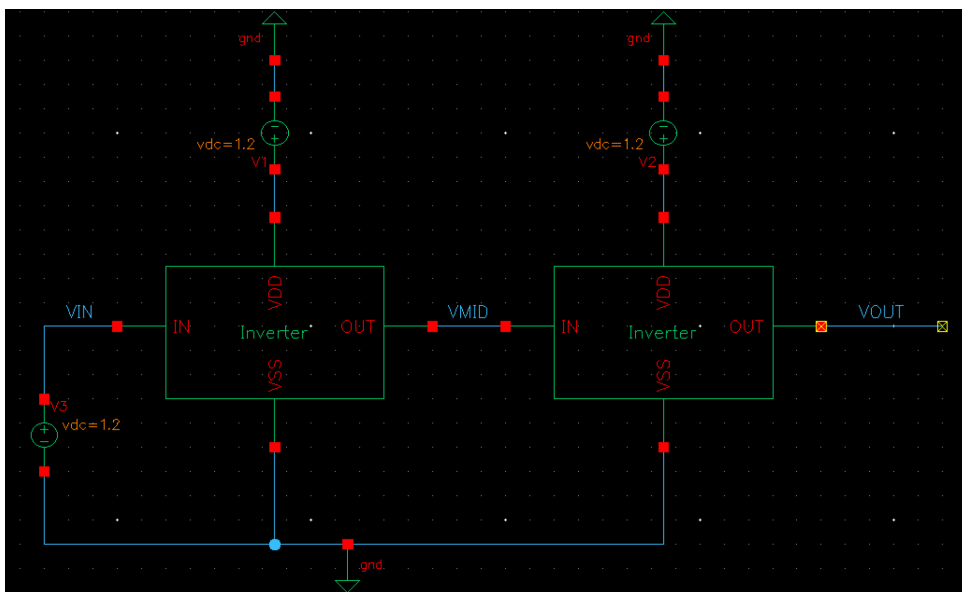
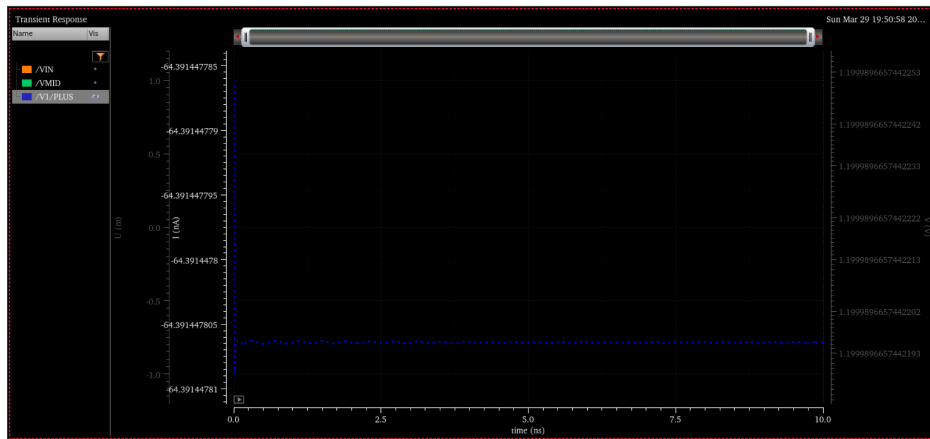
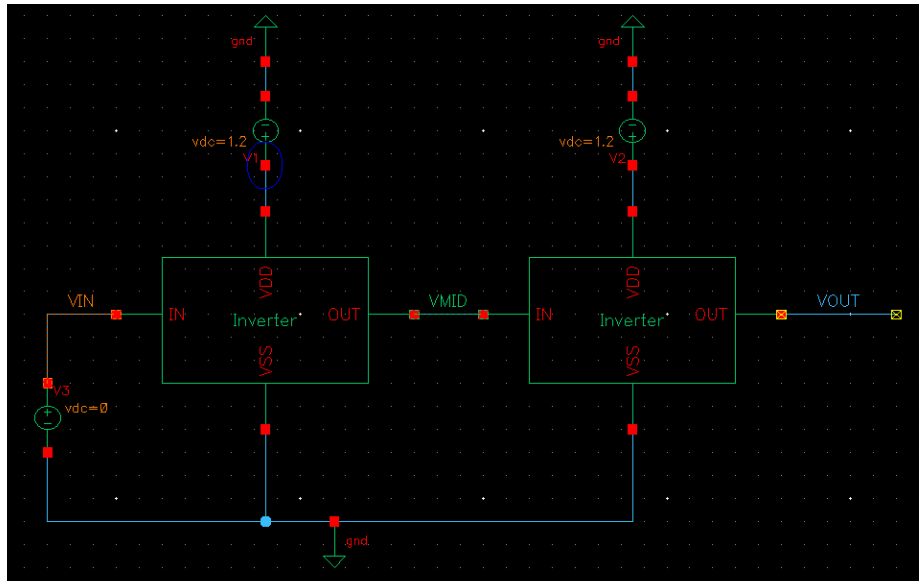
where C_L is the input capacitance of the second inverter.

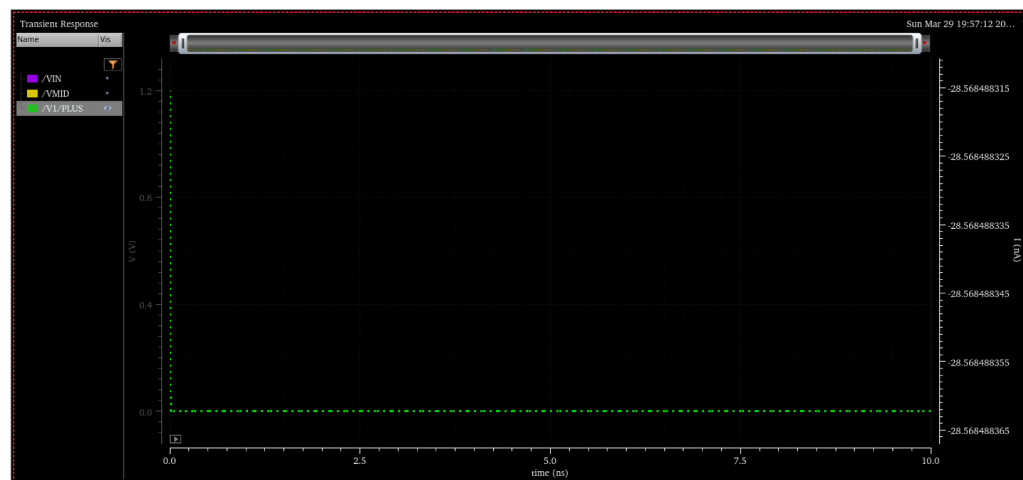
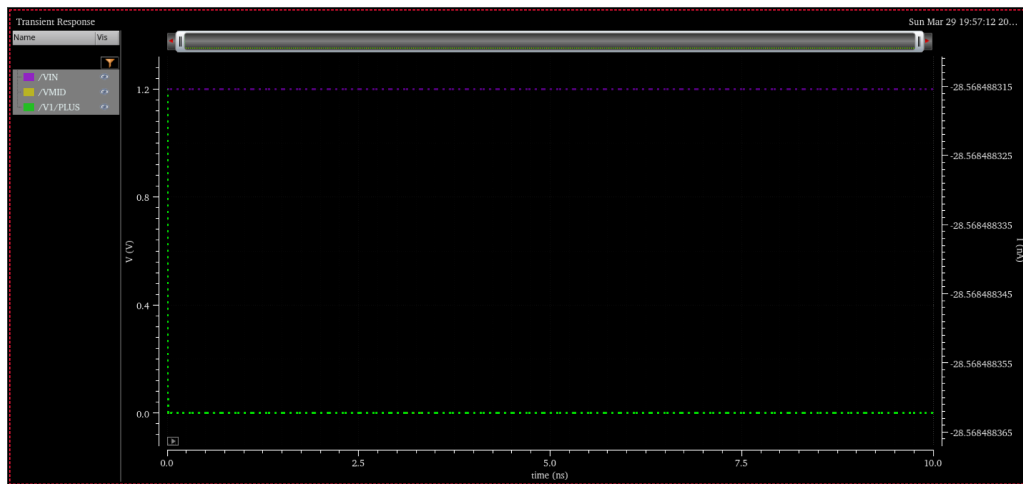
Step 6: Short-circuit power was obtained using:

$$P_{\text{sc}} = P_{\text{total}} - P_{\text{sw}} - P_{\text{leak}}$$









Results

Leakage current for $V_{IN} = 0$ V:

$I_{leak,0} = -64.39145$ nA

$P_{leak,0} = 77.27$ nW

Leakage current for $V_{IN} = 1.2$ V:

$I_{leak,1} = -28.56849$ nA

$P_{leak,1} = 34.28$ nW

Average leakage power: $P_{leak} = 55.78$ nW = 0.05578 uW

Estimated load capacitance: $C_L \approx 2.18$ fF

Estimated switching power: $P_{sw} \approx 0.196$ uW

Estimated short-circuit power: $P_{sc} \approx 0.5209$ uW

Discussion

The transient response of the 2-stage inverter chain was correct, with the first inverter driving the input capacitance of the second inverter. The leakage power was much smaller than the total power, showing that most of the power consumption was dynamic. The short-circuit power was significant because both NMOS and PMOS were ON simultaneously during the input transition.

Conclusion

The 2-stage 1X CMOS inverter chain was successfully characterized using transient simulations. The first-stage inverter showed a total average power of 0.7727 uW, with leakage power of 0.05578 uW,

switching power of about 0.196 μW , and short-circuit power of about 0.5209 μW . These results confirm correct transient operation and show that leakage power is much smaller than the dynamic power components.

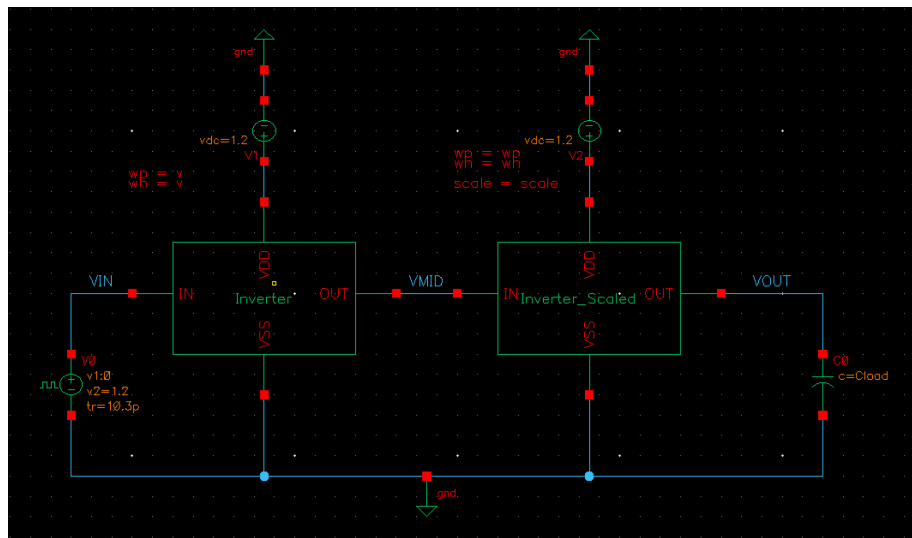
Q6 CLOAD Optimization for Inverter Chain

Objective

To determine the output load capacitance (CLOAD) at the output of the final stage of a two-stage inverter chain such that the propagation delay of the first stage is within 1% of the propagation delay of the second stage, while varying the final-stage inverter strength from 1X to 4X in steps of 0.5X.

Setup

A FreePDK45 CMOS two-stage inverter chain was used. The first stage was fixed at 1X. The final stage strength was varied as 1X, 1.5X, 2X, 2.5X, 3X, 3.5X, and 4X. The same input pulse as in Question 5(a) was applied. VDD was 1.2 V. Delay was measured at the 50% voltage level, i.e. 0.6 V. For each strength, CLOAD was swept in rounds to find the matching value.



Method

For each strength, the propagation delays of stage 1 and stage 2 were measured from the waveform markers. The average propagation delay was taken as:

$$t_p = (t_{pHL} + t_{pLH}) / 2$$

The percentage mismatch was calculated as:

$$\% \text{ mismatch} = |t_{p1} - t_{p2}| / t_{p2} \times 100$$

The CLOAD value was accepted when the mismatch became less than or equal to 1%.

Results

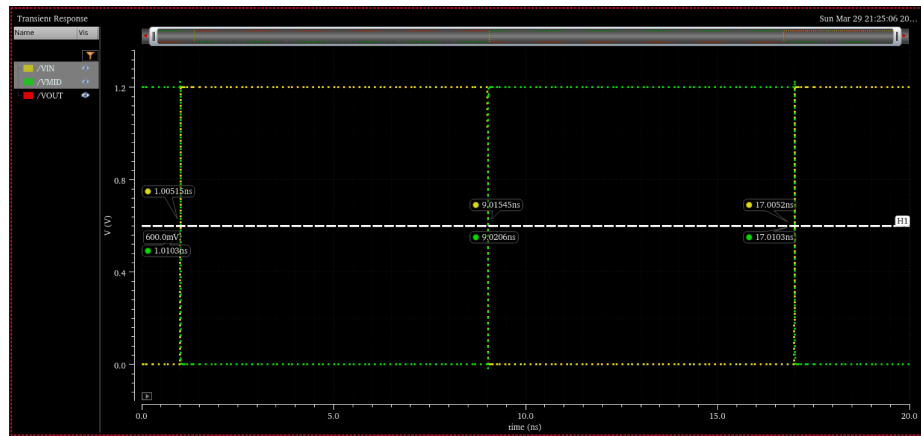
As the final-stage inverter strength increased, the required CLOAD also increased. This is because a stronger inverter has lower effective output resistance and therefore needs a larger load capacitance to maintain delay matching with the first stage.

XX

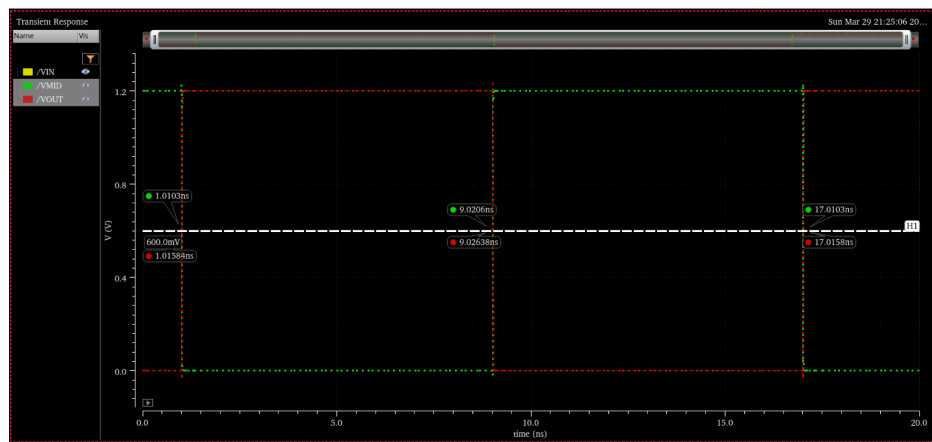
Strength = 1X

Cload = 5f

Stage 1:

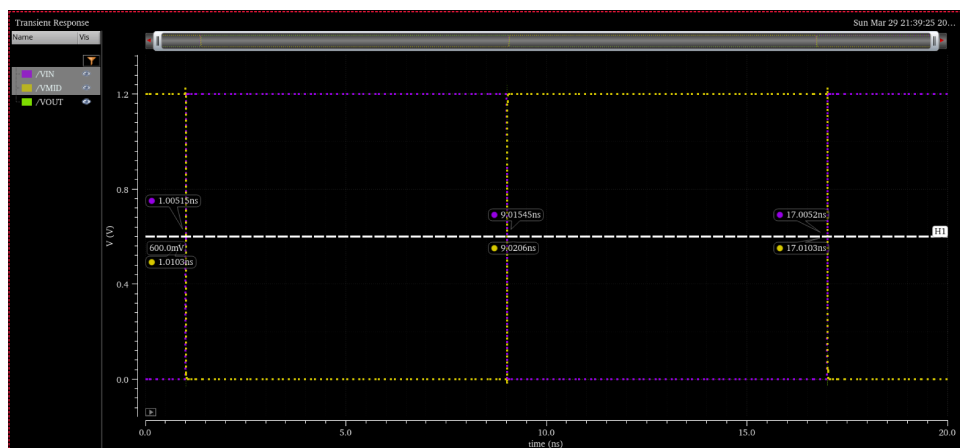


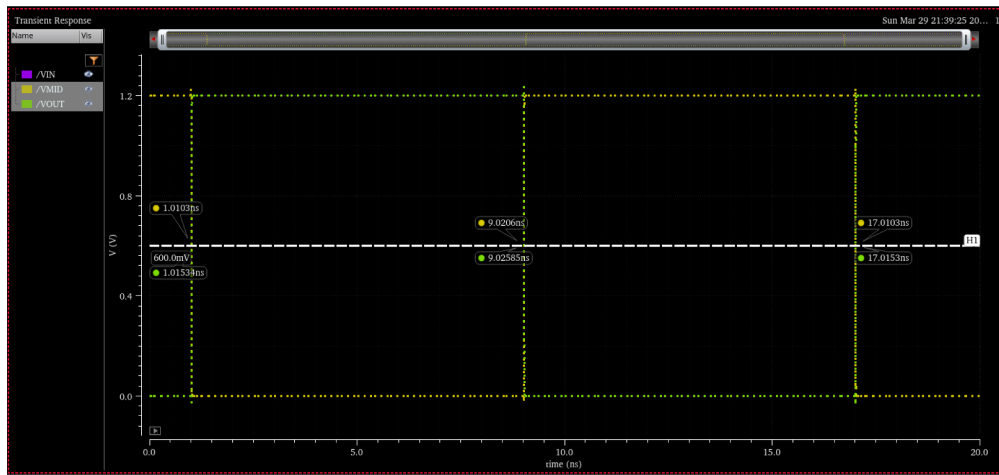
Stage 2:



Cloud = 4f

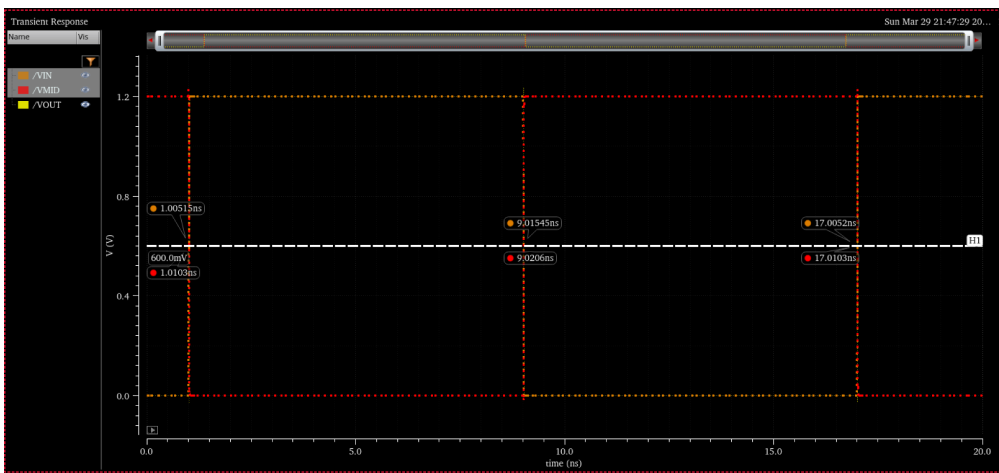
Stage1:



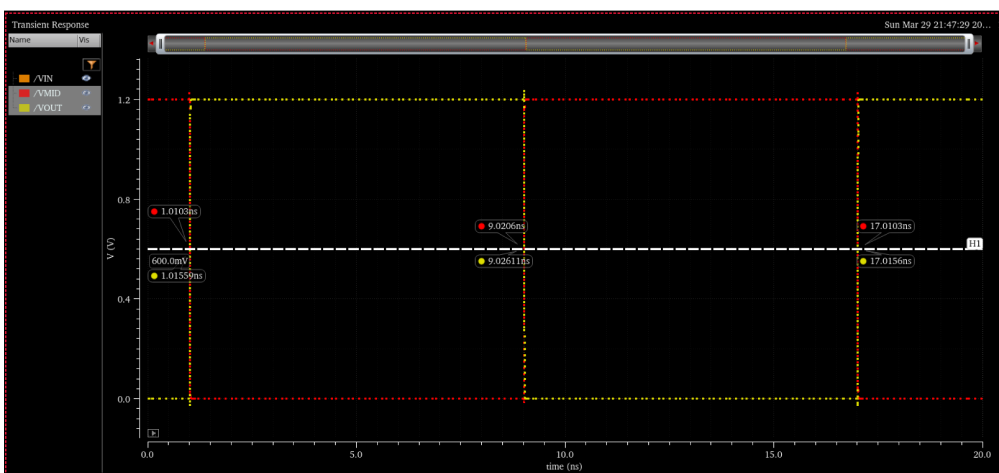


Cloud=4.5f

Stage1:

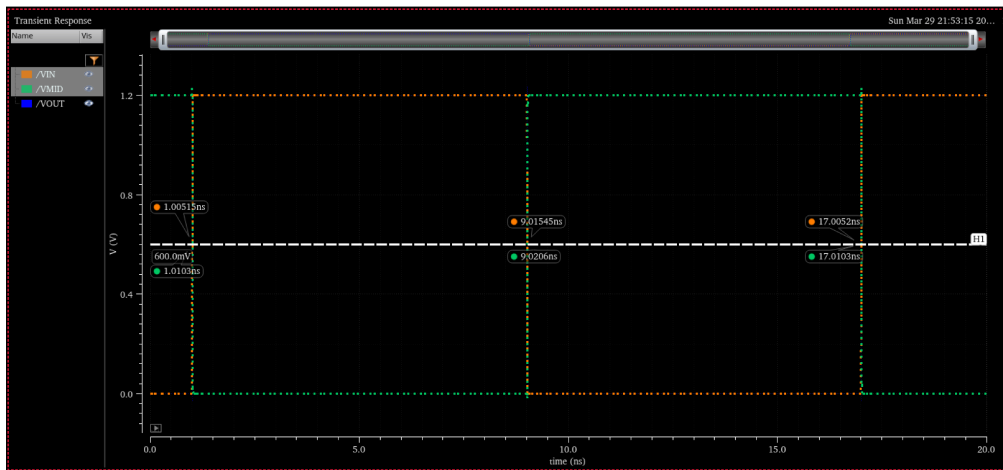


Stage 2:

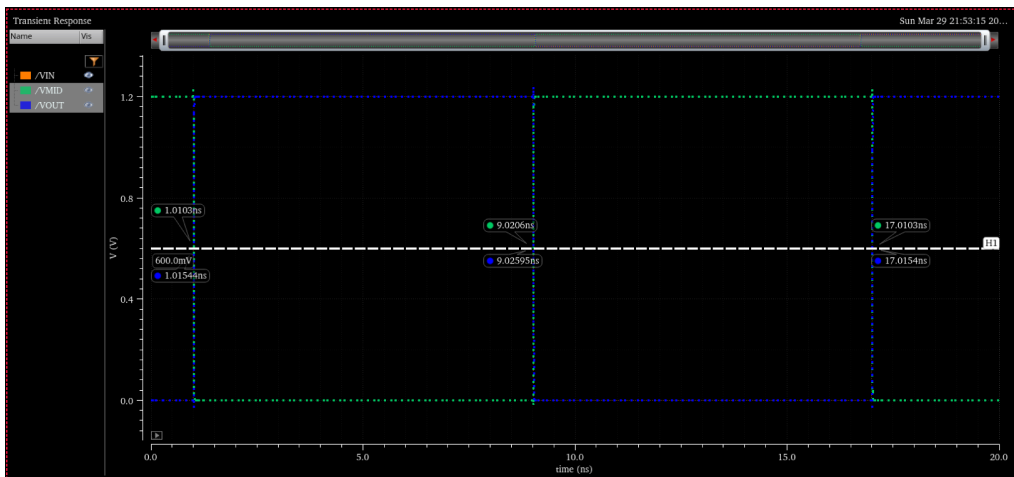


Cloud=4.2f

Stage 1:



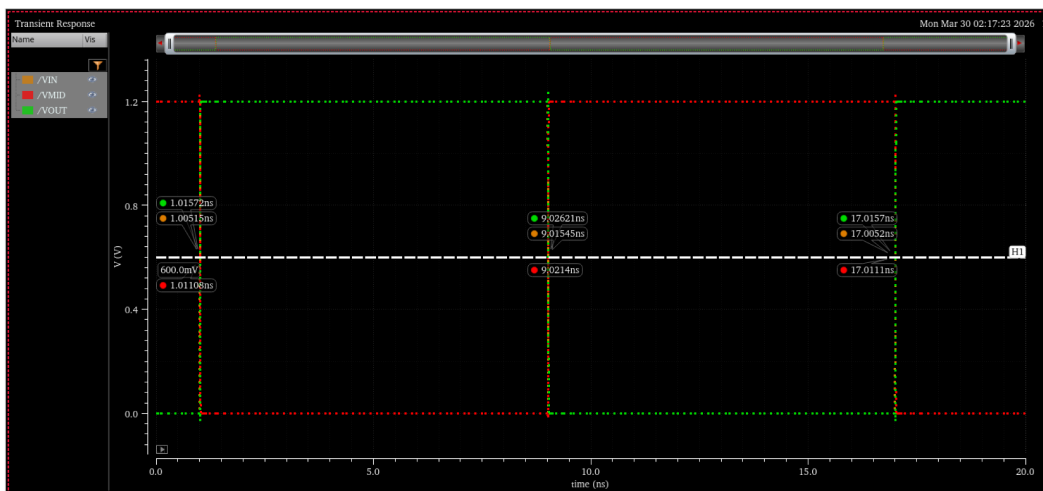
Stage 2:



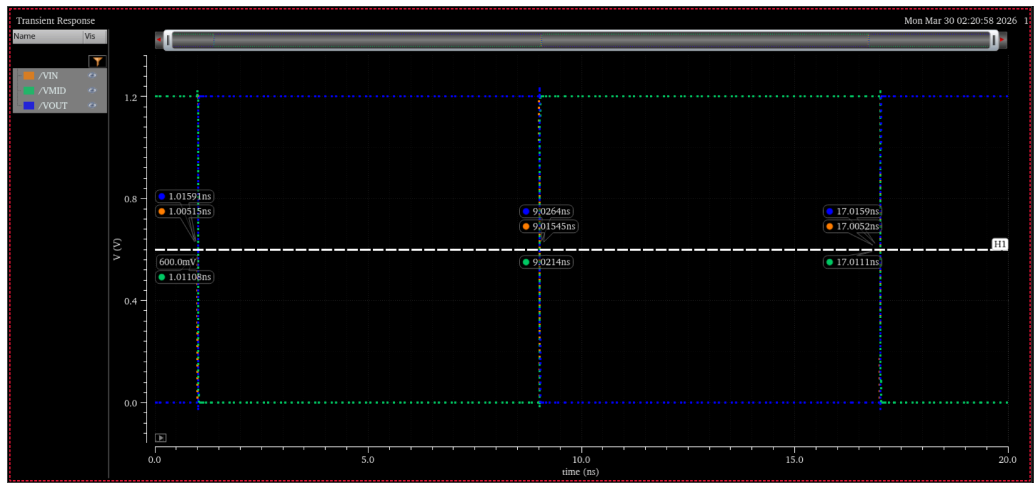
XX

Strength: 1.5X

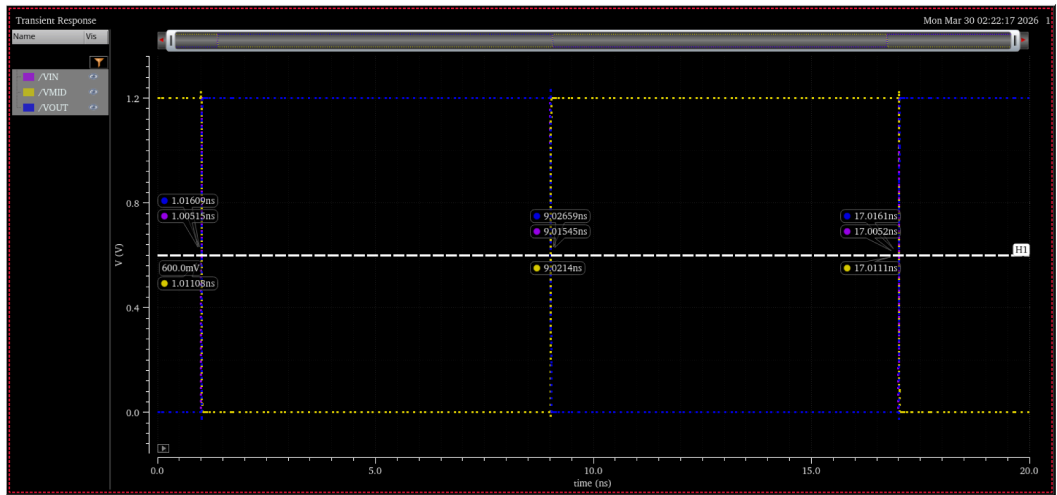
Load: 4f F



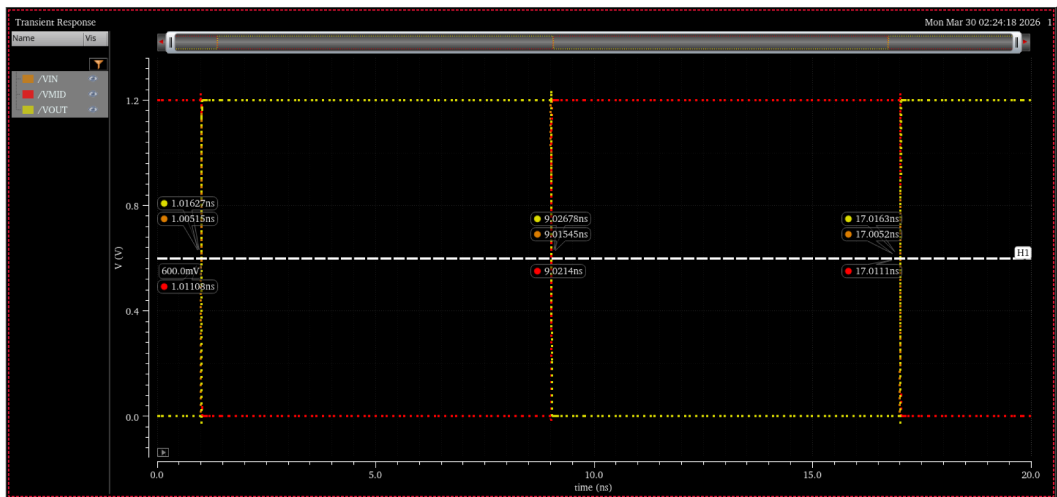
Load: 4.5f



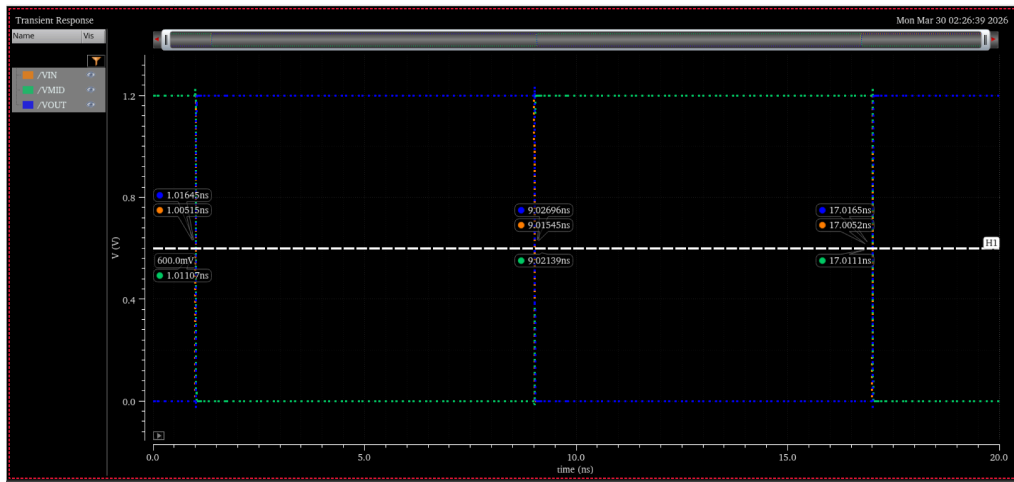
Cloud: 5f



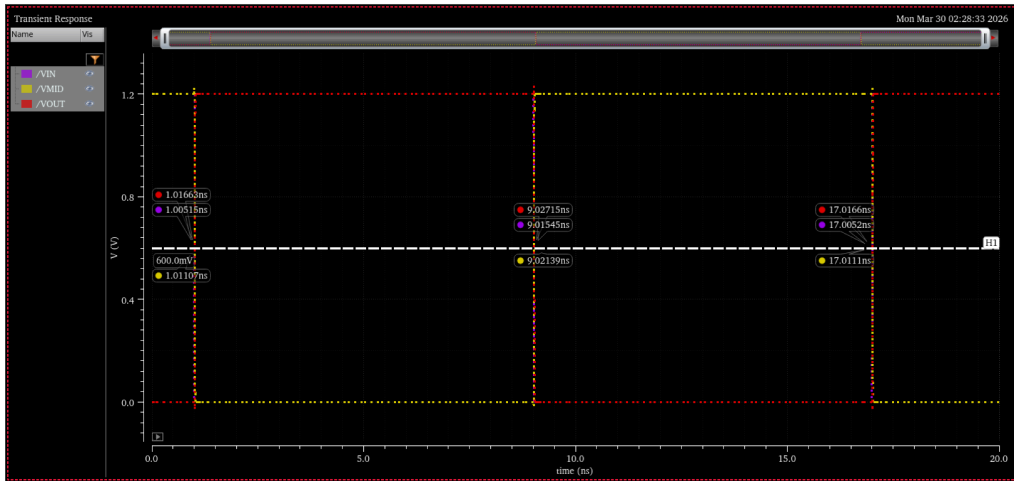
Cloud: 5.5f



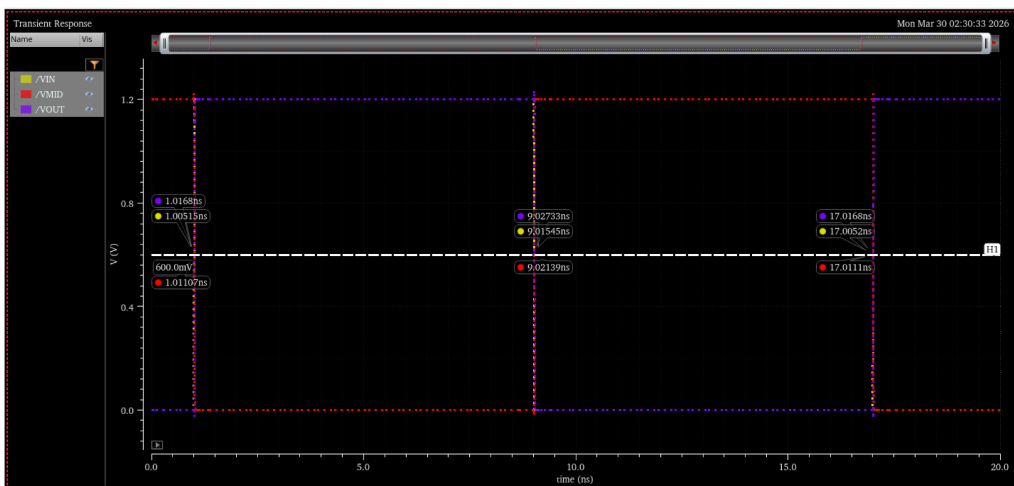
Cloud: 6f



Cloud: 6.5f

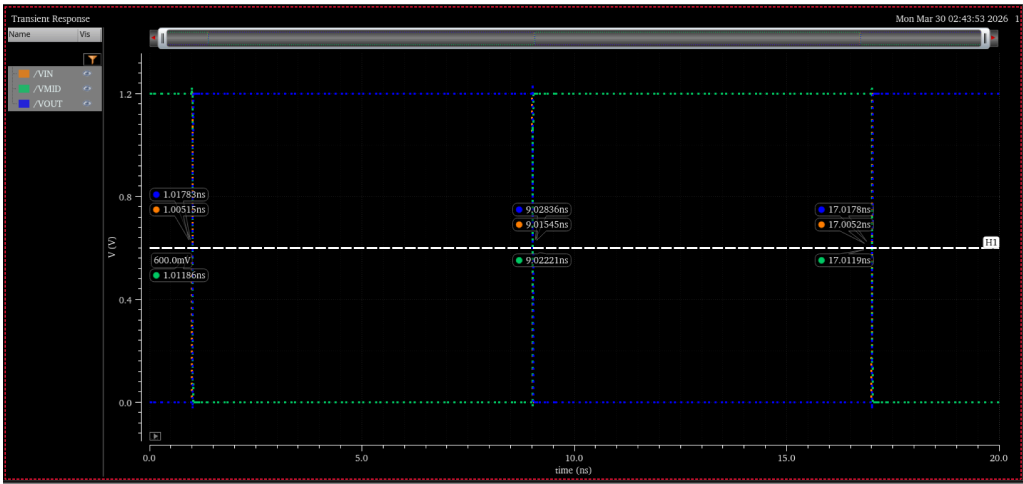


Cloud: 7f

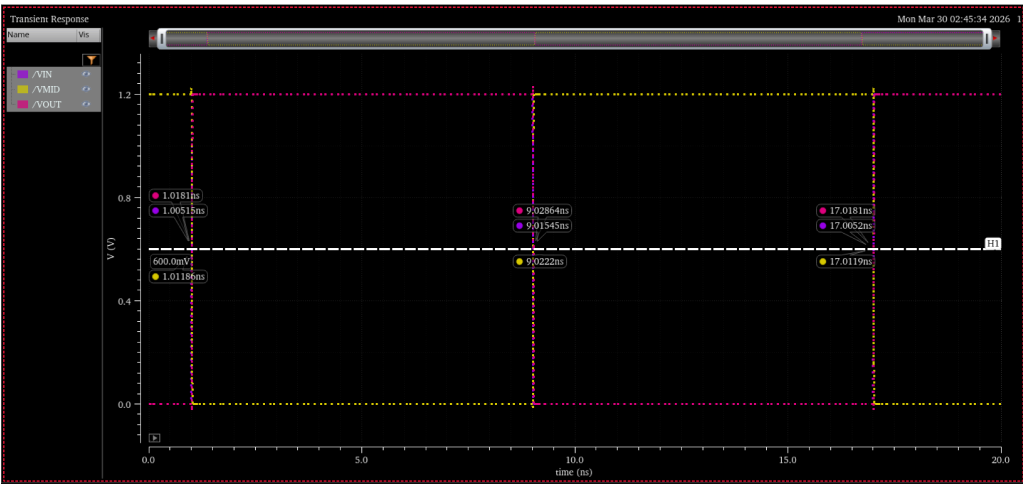


Cloud: 7.5f

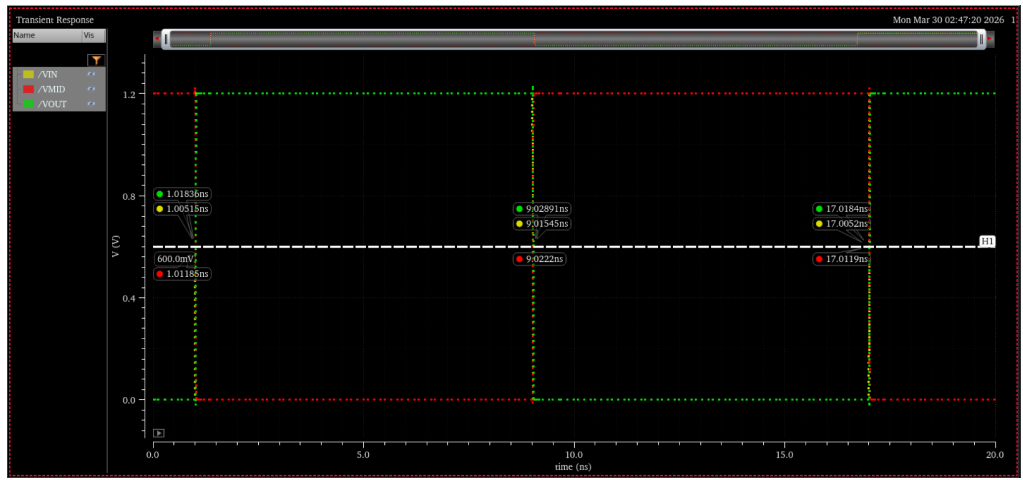
Cload: 9f



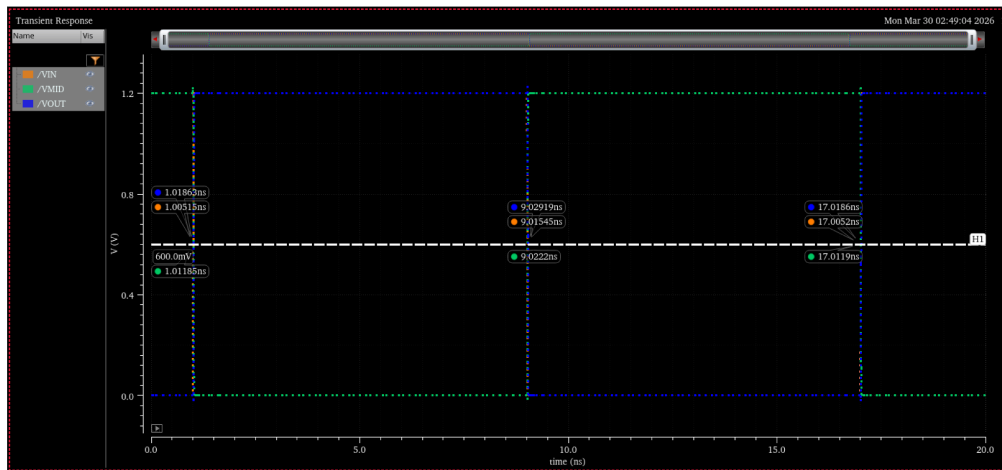
Cload:10f



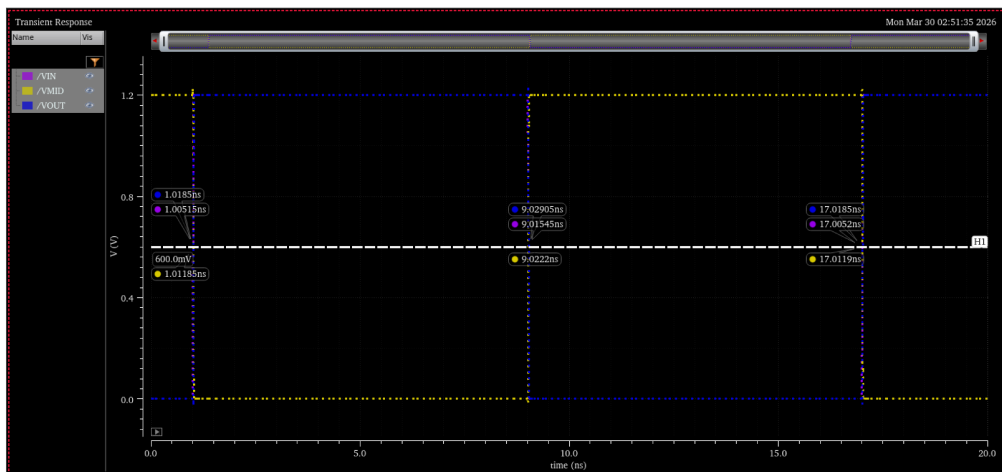
Cload:11f



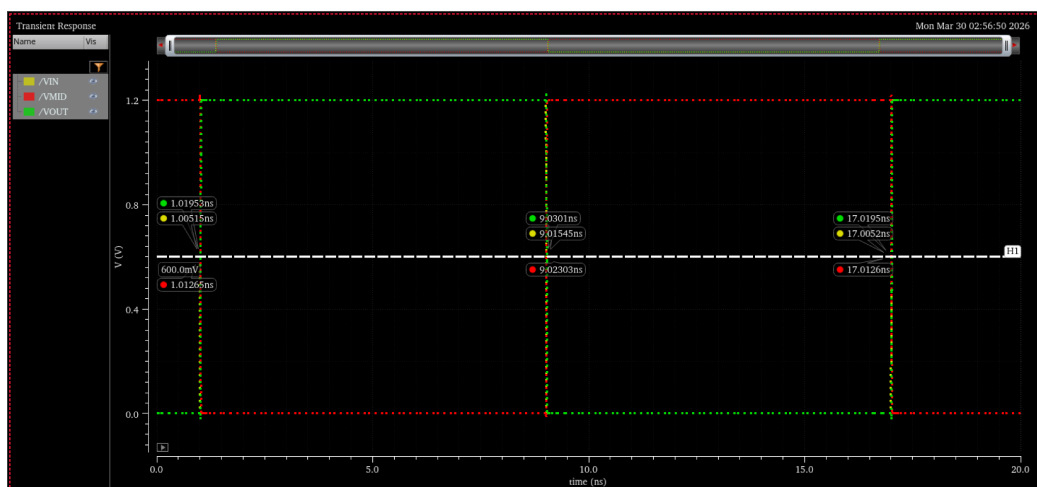
Cload:12f



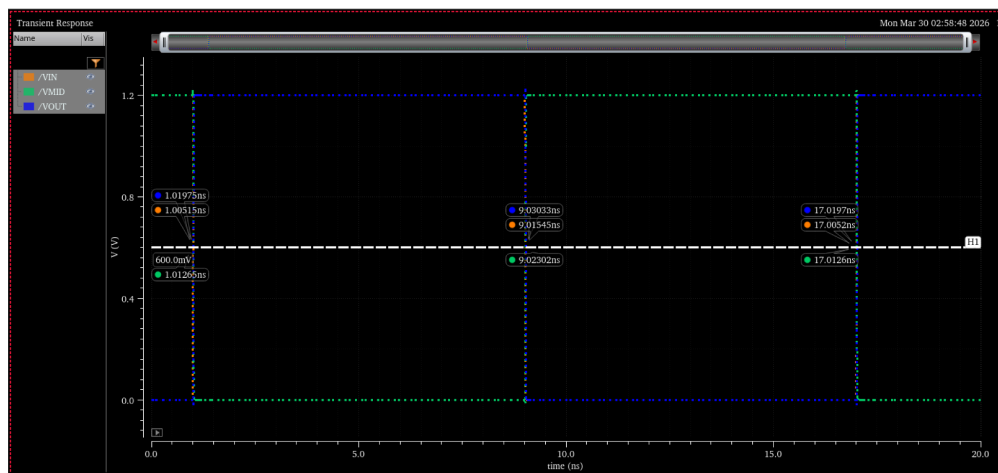
Round 2:
Cload:11.5f

[illegible]

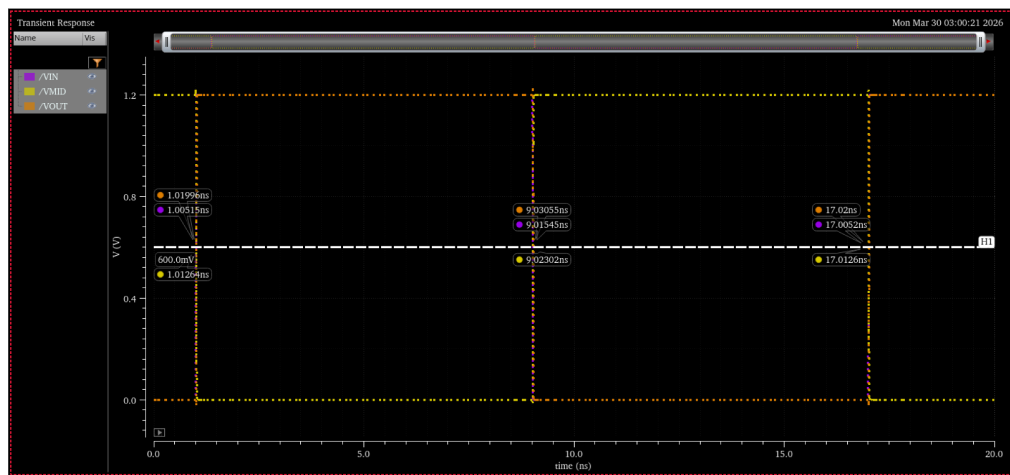
Strength: 2.5X
Clod:14f



Cload:15f

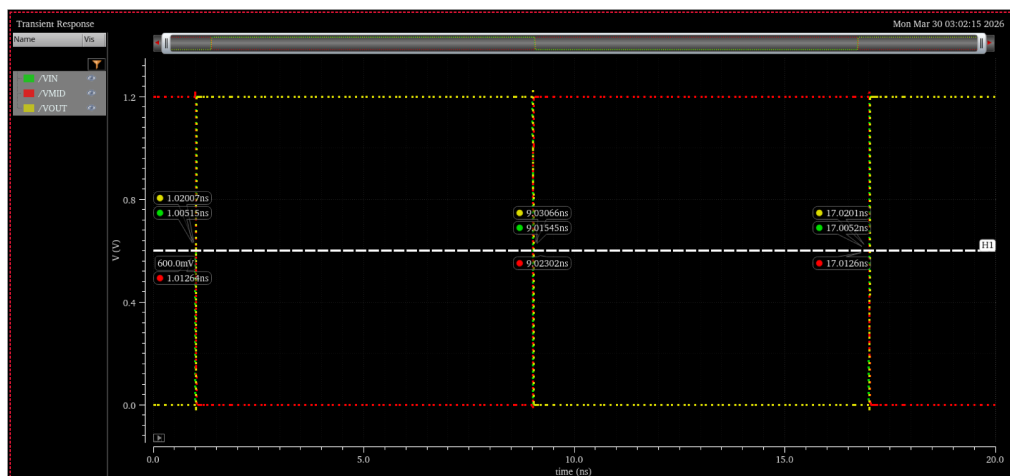


Cload:16f



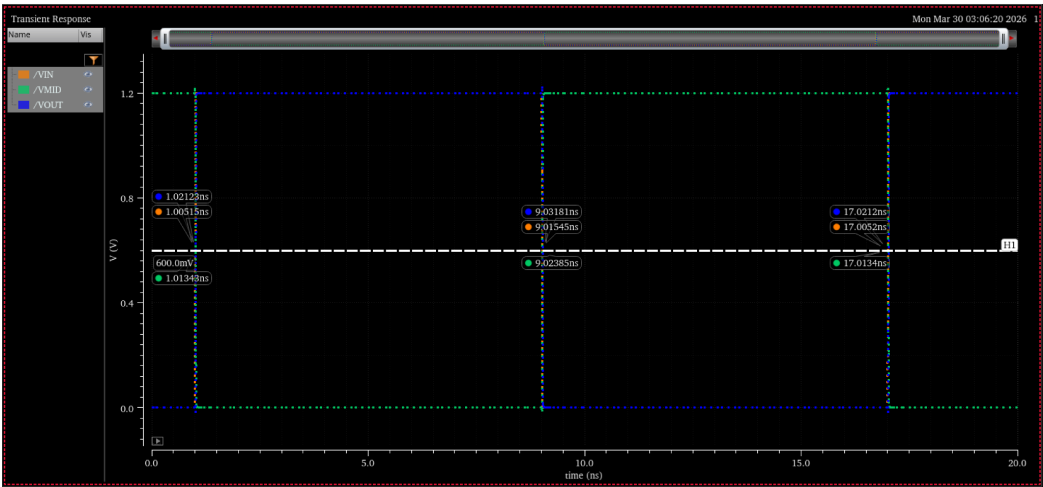
Round 2:

Clod: 16.5f

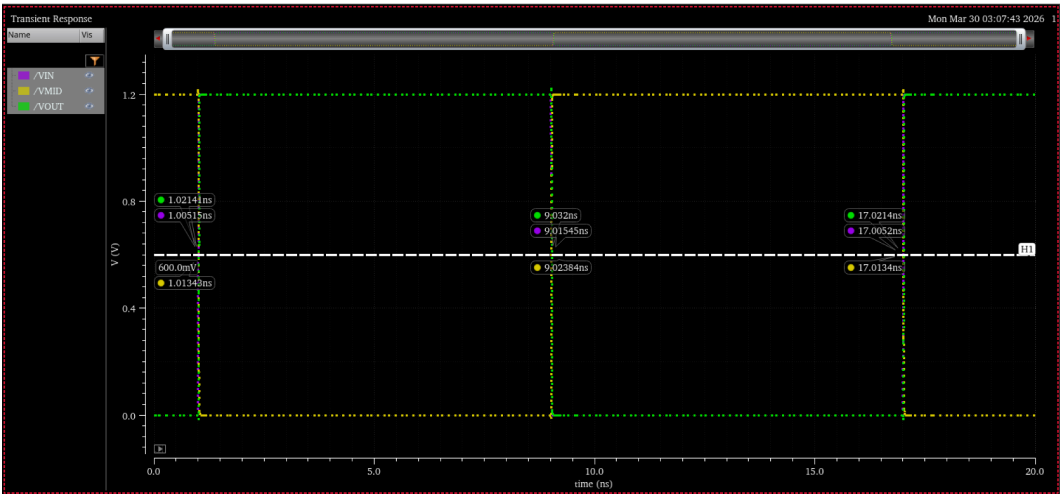
[illegible]

Strength:3X

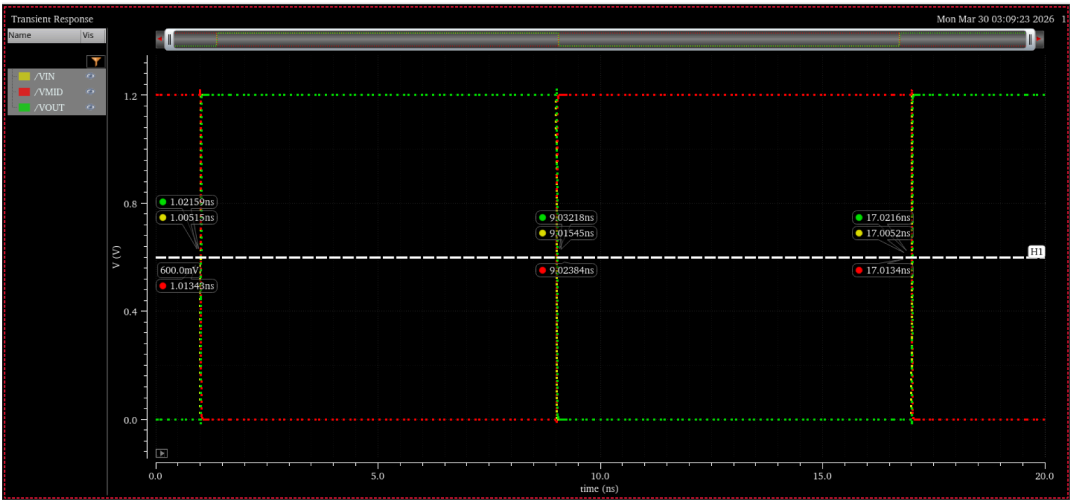
Round1:
Load:20f



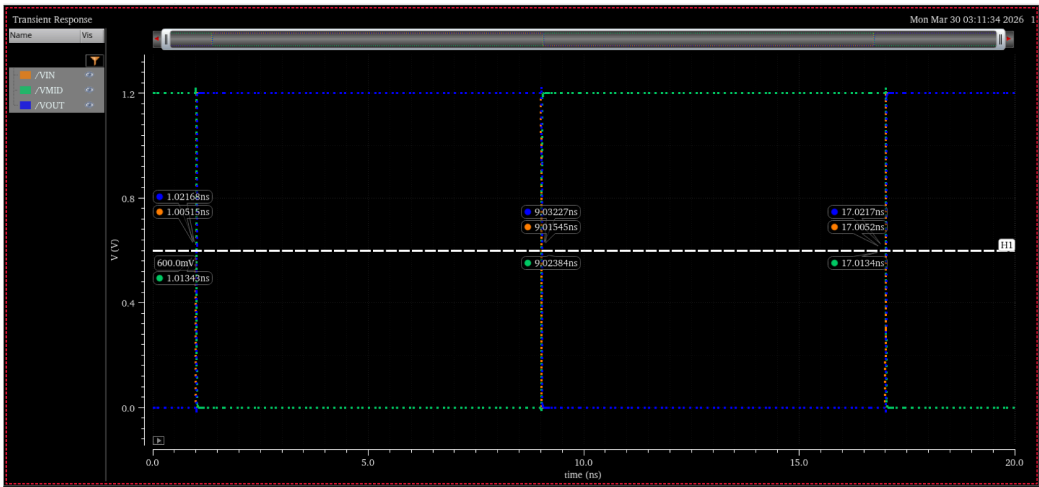
Load:21f



Load:22f



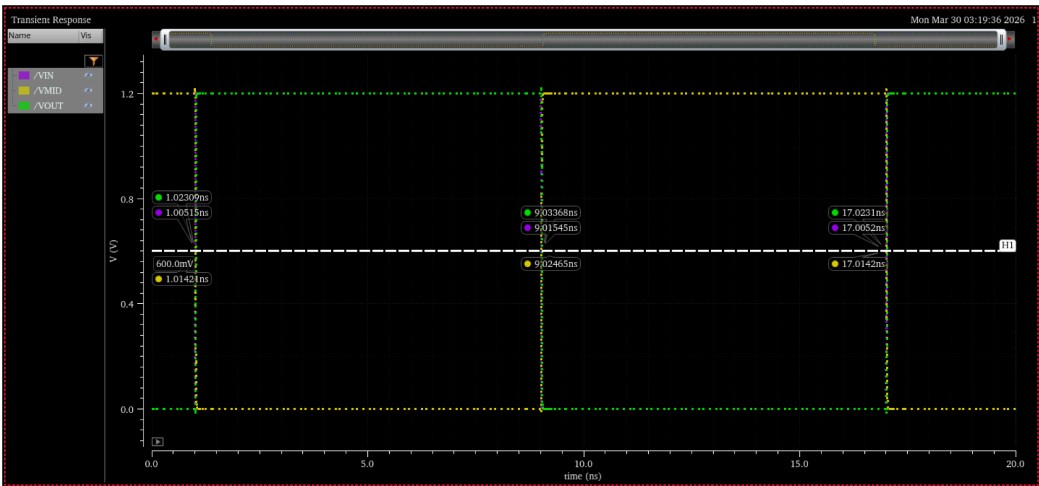
Round2:
Cloud:22.5f



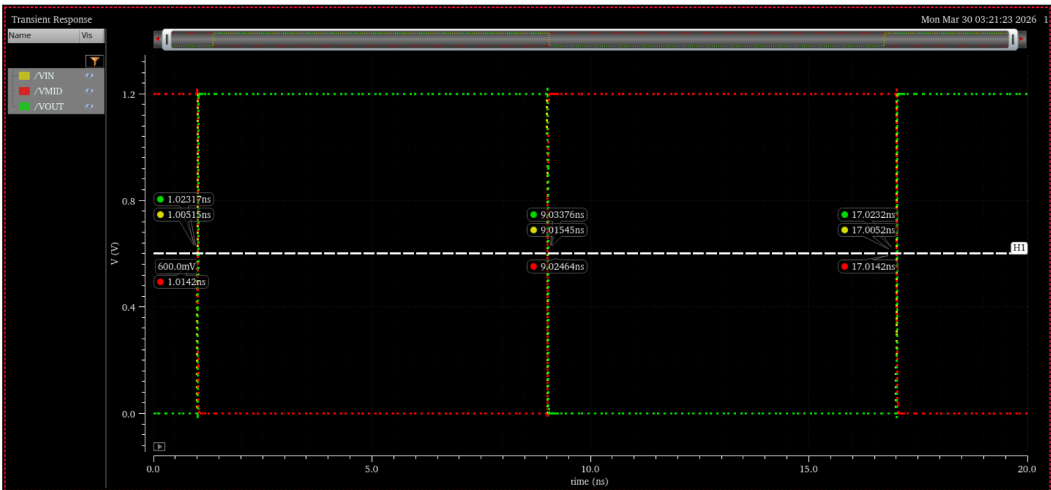
XX

Strength:3.5X

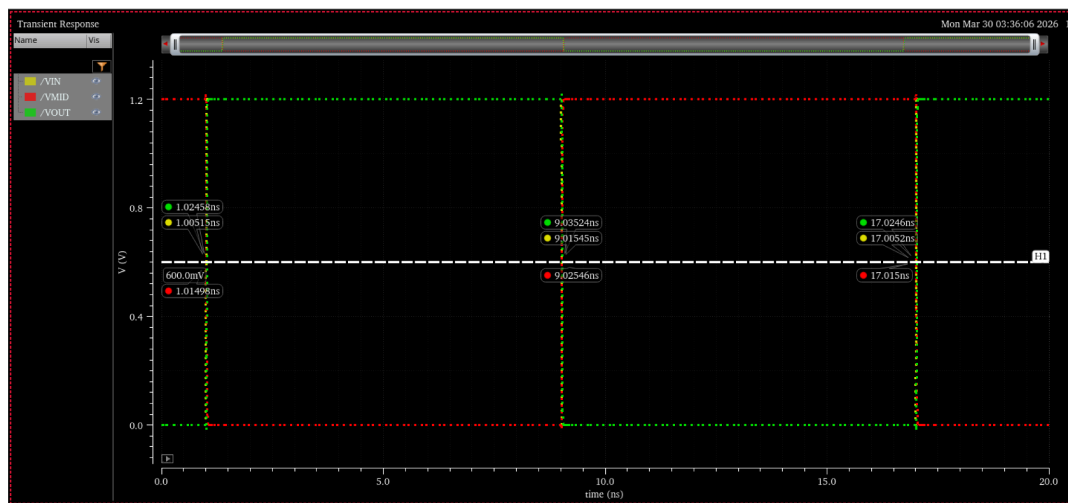
Cloud:28f



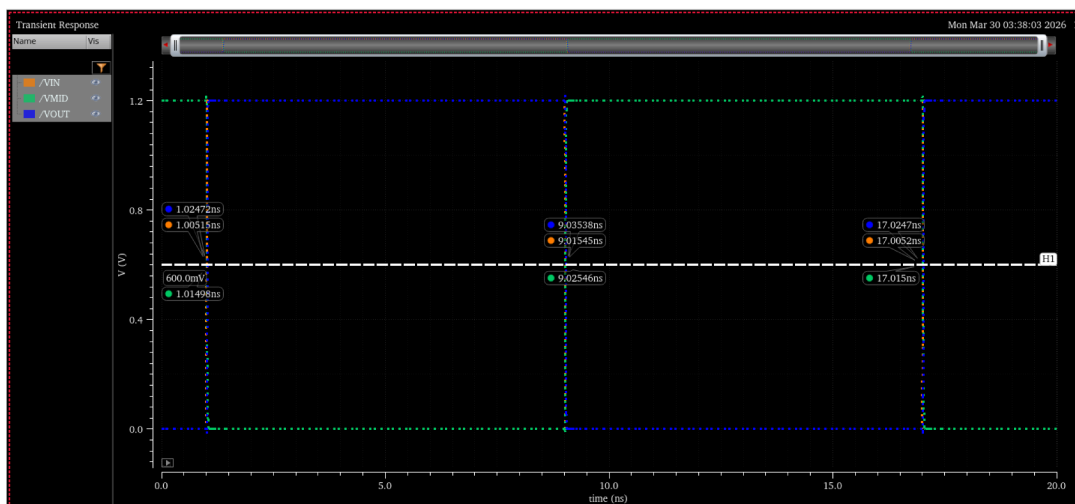
Cloud:28.5f



Clod: 35f



Cload:36f



Strength	Round	Cload tried (fF)	tp1 (ps)	tp2 (ps)	%Difference	Observation
1X	1	5	5.15	5.66	9.01	Rejected, Stage 1 faster
1X	1	4	5.15	5.24	1.72	Very close, Stage 1 slightly faster
1X	2	4.2	5.15	5.14	0.19	Accepted, Stage 2 slightly faster

1.5X	1	4	5.94	4.725	25.7	Stage 2 too fast
1.5X	1	4.5	5.93	4.915	20.9	Stage 2 is fast
1.5X	1	5	5.93	5.1	16.5	Stage 2 is fast
1.5X	1	5.5	5.93	5.285	12.4	Stage 2 is fast
1.5X	1	6	5.93	5.475	8.31	Stage 2 is fast
1.5X	1	6.5	5.93	5.66	4.77	Stage 2 slightly fast
1.5X	1	7	5.93	5.835	1.63	Very close, Stage 2 is fast
1.5X	1	7.5	5.93	6.015	1.41	Very close, Stage 2 is slow
1.5X	2	7.25	5.93	5.925	0.08	Accepted
2X	1	8	6.735	5.775	16.6	Stage 2 is fast
2X	1	9	6.735	6.06	11.1	Stage 2 is fast
2X	1	10	6.73	6.34	6.15	Stage 2 is fast
2X	1	11	6.725	6.61	1.74	Very close, Stage 2 slightly fast
2X	1	12	6.725	6.885	2.32	Very close, Stage 2 slightly slow
2X	2	11.5	6.725	6.75	0.37	Accepted
2.5X	1	14	7.54	6.975	8.10	Stage 2 too fast
2.5X	1	15	7.535	7.205	4.58	Stage 2 is fast
2.5X	1	16	7.53	7.425	1.41	Very close, Stage 2 slightly fast
2.5X	2	16.5	7.53	7.535	0.07	Accepted

3X	1	20	8.34	7.88	5.84	Stage 2 too fast
3X	1	21	8.335	8.07	3.28	Stage 2 too fast
3X	1	22	8.335	8.25	1.03	Borderline, sightly fast
3X	2	22.5	8.335	8.34	0.06	Accepted
3.5X	1	28	9.13	8.955	1.95	Very close, Stage 2 slightly fast
3.5X	2	28.5	9.12	9.045	0.83	Accepted
4X	1	35	9.92	9.69	2.37	Rejected, Stage 2 slightly fast
4X	2	36	9.92	9.83	0.92	Accepted

Strength (X)	Final Chosen Load (fF)
1	4.2
1.5	7.25
2	11.5
2.5	16.5
3	22.5
3.5	28.5
4	36

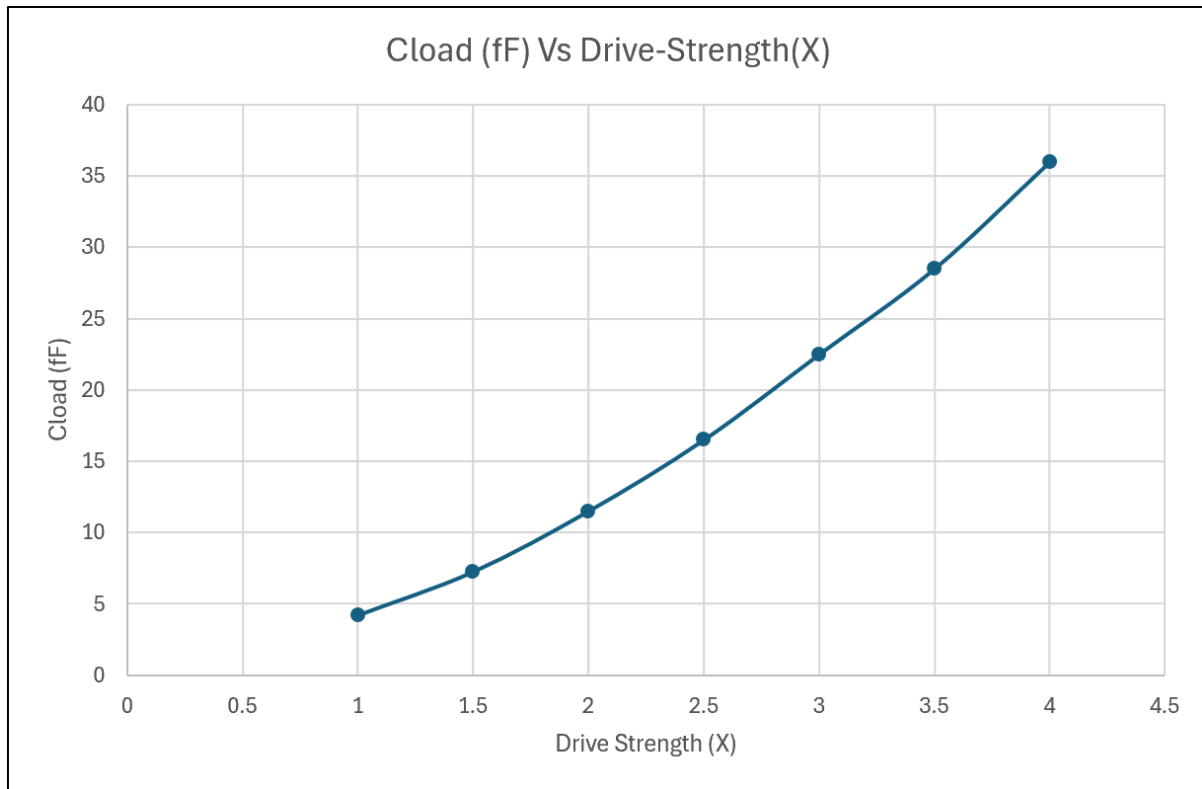


Figure shows that the required CLOAD increases with final-stage drive strength, with a slightly nonlinear upward trend.

Discussion

- The plot shows that the required CLOAD increases as the drive strength of the final-stage inverter increases from 1X to 4X. This trend is expected because increasing the inverter size reduces its effective output resistance and allows the second stage to switch faster. To keep the propagation delay of the second stage close to that of the first stage, a larger output load capacitance must therefore be added.
- The increase in CLOAD is not perfectly linear. At lower strengths, the required increase is moderate, while at higher strengths the required CLOAD rises more rapidly. This indicates that the delay-matching condition depends on the combined effect of inverter resistance, parasitic capacitances, and external load. Overall, the graph confirms the expected RC-based delay balancing behavior of an inverter chain.

Conclusion.

The required output load capacitance for delay matching was successfully extracted for final-stage inverter strengths from 1X to 4X. The final chosen CLOAD values were 4.2 fF, 7.25 fF, 11.5 fF, 16.5 fF, 22.5 fF, 28.5 fF, and 36.0 fF for 1X, 1.5X, 2X, 2.5X, 3X, 3.5X, and 4X respectively. The plot shows a clear increasing trend of CLOAD with drive strength, confirming that a stronger inverter requires a larger capacitive load to achieve delay matching with the first stage. Thus, the simulation results agree well with inverter chain optimization principles.

APPENDIX

```
// Point Netlist Generated on: Mar 27 00:05:52 2026

// Generated for: spectre

// Design Netlist Generated on: Mar 26 22:07:43 2026

// Design library name: HW01

// Design cell name: CMOS_Inverter

// Design view name: schematic

simulator lang=spectre

global 0

parameters temperature=27 Vin=0 wp=1.563u

include "/app11/FreePDK45/1.4/ncsu_basekit/models/hspice/hspice_nom.include"


// Library name: HW01

// Cell name: CMOS_Inverter

// View name: schematic

M1 (Vout Vin net1 net1) PMOS_VTG w=wp l=50n as=1.05e-13 ad=1.05e-13 \
    ps=1.21u pd=1.21u ld=105n ls=105n m=1

M0 (Vout Vin 0 0) NMOS_VTG w=1u l=50n as=1.05e-13 ad=1.05e-13 ps=1.21u \
    pd=1.21u ld=105n ls=105n m=1

V1 (net1 0) vsource dc=1.2 type=dc

V0 (Vin 0) vsource dc=Vin type=dc

simulatorOptions options psfversion="1.4.0" reltol=1e-3 vabstol=1e-6 \
    iabstol=1e-12 temp=27 tnom=27 scalem=1.0 scale=1.0 gmin=1e-12 rforce=1 \
    maxnotes=5 maxwarns=5 digits=5 cols=80 pivrel=1e-3 \
    sensfile="../psf/sens.output" checklimitdest=psf

dc dc param=Vin start=0 stop=1.2 step=0.001 write="spectre.dc" \
    oppoint=rawfile maxiters=150 maxsteps=10000 annotate=status

modelParameter info what=models where=rawfile

element info what=inst where=rawfile

outputParameter info what=output where=rawfile
```

designParamVals info what=parameters where=rawfile

primitives info what=primitives where=rawfile

subckts info what=subckts where=rawfile

save V1:p

saveOptions options save=allpub

XX

// Point Netlist Generated on: Mar 27 00:05:52 2026

// Generated for: spectre

// Design Netlist Generated on: Mar 26 22:07:43 2026

// Design library name: HW01

// Design cell name: CMOS_Inverter

// Design view name: schematic

simulator lang=spectre

global 0

parameters temperature=27 Vin=0 wp=1.563u

include "/app11/FreePDK45/1.4/ncsu_basekit/models/hspice/hspice_nom.include"

// Library name: HW01

// Cell name: CMOS_Inverter

// View name: schematic

M1 (Vout Vin net1 net1) PMOS_VTG w=wp l=50n as=1.05e-13 ad=1.05e-13 \

ps=1.21u pd=1.21u ld=105n ls=105n m=1

M0 (Vout Vin 0 0) NMOS_VTG w=1u l=50n as=1.05e-13 ad=1.05e-13 ps=1.21u \

pd=1.21u ld=105n ls=105n m=1

V1 (net1 0) vsource dc=1.2 type=dc

V0 (Vin 0) vsource dc=Vin type=dc

simulatorOptions options psfversion="1.4.0" reltol=1e-3 vabstol=1e-6 \

iabstol=1e-12 temp=27 tnom=27 scalem=1.0 scale=1.0 gmin=1e-12 rforce=1 \

maxnotes=5 maxwarns=5 digits=5 cols=80 pivrel=1e-3 \

sensfile="../psf/sens.output" checklimitdest=psf

```
dc dc param=Vin start=0 stop=1.2 step=0.001 write="spectre.dc" \
    oppoint=rawfile maxiters=150 maxsteps=10000 annotate=status
modelParameter info what=models where=rawfile
element info what=inst where=rawfile
outputParameter info what=output where=rawfile
designParamVals info what=parameters where=rawfile
primitives info what=primitives where=rawfile
subckts info what=subckts where=rawfile
save V1:p
saveOptions options save=allpub
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
// Point Netlist Generated on: Mar 28 11:11:37 2026
// Generated for: spectre
// Design Netlist Generated on: Mar 28 11:11:37 2026
// Design library name: HW01
// Design cell name: Inverter
// Design view name: schematic
simulator lang=spectre
global 0
parameters temperature=27
include "/app11/FreePDK45/1.4/ncsu_basekit/models/hspice/hspice_nom.include"

// Library name: HW01
// Cell name: Inverter
// View name: schematic
M0 (OUT IN VDD VDD) PMOS_VTG w=1.5625u l=50n as=1.64063e-13 ad=1.64063e-13 \
    ps=1.7725u pd=1.7725u ld=105n ls=105n m=1
M1 (OUT IN VSS VSS) NMOS_VTG w=1u l=50n as=1.05e-13 ad=1.05e-13 ps=1.21u \
    pd=1.21u ld=105n ls=105n m=1
simulatorOptions options psfversion="1.4.0" reltol=1e-3 vabstol=1e-6 \
```

```
iabstol=1e-12 temp=27 tnom=27 scalem=1.0 scale=1.0 gmin=1e-12 rforce=1 \
maxnotes=5 maxwarns=5 digits=5 cols=80 pivrel=1e-3 \
sensfile="../psf/sens.output" checklimitdest=psf
tran tran stop=20n write="spectre.ic" writefinal="spectre.fc" \
    annotate=status maxiters=5
finalTimeOP info what=oppoint where=rawfile
modelParameter info what=models where=rawfile
element info what=inst where=rawfile
outputParameter info what=output where=rawfile
designParamVals info what=parameters where=rawfile
primitives info what=primitives where=rawfile
subckts info what=subckts where=rawfile
saveOptions options save=allpub
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
// Point Netlist Generated on: Mar 28 11:11:37 2026
// Generated for: spectre
// Design Netlist Generated on: Mar 28 11:11:37 2026
// Design library name: HW01
// Design cell name: Inverter
// Design view name: schematic
simulator lang=spectre
global 0
parameters temperature=27
include "/app11/FreePDK45/1.4/ncsu_basekit/models/hspice/hspice_nom.include"

// Library name: HW01
// Cell name: Inverter
// View name: schematic
M0 (OUT IN VDD VDD) PMOS_VTG w=1.5625u l=50n as=1.64063e-13 ad=1.64063e-13 \
    ps=1.7725u pd=1.7725u ld=105n ls=105n m=1
```

```
M1 (OUT IN VSS VSS) NMOS_VTG w=1u l=50n as=1.05e-13 ad=1.05e-13 ps=1.21u \
    pd=1.21u ld=105n ls=105n m=1
simulatorOptions options psfversion="1.4.0" reltol=1e-3 vabstol=1e-6 \
    iabstol=1e-12 temp=27 tnom=27 scalem=1.0 scale=1.0 gmin=1e-12 rforce=1 \
    maxnotes=5 maxwarns=5 digits=5 cols=80 pivrel=1e-3 \
    sensfile="../psf/sens.output" checklimitdest=psf
tran tran stop=20n write="spectre.ic" writefinal="spectre.fc" \
    annotate=status maxiters=5
finalTimeOP info what=oppoint where=rawfile
modelParameter info what=models where=rawfile
element info what=inst where=rawfile
outputParameter info what=output where=rawfile
designParamVals info what=parameters where=rawfile
primitives info what=primitives where=rawfile
subckts info what=subckts where=rawfile
saveOptions options save=allpub
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
// Point Netlist Generated on: Mar 26 17:19:26 2026
// Generated for: spectre
// Design Netlist Generated on: Mar 26 16:22:56 2026
// Design library name: HW01
// Design cell name: q1_pmos_dc
// Design view name: schematic
simulator lang=spectre
global 0
parameters temperature=27 Vd=1 Vg=0
include "/app11/FreePDK45/1.4/ncsu_basekit/models/hspice/hspice_nom.include"

// Library name: HW01
// Cell name: q1_pmos_dc
```

```
// View name: schematic

M0 (net4 net2 net1 net1) PMOS_VTG w=1u l=50n as=1.05e-13 ad=1.05e-13 \
    ps=1.21u pd=1.21u ld=105n ls=105n m=1

V2 (net1 0) vsource dc=1.3 type=dc

V1 (net4 0) vsource dc=Vd type=dc

V0 (net2 0) vsource dc=Vg type=dc

simulatorOptions options psfversion="1.4.0" reltol=1e-3 vabstol=1e-6 \
    iabstol=1e-12 temp=27 tnom=27 scalem=1.0 scale=1.0 gmin=1e-12 rforce=1 \
    maxnotes=5 maxwarns=5 digits=5 cols=80 pivrel=1e-3 \
    sensfile="../psf/sens.output" checklimitdest=psf

dc dc param=Vd start=1.3 stop=0 step=-0.01 write="spectre.dc" \
    oppoint=rawfile maxiters=150 maxsteps=10000 annotate=status

modelParameter info what=models where=rawfile

element info what=inst where=rawfile

outputParameter info what=output where=rawfile

designParamVals info what=parameters where=rawfile

primitives info what=primitives where=rawfile

subckts info what=subckts where=rawfile

save M0:d

saveOptions options save=allpub

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

// Point Netlist Generated on: Mar 26 17:19:26 2026

// Generated for: spectre

// Design Netlist Generated on: Mar 26 16:22:56 2026

// Design library name: HW01

// Design cell name: q1_pmos_dc

// Design view name: schematic

simulator lang=spectre

global 0

parameters temperature=27 Vd=1 Vg=0
```

```
include "/app11/FreePDK45/1.4/ncsu_basekit/models/hspice/hspice_nom.include"

// Library name: HW01

// Cell name: q1_pmos_dc

// View name: schematic

M0 (net4 net2 net1 net1) PMOS_VTG w=1u l=50n as=1.05e-13 ad=1.05e-13 \
    ps=1.21u pd=1.21u ld=105n ls=105n m=1

V2 (net1 0) vsource dc=1.3 type=dc

V1 (net4 0) vsource dc=Vd type=dc

V0 (net2 0) vsource dc=Vg type=dc

simulatorOptions options psfversion="1.4.0" reltol=1e-3 vabstol=1e-6 \
    iabstol=1e-12 temp=27 tnom=27 scalem=1.0 scale=1.0 gmin=1e-12 rforce=1 \
    maxnotes=5 maxwarns=5 digits=5 cols=80 pivrel=1e-3 \
    sensfile="../psf/sens.output" checklimitdest=psf

dc dc param=Vd start=1.3 stop=0 step=-0.01 write="spectre.dc" \
    oppoint=rawfile maxiters=150 maxsteps=10000 annotate=status

modelParameter info what=models where=rawfile

element info what=inst where=rawfile

outputParameter info what=output where=rawfile

designParamVals info what=parameters where=rawfile

primitives info what=primitives where=rawfile

subckts info what=subckts where=rawfile

save M0:d

saveOptions options save=allpub

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

// Point Netlist Generated on: Mar 28 11:14:05 2026

// Generated for: spectre

// Design Netlist Generated on: Mar 27 17:06:12 2026

// Design library name: HW01

// Design cell name: TestBench_Inverter
```

```

// Design view name: schematic

simulator lang=spectre

global 0

parameters temperature=27

include "/app11/FreePDK45/1.4/ncsu_basekit/models/hspice/hspice_nom.include"


// Library name: HW01

// Cell name: Inverter

// View name: schematic

subckt Inverter IN OUT VDD VSS

    M0 (OUT IN VDD VDD) PMOS_VTG w=1.5625u l=50n as=1.64063e-13 \
        ad=1.64063e-13 ps=1.7725u pd=1.7725u ld=105n ls=105n m=1

    M1 (OUT IN VSS VSS) NMOS_VTG w=1u l=50n as=1.05e-13 ad=1.05e-13 \
        ps=1.21u pd=1.21u ld=105n ls=105n m=1

ends Inverter

// End of subcircuit definition.


// Library name: HW01

// Cell name: TestBench_Inverter

// View name: schematic

I2 (IN OUT net3 0) Inverter

V0 (net3 0) vsource dc=1.2 type=dc

V1 (IN 0) vsource type=pulse val0=0 val1=1.2 period=8n delay=1n \
    rise=10.00f fall=10.00f width=4n

simulatorOptions options psfversion="1.4.0" reltol=1e-3 vabstol=1e-6 \
    iabstol=1e-12 temp=27 tnom=27 scalem=1.0 scale=1.0 gmin=1e-12 rforce=1 \
    maxnotes=5 maxwarns=5 digits=5 cols=80 pivrel=1e-3 \
    sensfile="../psf/sens.output" checklimitdest=psf

tran tran stop=20n write="spectre.ic" writefinal="spectre.fc" \
    annotate=status maxiters=5

```

finalTimeOP info what=oppoint where=rawfile

modelParameter info what=models where=rawfile

element info what=inst where=rawfile

outputParameter info what=output where=rawfile

designParamVals info what=parameters where=rawfile

primitives info what=primitives where=rawfile

subckts info what=subckts where=rawfile

saveOptions options save=allpub

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// Point Netlist Generated on: Mar 28 11:14:05 2026

// Generated for: spectre

// Design Netlist Generated on: Mar 27 17:06:12 2026

// Design library name: HW01

// Design cell name: TestBench_Inverter

// Design view name: schematic

simulator lang=spectre

global 0

parameters temperature=27

include "/app11/FreePDK45/1.4/ncsu_basekit/models/hspice/hspice_nom.include"

// Library name: HW01

// Cell name: Inverter

// View name: schematic

subckt Inverter IN OUT VDD VSS

M0 (OUT IN VDD VDD) PMOS_VTG w=1.5625u l=50n as=1.64063e-13 \

ad=1.64063e-13 ps=1.7725u pd=1.7725u ld=105n ls=105n m=1

M1 (OUT IN VSS VSS) NMOS_VTG w=1u l=50n as=1.05e-13 ad=1.05e-13 \

ps=1.21u pd=1.21u ld=105n ls=105n m=1

ends Inverter

// End of subcircuit definition.

```
// Library name: HW01

// Cell name: TestBench_Inverter

// View name: schematic

I2 (IN OUT net3 0) Inverter

V0 (net3 0) vsource dc=1.2 type=dc

V1 (IN 0) vsource type=pulse val0=0 val1=1.2 period=8n delay=1n \
    rise=10.00f fall=10.00f width=4n

simulatorOptions options psfversion="1.4.0" reltol=1e-3 vabstol=1e-6 \
    iabstol=1e-12 temp=27 tnom=27 scalem=1.0 scale=1.0 gmin=1e-12 rforce=1 \
    maxnotes=5 maxwarns=5 digits=5 cols=80 pivrel=1e-3 \
    sensfile="../psf/sens.output" checklimitdest=psf

tran tran stop=20n write="spectre.ic" writefinal="spectre.fc" \
    annotate=status maxiters=5

finalTimeOP info what=oppoint where=rawfile

modelParameter info what=models where=rawfile

element info what=inst where=rawfile

outputParameter info what=output where=rawfile

designParamVals info what=parameters where=rawfile

primitives info what=primitives where=rawfile

subckts info what=subckts where=rawfile

saveOptions options save=allpub

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// Point Netlist Generated on: Mar 30 19:42:40 2026

// Generated for: spectre

// Design Netlist Generated on: Mar 30 19:34:47 2026

// Design library name: HW01

// Design cell name: TestBench_InverterChain

// Design view name: schematic

simulator lang=spectre
```

global 0

parameters temperature=27 tf=10.25p tr=10.25p

include "/app11/FreePDK45/1.4/ncsu_basekit/models/hspice/hspice_nom.include"

// Library name: HW01

// Cell name: Inverter_2X

// View name: schematic

subckt Inverter_2X IN OUT VDD VSS

M0 (OUT IN VDD VDD) PMOS_VTG w=3.125u l=50n as=3.28125e-13 \

ad=3.28125e-13 ps=3.335u pd=3.335u ld=105n ls=105n m=1

M1 (OUT IN VSS VSS) NMOS_VTG w=2u l=50n as=2.1e-13 ad=2.1e-13 ps=2.21u \

pd=2.21u ld=105n ls=105n m=1

ends Inverter_2X

// End of subcircuit definition.

// Library name: HW01

// Cell name: TestBench_InverterChain

// View name: schematic

V0 (VIN 0) vsource type=pulse val0=1.2 val1=0 period=16n delay=1n rise=tr \

fall=tf width=8n

V2 (net2 0) vsource dc=1.2 type=dc

V1 (net1 0) vsource dc=1.2 type=dc

I29 (VMID VOUT net2 0) Inverter_2X

I28 (VIN VMID net1 0) Inverter_2X

simulatorOptions options psfversion="1.4.0" reltol=1e-3 vabstol=1e-6 \

iabstol=1e-12 temp=27 tnom=27 scalem=1.0 scale=1.0 gmin=1e-12 rforce=1 \

maxnotes=5 maxwarns=5 digits=5 cols=80 pivrel=1e-3 \

sensfile="../psf/sens.output" checklimitdest=psf

tran tran stop=20n write="spectre.ic" writefinal="spectre.fc" \

annotate=status maxiters=5

finalTimeOP info what=oppoint where=rawfile

modelParameter info what=models where=rawfile

element info what=inst where=rawfile

outputParameter info what=output where=rawfile

designParamVals info what=parameters where=rawfile

primitives info what=primitives where=rawfile

subckts info what=subckts where=rawfile

saveOptions options save=allpub

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// Point Netlist Generated on: Mar 30 19:42:40 2026

// Generated for: spectre

// Design Netlist Generated on: Mar 30 19:34:47 2026

// Design library name: HW01

// Design cell name: TestBench_InverterChain

// Design view name: schematic

simulator lang=spectre

global 0

parameters temperature=27 tf=10.25p tr=10.25p

include "/app11/FreePDK45/1.4/ncsu_basekit/models/hspice/hspice_nom.include"

// Library name: HW01

// Cell name: Inverter_2X

// View name: schematic

subckt Inverter_2X IN OUT VDD VSS

M0 (OUT IN VDD VDD) PMOS_VTG w=3.125u l=50n as=3.28125e-13 \

ad=3.28125e-13 ps=3.335u pd=3.335u ld=105n ls=105n m=1

M1 (OUT IN VSS VSS) NMOS_VTG w=2u l=50n as=2.1e-13 ad=2.1e-13 ps=2.21u \

pd=2.21u ld=105n ls=105n m=1

ends Inverter_2X

// End of subcircuit definition.

```
// Library name: HW01

// Cell name: TestBench_InverterChain

// View name: schematic

V0 (VIN 0) vsource type=pulse val0=1.2 val1=0 period=16n delay=1n rise=tr \
    fall=tf width=8n

V2 (net2 0) vsource dc=1.2 type=dc

V1 (net1 0) vsource dc=1.2 type=dc

I29 (VMID VOUT net2 0) Inverter_2X

I28 (VIN VMID net1 0) Inverter_2X

simulatorOptions options psfversion="1.4.0" reltol=1e-3 vabstol=1e-6 \
    iabstol=1e-12 temp=27 tnom=27 scalem=1.0 scale=1.0 gmin=1e-12 rforce=1 \
    maxnotes=5 maxwarns=5 digits=5 cols=80 pivrel=1e-3 \
    sensfile="../psf/sens.output" checklimitdest=psf

tran tran stop=20n write="spectre.ic" writefinal="spectre.fc" \
    annotate=status maxiters=5

finalTimeOP info what=oppoint where=rawfile

modelParameter info what=models where=rawfile

element info what=inst where=rawfile

outputParameter info what=output where=rawfile

designParamVals info what=parameters where=rawfile

primitives info what=primitives where=rawfile

subckts info what=subckts where=rawfile

saveOptions options save=allpub

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// Point Netlist Generated on: Mar 31 00:36:53 2026

// Generated for: spectre

// Design Netlist Generated on: Mar 31 00:36:52 2026

// Design library name: HW01

// Design cell name: q1_nmos_dc
```

